US Patent & Trademark Office
Patent Public Search | Text View

United States Patent	12408338
Kind Code	B2
Date of Patent	September 02, 2025
Inventor(s)	Zhu; Hongbin et al.

Memory devices having vertical transistors and methods for forming the same

Abstract

In certain aspects, a three-dimensional (3D) memory device includes a first semiconductor structure, a second semiconductor structure, a third semiconductor structure, a first bonding interface between the first semiconductor structure and the second semiconductor structure, and a second bonding interface between the second semiconductor structure and the third semiconductor structure. The first semiconductor structure includes a peripheral circuit. The second semiconductor structure includes a first array of memory cells. The third semiconductor structure includes a second array of memory cells. Each of the memory cells of the first and second arrays includes a vertical transistor extending in a first direction, and a storage unit coupled to the vertical transistor. The first array of memory cells is coupled to the peripheral circuit across the first bonding interface. The second array of memory cells is coupled to the peripheral circuit across the first bonding interface and the second bonding interfaces.

Inventors: **Zhu; Hongbin (Wuhan, CN), Liu; Wei (Wuhan, CN), Wang; Yanhong (Wuhan, CN)**

Applicant: **YANGTZE MEMORY TECHNOLOGIES CO., LTD.** (Wuhan, CN)

Family ID: **85286282**

Assignee: **YANGTZE MEMORY TECHNOLOGIES CO., LTD. (Wuhan, CN)**

Appl. No.: **17/553763**

Filed: **December 16, 2021**

Prior Publication Data

Document Identifier	Publication Date
US 20230062083 A1	Mar. 02, 2023

Related U.S. Application Data

continuation parent-doc WO PCT/CN2021/115545 20210831 PENDING child-doc US 17553763

Publication Classification

Int. Cl.: **H10B41/40** (20230101); **G11C7/18** (20060101); **G11C8/14** (20060101); **H01L23/528** (20060101); **H10B41/10** (20230101); **H10B41/27** (20230101); **H10B43/10** (20230101); **H10B43/27** (20230101); **H10B43/40** (20230101)

U.S. Cl.:

CPC **H10B41/40** (20230201); **G11C7/18** (20130101); **G11C8/14** (20130101); **H01L23/5283** (20130101); **H10B41/10** (20230201); **H10B41/27** (20230201); **H10B43/10** (20230201); **H10B43/27** (20230201); **H10B43/40** (20230201);

Field of Classification Search

USPC: None

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
10074667	12/2017	Kwon	N/A	N/A
10181472	12/2018	Huang	N/A	N/A
12176310	12/2023	Liu	N/A	H01L 25/0657
2005/0275006	12/2004	Tang	N/A	N/A
2005/0280061	12/2004	Le	N/A	N/A
2006/0172483	12/2005	Forbes	N/A	N/A
2008/0061340	12/2007	Heineck et al.	N/A	N/A
2009/0072291	12/2008	Takaishi	N/A	N/A
2010/0102361	12/2009	Chen	N/A	N/A
2012/0033487	12/2011	Inoue et al.	N/A	N/A
2013/0126954	12/2012	Wu et al.	N/A	N/A
2013/0320397	12/2012	Sze et al.	N/A	N/A
2014/0374688	12/2013	Wu et al.	N/A	N/A
2016/0049406	12/2015	Sandhu	N/A	N/A
2017/0213821	12/2016	Or-Bach	N/A	N/A
2018/0061834	12/2017	Derner et al.	N/A	N/A
2018/0061835	12/2017	Yang et al.	N/A	N/A
2018/0204931	12/2017	Reboh	N/A	N/A
2018/0261575	12/2017	Tagami et al.	N/A	N/A
2018/0308559	12/2017	Kim et al.	N/A	N/A
2019/0081017	12/2018	Nakajima	N/A	N/A
2019/0237581	12/2018	Saito et al.	N/A	N/A
2019/0326292	12/2018	Mathew et al.	N/A	N/A
2020/0111800	12/2019	Ramaswamy	N/A	N/A
2020/0144330	12/2019	Majhi	N/A	N/A
2020/0258897	12/2019	Yan et al.	N/A	N/A

2020/0388336	12/2019	Yun et al.	N/A	N/A
2021/0013210	12/2020	Lee	N/A	N/A
2021/0028174	12/2020	Lee	N/A	N/A
2021/0028176	12/2020	Derner et al.	N/A	N/A
2021/0066278	12/2020	Kim et al.	N/A	N/A
2021/0066281	12/2020	Kwon	N/A	N/A
2021/0351182	12/2020	Yang et al.	N/A	N/A
2021/0399133	12/2020	Li	N/A	N/A
2023/0060149	12/2022	Zhu	N/A	N/A
2023/0062083	12/2022	Zhu	N/A	N/A
2023/0062524	12/2022	Liu	N/A	N/A
2023/0064388	12/2022	Liu	N/A	N/A
2023/0065806	12/2022	Zhu	N/A	N/A
2023/0066312	12/2022	Zhu	N/A	N/A
2023/0228739	12/2022	Naj	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	**Application Date**	**Country**	**CPC**
1434515	12/2002	CN	N/A
1819205	12/2005	CN	N/A
101090117	12/2006	CN	N/A
101789433	12/2009	CN	N/A
102522407	12/2011	CN	N/A
108807413	12/2017	CN	N/A
108807660	12/2017	CN	N/A
109155311	12/2018	CN	N/A
109411473	12/2018	CN	N/A
109417075	12/2018	CN	N/A
109461737	12/2018	CN	N/A
109461738	12/2018	CN	N/A
208655644	12/2018	CN	N/A
110024133	12/2018	CN	N/A
110192269	12/2018	CN	N/A
110265292	12/2018	CN	N/A
110265397	12/2018	CN	N/A
110291631	12/2018	CN	N/A
110476209	12/2018	CN	N/A
110537260	12/2018	CN	N/A
110720145	12/2019	CN	N/A
110731012	12/2019	CN	N/A
110770898	12/2019	CN	N/A
110770901	12/2019	CN	N/A
110870062	12/2019	CN	N/A
110896074	12/2019	CN	N/A
110914987	12/2019	CN	N/A
110945652	12/2019	CN	N/A
110998844	12/2019	CN	N/A
111727503	12/2019	CN	N/A
112164674	12/2020	CN	N/A
112310079	12/2020	CN	N/A

112470274	12/2020	CN	N/A
112864158	12/2020	CN	N/A
112997319	12/2020	CN	N/A
113169182	12/2020	CN	N/A
113206099	12/2020	CN	N/A
113506736	12/2020	CN	N/A
113506737	12/2020	CN	N/A
113517346	12/2020	CN	N/A
113611665	12/2020	CN	N/A
113611667	12/2020	CN	N/A
113629013	12/2020	CN	N/A
20140106903	12/2013	KR	N/A
2013091374	12/2012	WO	N/A
2020211272	12/2019	WO	N/A

OTHER PUBLICATIONS

CN 110945652, Stacked Three-dimensional Heterogenous Memory Device and Forming Method Thereof, Mar. 31, 2020 (Year: 2020). cited by examiner
CN 208655644, Zhu, Memory and Semiconductor Device, Aug. 31, 2018. (Year: 2018). cited by examiner
International Search Report issued in corresponding International Application No. PCT/CN2021/115594, mailed May 27, 2022, 4 pages. cited by applicant
International Search Report issued in corresponding International Application No. PCT/CN2021/115545, mailed May 31, 2022, 5 pages. cited by applicant
International Search Report issued in corresponding International Application No. PCT/CN2021/115704, mailed May 31, 2022, 5 pages. cited by applicant
International Search Report issued in corresponding International Application No. PCT/CN2021/115743, mailed May 27, 2022, 4 pages. cited by applicant
International Search Report issued in corresponding International Application No. PCT/CN2021/115775, mailed Jun. 6, 2022, 4 pages. cited by applicant
International Search Report issued in corresponding International Application No. PCT/CN2021/115820, mailed May 25, 2022, 4 pages. cited by applicant
International Search Report issued in corresponding International Application No. PCT/CN2021/115652, mailed Apr. 28, 2022, 4 pages. cited by applicant

---

Primary Examiner: Ghyka; Alexander G

Attorney, Agent or Firm: BAYES PLLC

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of International Application No. PCT/CN2021/115545, filed on Aug. 31, 2021, entitled “MEMORY DEVICES HAVING VERTICAL TRANSISTORS AND METHODS FOR FORMING THE SAME,” which is hereby incorporated by reference in its entirety. This application is also related to co-pending U.S. application Ser. No. 17/553,759, filed on Dec. 16, 2021, entitled “MEMORY DEVICES HAVING VERTICAL TRANSISTORS AND METHODS FOR FORMING THE SAME,” co-pending U.S. application Ser. No. 17/553,765, filed on Dec. 16, 2021 even date, entitled “MEMORY DEVICES HAVING VERTICAL TRANSISTORS AND METHODS FOR

FORMING THE SAME," co-pending U.S. application Ser. No. 17/553,772, filed on Dec. 16, 2021, entitled "MEMORY DEVICES HAVING VERTICAL TRANSISTORS AND METHODS FOR FORMING THE SAME," co-pending U.S. application Ser. No. 17/553,773, filed on Dec. 16, 2021, entitled "MEMORY DEVICES HAVING VERTICAL TRANSISTORS AND METHODS FOR FORMING THE SAME," co-pending U.S. application Ser. No. 17/553,776, filed on Dec. 16, 2021, entitled "MEMORY DEVICES HAVING VERTICAL TRANSISTORS AND METHODS FOR FORMING THE SAME," and co-pending U.S. application Ser. No. 17/553,781, filed on Dec. 16, 2021, entitled "memory devices having vertical transistors and methods for forming the same," all of which are hereby incorporated by reference in their entireties.

BACKGROUND

(1) The present disclosure relates to memory devices and fabrication methods thereof.

(2) Planar memory cells are scaled to smaller sizes by improving process technology, circuit design, programming algorithm, and fabrication process. However, as feature sizes of the memory cells approach a lower limit, planar process and fabrication techniques become challenging and costly. As a result, memory density for planar memory cells approaches an upper limit.

(3) A three-dimensional (3D) memory architecture can address the density limitation in planar memory cells. The 3D memory architecture includes a memory array and peripheral circuits for facilitating operations of the memory array.

SUMMARY

(4) In one aspect, a 3D memory device includes a first semiconductor structure, a second semiconductor structure, a third semiconductor structure, a first bonding interface between the first semiconductor structure and the second semiconductor structure, and a second bonding interface between the second semiconductor structure and the third semiconductor structure. The first semiconductor structure includes a peripheral circuit. The second semiconductor structure includes a first array of memory cells. The third semiconductor structure includes a second array of memory cells. Each of the memory cells of the first and second arrays includes a vertical transistor extending in a first direction, and a storage unit coupled to the vertical transistor. The first array of memory cells is coupled to the peripheral circuit across the first bonding interface. The second array of memory cells is coupled to the peripheral circuit across the first bonding interface and the second bonding interfaces.

(5) In another aspect, a memory system includes a memory device configured to store data and a memory controller coupled to the memory device. The memory device includes a first semiconductor structure, a second semiconductor structure, a third semiconductor structure, a first bonding interface between the first semiconductor structure and the second semiconductor structure, and a second bonding interface between the second semiconductor structure and the third semiconductor structure. The first semiconductor structure includes a peripheral circuit. The second semiconductor structure includes a first array of memory cells. The third semiconductor structure includes a second array of memory cells. Each of the memory cells of the first and second arrays includes a vertical transistor extending in a first direction, and a storage unit coupled to the vertical transistor. The first array of memory cells is coupled to the peripheral circuit across the first bonding interface. The second array of memory cells is coupled to the peripheral circuit across the first bonding interface and the second bonding interfaces. The memory controller is configured to control the first and second arrays of memory cells through the peripheral circuit.

(6) In still another aspect, a method for forming a 3D memory device is disclosed. A first semiconductor structure including a peripheral circuit is formed. A second semiconductor structure including a first array of memory cells is formed. Each of the memory cells of the first array includes a vertical transistor extending in the first direction, and a storage unit coupled to the vertical transistor. A third semiconductor structure including a second array of memory cells is formed. Each of the memory cells of the second array includes a vertical transistor extending in the

first direction, and a storage unit coupled to the vertical transistor. The first semiconductor structure and the second semiconductor structure are bonded in a face-to-face manner, such that the first array of memory cells is coupled to the peripheral circuit across a bonding interface. The second semiconductor structure and the third semiconductor structure are bonded in a face-to-face manner.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The accompanying drawings, which are incorporated herein and form a part of the specification, illustrate aspects of the present disclosure and, together with the description, further serve to explain the principles of the present disclosure and to enable a person skilled in the pertinent art to make and use the present disclosure.

(2) FIG. **1**A illustrates a schematic view of a cross-section of a 3D memory device, according to some aspects of the present disclosure.

(3) FIG. **1**B illustrates a schematic view of a cross-section of another 3D memory device, according to some aspects of the present disclosure.

(4) FIG. **2** illustrates a schematic diagram of a memory device including peripheral circuits and an array of memory cells each having a vertical transistor, according to some aspects of the present disclosure.

(5) FIG. **3** illustrates a schematic circuit diagram of a memory device including peripheral circuits and an array of dynamic random-access memory (DRAM) cells, according to some aspects of the present disclosure.

(6) FIG. **4** illustrates a schematic circuit diagram of a memory device including peripheral circuits and an array of phase-change memory (PCM) cells, according to some aspects of the present disclosure.

(7) FIG. **5** illustrates a plan view of an array of memory cells each including a vertical transistor in a memory device, according to some aspects of the present disclosure.

(8) FIG. **6**A illustrates a side view of a cross-section of a 3D memory device including vertical transistors, according to some aspects of the present disclosure.

(9) FIG. **6**B illustrates a side view of a cross-section of another 3D memory device including vertical transistors, according to some aspects of the present disclosure.

(10) FIG. **6**C illustrates a side view of a cross-section of still another 3D memory device including vertical transistors, according to some aspects of the present disclosure.

(11) FIG. **6**D illustrates a side view of a cross-section of yet another 3D memory device including vertical transistors, according to some aspects of the present disclosure.

(12) FIG. **6**E illustrates a side view of a cross-section of yet another 3D memory device including vertical transistors, according to some aspects of the present disclosure.

(13) FIG. **7** illustrates a side view of a cross-section of yet another 3D memory device including vertical transistors, according to some aspects of the present disclosure.

(14) FIG. **8** illustrates a plan view of another array of memory cells each including a vertical transistor in a memory device, according to some aspects of the present disclosure.

(15) FIG. **9** illustrates a side view of a cross-section of yet another 3D memory device including vertical transistors, according to some aspects of the present disclosure.

(16) FIGS. **10**A-**10**M illustrate a fabrication process for forming a 3D memory device including vertical transistors, according to some aspects of the present disclosure.

(17) FIGS. **11**A-**11**I illustrate a fabrication process for forming another 3D memory device including vertical transistors, according to some aspects of the present disclosure.

(18) FIGS. **12**A-**12**H illustrate a fabrication process for forming still another 3D memory device including vertical transistors, according to some aspects of the present disclosure.

(19) FIGS. **13**A-**13**H illustrate a fabrication process for forming yet another 3D memory device including vertical transistors, according to some aspects of the present disclosure.
(20) FIGS. **14**A-**14**E illustrate a fabrication process for forming yet another 3D memory device including vertical transistors, according to some aspects of the present disclosure.
(21) FIGS. **15**A-**15**D illustrate a fabrication process for forming yet another 3D memory device including vertical transistors, according to some aspects of the present disclosure.
(22) FIG. **16** illustrates a plan view of still another array of memory cells each including a vertical transistor in a memory device, according to some aspects of the present disclosure.
(23) FIG. **17** illustrates a side view of a cross-section of yet another 3D memory device including vertical transistors, according to some aspects of the present disclosure.
(24) FIG. **18** illustrates a perspective view of an array of vertical transistors, according to some aspects of the present disclosure.
(25) FIGS. **19**A-**19**M illustrate a fabrication process for forming yet another 3D memory device including vertical transistors, according to some aspects of the present disclosure.
(26) FIG. **20** illustrates a plan view of yet another array of memory cells each including a vertical transistor in a memory device, according to some aspects of the present disclosure.
(27) FIG. **21** illustrates a side view of a cross-section of yet another 3D memory device including vertical transistors, according to some aspects of the present disclosure.
(28) FIGS. **22**A-**22**M illustrate a fabrication process for forming yet another 3D memory device including vertical transistors, according to some aspects of the present disclosure.
(29) FIG. **23** illustrates a flowchart of a method for forming a 3D memory device including vertical transistors, according to some aspects of the present disclosure.
(30) FIG. **24** illustrates a flowchart of a method for forming an array of memory cells each including a vertical transistor, according to some aspects of the present disclosure.
(31) FIG. **25** illustrates a flowchart of a method for forming another array of memory cells each including a vertical transistor, according to some aspects of the present disclosure.
(32) FIG. **26** illustrates a flowchart of a method for forming still another array of memory cells each including a vertical transistor, according to some aspects of the present disclosure.
(33) FIG. **27** illustrates a block diagram of an exemplary system having a memory device, according to some aspects of the present disclosure.
(34) The present disclosure will be described with reference to the accompanying drawings.

DETAILED DESCRIPTION

(35) Although specific configurations and arrangements are discussed, it should be understood that this is done for illustrative purposes only. As such, other configurations and arrangements can be used without departing from the scope of the present disclosure. Also, the present disclosure can also be employed in a variety of other applications. Functional and structural features as described in the present disclosures can be combined, adjusted, and modified with one another and in ways not specifically depicted in the drawings, such that these combinations, adjustments, and modifications are within the scope of the present disclosure.
(36) In general, terminology may be understood at least in part from usage in context. For example, the term “one or more” as used herein, depending at least in part upon context, may be used to describe any feature, structure, or characteristic in a singular sense or may be used to describe combinations of features, structures or characteristics in a plural sense. Similarly, terms, such as “a,” “an,” or “the,” again, may be understood to convey a singular usage or to convey a plural usage, depending at least in part upon context. In addition, the term “based on” may be understood as not necessarily intended to convey an exclusive set of factors and may, instead, allow for existence of additional factors not necessarily expressly described, again, depending at least in part on context.
(37) It should be readily understood that the meaning of “on,” “above,” and “over” in the present disclosure should be interpreted in the broadest manner such that “on” not only means “directly on”

something but also includes the meaning of "on" something with an intermediate feature or a layer therebetween, and that "above" or "over" not only means the meaning of "above" or "over" something but can also include the meaning it is "above" or "over" something with no intermediate feature or layer therebetween (directly on something).

(38) Further, spatially relative terms, such as "beneath," "below," "lower," "above," "upper," and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations), and the spatially relative descriptors used herein may likewise be interpreted accordingly.

(39) As used herein, the term "substrate" refers to a material onto which subsequent material layers are added. The substrate itself can be patterned. Materials added on top of the substrate can be patterned or can remain unpatterned. Furthermore, the substrate can include a wide array of semiconductor materials, such as silicon, germanium, gallium arsenide, indium phosphide, etc. Alternatively, the substrate can be made from an electrically non-conductive material, such as a glass, a plastic, or a sapphire wafer.

(40) As used herein, the term "layer" refers to a material portion including a region with a thickness. A layer can extend over the entirety of an underlying or overlying structure or may have an extent less than the extent of an underlying or overlying structure. Further, a layer can be a region of a homogeneous or inhomogeneous continuous structure that has a thickness less than the thickness of the continuous structure. For example, a layer can be located between any pair of horizontal planes between, or at, a top surface and a bottom surface of the continuous structure. A layer can extend horizontally, vertically, and/or along a tapered surface. A substrate can be a layer, can include one or more layers therein, and/or can have one or more layers thereupon, thereabove, and/or therebelow. A layer can include multiple layers. For example, an interconnect layer can include one or more conductors and contact layers (in which interconnect lines and/or vertical interconnect access (via) contacts are formed) and one or more dielectric layers.

(41) Transistors are used as the switch or selecting devices in the memory cells of some memory devices, such as DRAM, PCM, and ferroelectric DRAM (FRAM). However, the planar transistors commonly used in existing memory cells usually have a horizontal structure with buried word lines in the substrate and bit lines above the substrate. Since the source and drain of a planar transistor are disposed laterally at different locations, which increases the area occupied by the transistor. The design of planar transistors also complicates the arrangement of interconnected structures, such as word lines and bit lines, coupled to the memory cells, for example, limiting the pitches of the word lines and/or bit lines, thereby increasing the fabrication complexity and reducing the production yield. Moreover, because the bit lines and the storage units (e.g., capacitors or PCM elements) are arranged on the same side of the planar transistors (above the transistors and substrate), the bit line process margin is limited by the storage units, and the coupling capacitance between the bit lines and storage units, such as capacitors, are increased. Planar transistors may also suffer from a high leakage current as the saturated drain current keeps increasing, which is undesirable for the performance of memory devices.

(42) On the other hand, the memory cell array and the peripheral circuits for controlling the memory cell array are usually arranged side-by-side in the same plane. As the number of memory cells keeps increasing, to maintain the same chip size, the dimensions of the components in the memory cell array, such as transistors, word lines, and/or bit lines, need to keep decreasing in order not to significantly reduce the memory cell array efficiency.

(43) To address one or more of the aforementioned issues, the present disclosure introduces a solution in which vertical transistors replace the planar transistors as the switch and selecting devices in a memory cell array of memory devices (e.g., DRAM, PCM, and FRAM). Compared

with planar transistors, the vertically arranged transistors (e.g., the drain and source are overlapped in the plan view) can reduce the area of the transistor as well as simplify the layout of the interconnect structures, e.g., metal wiring the word lines and bit lines, which can reduce the fabrication complexity and improve the yield. For example, the pitches of word lines and/or bit lines can be reduced for ease of fabrication. The vertical structures of the transistors also allow the bit lines and storage units, such as capacitors, to be arranged on opposite sides of the transistors in the vertical direction (e.g., one above and on below the transistors), such that the process margin of the bit lines can be increased and the coupling capacitance between the bit lines and the storage units can be decreased.

(44) Consistent with the scope of the present disclosure, according to some aspects of the present disclosure, the memory cell array having vertical transistors and the peripheral circuits of the memory cell array can be formed on different wafers and bonded together in a face-to-face manner. Thus, the thermal budget of fabricating the memory cell array does not affect the fabrication of the peripheral circuits. The stacked memory cell array and peripheral circuits can also reduce the chip size compared with the side-by-side arrangement, thereby improving the array efficiency. In some implementations, more than one memory cell array is stacked over one another using bonding techniques to further increase the array efficiency. In some implementations, the word lines and bit lines are disposed close to the bonding interface due to the vertically arranged transistors, which can be coupled to the peripheral circuits through a large number (e.g., millions) of parallel bonding contacts across the bonding interface can make direct, short-distance (e.g., micron-level) electrical connections between the memory cell array and peripheral circuits to increase the throughput and input/output (I/O) speed of the memory devices.

(45) In some implementations, the vertical transistors disclosed herein include multi-gate transistors (e.g., gate-all-around (GAA) transistors, tri-gate transistors, or double-gate transistors), which can have a larger gate control area to achieve better channel control with a smaller subthreshold swing. Since the channel is fully depleted, the leakage current of multi-gate transistors can be significantly reduced as well. Thus, using multi-gate transistors instead of planar transistors can achieve a much better speed (saturated drain current)/leakage current performance.

(46) In some implementations, the vertical transistors disclosed herein include single-gate transistors (a.k.a. single-side gate transistors) in a mirror-symmetric arrangement with respect to adjacent transistors in the bit line direction as a result of splitting multi-gate transistors (e.g., double-gate transistors) using trench isolations extending along the word line direction. Thus, the memory cell density in the bit line direction can be significantly increased (e.g., doubled) without unduly complicating the fabrication process compared with using processes, such as self-aligned double patterning (SADP). Also, the mirror-symmetric single-gate transistors have a larger process window for word line, bit line, and transistor pitch reduction, compared to either planar transistors or multi-gate vertical transistors, for example, with dual-side or all-around gates.

(47) FIG. **1**A illustrates a schematic view of a cross-section of a 3D memory device **100**, according to some aspects of the present disclosure. 3D memory device **100** represents an example of a bonded chip. The components of 3D memory device **100** (e.g., memory cell array and peripheral circuits) can be formed separately on different substrates and then jointed to form a bonded chip. 3D memory device **100** can include a first semiconductor structure **102** including the peripheral circuits of a memory cell array. 3D memory device **100** can also include a second semiconductor structure **104** including the memory cell array. The peripheral circuits (a.k.a. control and sensing circuits) can include any suitable digital, analog, and/or mixed-signal circuits used for facilitating the operations of the memory cell array. For example, the peripheral circuit can include one or more of a page buffer, a decoder (e.g., a row decoder and a column decoder), a sense amplifier, a driver (e.g., a word line driver), an input/output (I/O) circuit, a charge pump, a voltage source or generator, a current or voltage reference, any portions (e.g., a sub-circuit) of the functional circuits mentioned above, or any active or passive components of the circuit (e.g., transistors, diodes,

resistors, or capacitors). The peripheral circuits in first semiconductor structure **102** use complementary metal-oxide-semiconductor (CMOS) technology, e.g., which can be implemented with logic processes (e.g., technology nodes of 90 nm, 65 nm, 60 nm, 45 nm, 32 nm, 28 nm, 22 nm, 20 nm, 16 nm, 14 nm, 10 nm, 7 nm, 5 nm, 3 nm, 2 nm, etc.), according to some implementations.

(48) As shown in FIG. **1**A, 3D memory device **100** can also include first semiconductor structure **104** including an array of memory cells (memory cell array) that can use transistors as the switch and selecting devices. In some implementations, the memory cell array includes an array of DRAM cells. For ease of description, a DRAM cell array may be used as an example for describing the memory cell array in the present disclosure. But it is understood that the memory cell array is not limited to DRAM cell array and may include any other suitable types of memory cell arrays that can use transistors as the switch and selecting devices, such as PCM cell array, static random-access memory (SRAM) cell array, FRAM cell array, resistive memory cell array, magnetic memory cell array, spin transfer torque (STT) memory cell array, to name a few, or any combination thereof.

(49) Second semiconductor structure **104** can be a DRAM device in which memory cells are provided in the form of an array of DRAM cells. In some embodiments, each DRAM cell includes a capacitor for storing a bit of data as a positive or negative electrical charge as well as one or more transistors (a.k.a. pass transistors) that control (e.g., switch and selecting) access to it. In some implementations, each DRAM cell is a one-transistor, one-capacitor (1T1C) cell. Since transistors always leak a small amount of charge, the capacitors will slowly discharge, causing information stored in them to drain. As such, a DRAM cell has to be refreshed to retain data, for example, by the peripheral circuit in first semiconductor structure **102**, according to some implementation.

(50) As shown in FIG. **1**A, 3D memory device **100** further includes a bonding interface **106** vertically between (in the vertical direction, e.g., the z-direction in FIG. **1**A) first semiconductor structure **102** and second semiconductor structure **104**. As described below in detail, first and second semiconductor structures **102** and **104** can be fabricated separately (and in parallel in some implementations) such that the thermal budget of fabricating one of first and second semiconductor structures **102** and **104** does not limit the processes of fabricating another one of first and second semiconductor structures **102** and **104**. Moreover, a large number of interconnects (e.g., bonding contacts) can be formed through bonding interface **106** to make direct, short-distance (e.g., micron-level) electrical connections between first semiconductor structure **102** and second semiconductor structure **104**, as opposed to the long-distance (e.g., millimeter or centimeter-level) chip-to-chip data bus on the circuit board, such as printed circuit board (PCB), thereby eliminating chip interface delay and achieving high-speed I/O throughput with reduced power consumption. Data transfer between the memory cell array in second semiconductor structure **104** and the peripheral circuits in first semiconductor structure **102** can be performed through the interconnects (e.g., bonding contacts) across bonding interface **106**. By vertically integrating first and second semiconductor structures **102** and **104**, the chip size can be reduced, and the memory cell density can be increased.

(51) It is understood that the relative positions of stacked first and second semiconductor structures **102** and **104** are not limited. FIG. **1**B illustrates a schematic view of a cross-section of another exemplary 3D memory device **101**, according to some implementations. Different from 3D memory device **100** in FIG. **1**A in which second semiconductor structure **104** including the memory cell array is above first semiconductor structure **102** including the peripheral circuits, in 3D memory device **101** in FIG. **1**B, first semiconductor structure **102** including the peripheral circuit is above second semiconductor structure **104** including the memory cell array. Nevertheless, bonding interface **106** is formed vertically between first and second semiconductor structures **102** and **104** in 3D memory device **101**, and first and second semiconductor structures **102** and **104** are jointed vertically through bonding (e.g., hybrid bonding) according to some implementations.

Hybrid bonding, also known as “metal/dielectric hybrid bonding,” is a direct bonding technology (e.g., forming bonding between surfaces without using intermediate layers, such as solder or adhesives) and can obtain metal-metal (e.g., copper-to-copper) bonding and dielectric-dielectric (e.g., silicon oxide-to-silicon oxide) bonding simultaneously. Data transfer between the memory cell array in second semiconductor structure **104** and the peripheral circuits in first semiconductor structure **102** can be performed through the interconnects (e.g., bonding contacts) across bonding interface **106**.
(52) It is noted that x, y, and z axes are included in FIGS. **1**A and **1**B to further illustrate the spatial relationship of the components in 3D memory devices **100** and **101**. The substrate of the 3D memory device includes two lateral surfaces extending laterally in the x-y plane: a top surface on the front side of the wafer on which the semiconductor devices can be formed, and a bottom surface on the backside opposite to the front side of the wafer. The z-axis is perpendicular to both the x and y axes. As used herein, whether one component (e.g., a layer or a device) is “on,” “above,” or “below” another component (e.g., a layer or a device) of the 3D memory device is determined relative to the substrate of the 3D memory device in the z-direction (the vertical direction perpendicular to the x-y plane, e.g., the thickness direction of the substrate) when the substrate is positioned in the lowest plane of the 3D memory device in the z-direction. The same notion for describing the spatial relationships is applied throughout the present disclosure.
(53) FIG. **2** illustrates a schematic diagram of a memory device **200** including peripheral circuits and an array of memory cells each having a vertical transistor, according to some aspects of the present disclosure. Memory device **200** can include a memory cell array **201** and peripheral circuits **202** coupled to memory cell array **201**. 3D memory devices **100** and **101** may be examples of memory device **200** in which memory cell array **201** and peripheral circuits **202** may be included in second and first semiconductor structures **104** and **102**, respectively. Memory cell array **201** can be any suitable memory cell array in which each memory cell **208** includes a vertical transistor **210** and a storage unit **212** coupled to vertical transistor **210**. In some implementations, memory cell array **201** is a DRAM cell array, and storage unit **212** is a capacitor for storing charge as the binary information stored by the respective DRAM cell. In some implementations, memory cell array **201** is a PCM cell array, and storage unit **212** is a PCM element (e.g., including chalcogenide alloys) for storing binary information of the respective PCM cell based on the different resistivities of the PCM element in the amorphous phase and the crystalline phase. In some implementations, memory cell array **201** is a FRAM cell array, and storage unit **212** is a ferroelectric capacitor for storing binary information of the respective FRAM cell based on the switch between two polarization states of ferroelectric materials under an external electric field.
(54) As shown in FIG. **2**, memory cells **208** can be arranged in a two-dimensional (2D) array having rows and columns. Memory device **200** can include word lines **204** coupling peripheral circuits **202** and memory cell array **201** for controlling the switch of vertical transistors **210** in memory cells **208** located in a row, as well as bit lines **206** coupling peripheral circuits **202** and memory cell array **201** for sending data to and/or receiving data from memory cells **208** located in a column. That is, each word line **204** is coupled to a respective row of memory cells **208**, and each bit line is coupled to a respective column of memory cells **208**.
(55) Consistent with the scope of the present disclosure, vertical transistors **210**, such as vertical metal-oxide-semiconductor field-effect transistors (MOSFETs), can replace the planar transistors as the pass transistors of memory cells **208** to reduce the area occupied by the pass transistors, the coupling capacitance, as well as the interconnect routing complexity, as described below in detail. As shown in FIG. **2**, in some implementations, different from planar transistors in which the active regions are formed in the substrates, vertical transistor **210** includes a semiconductor body **214** extending vertically (in the z-direction) above the substrate (not shown). That is, semiconductor body **214** can extend above the top surface of the substrate to allow channels to be formed not only at the top surface of semiconductor body **214**, but also at one or more side surfaces thereof. As

shown in FIG. **2**, for example, semiconductor body **214** can have a cuboid shape to expose four sides thereof. It is understood that semiconductor body **214** may have any suitable 3D shape, such as polyhedron shapes or a cylinder shape. That is, the cross-section of semiconductor body **214** in the plan view (e.g., in the x-y plane) can have a square shape, a rectangular shape (or a trapezoidal shape), a circular (or an oval shape), or any other suitable shapes. It is understood that consistent with the scope of the present disclosure, for semiconductor bodies that have a circular or oval shape of their cross-sections in the plan view, the semiconductor bodies may still be considered as having multiple sides, such that the gate structures are in contact with more than one side of the semiconductor bodies. As described below with respect to the fabrication process, semiconductor body **214** can be formed from the substrate (e.g., by etching or epitaxy) and thus, has the same semiconductor material (e.g., silicon crystalline silicon) as the substrate (e.g., a silicon substrate).

(56) As shown in FIG. **2**, vertical transistor **210** can also include a gate structure **216** in contact with one or more sides of semiconductor body **214**, e.g., in one or more planes of the side surface(s) of the active region. In other words, the active region of vertical transistor **210**, e.g., semiconductor body **214**, can be at least partially surrounded by gate structure **216**. Gate structure **216** can include a gate dielectric **218** over one or more sides of semiconductor body **214**, e.g., in contact with four side surfaces of semiconductor body **214** as shown in FIG. **2**. Gate structure **216** can also include a gate electrode **220** over and in contact with gate dielectric **218**. Gate dielectric **218** can include any suitable dielectric materials, such as silicon oxide, silicon nitride, silicon oxynitride, or high-k dielectrics. For example, gate dielectric **218** may include silicon oxide, which is a form of gate oxide. Gate electrode **220** can include any suitable conductive materials, such as polysilicon, metals (e.g., tungsten (W), copper (Cu), aluminum (Al), etc.), metal compounds (e.g., titanium nitride (TiN), tantalum nitride (TaN), etc.), or silicides. For example, gate electrode **220** may include doped polysilicon, which is a form of a gate poly. In some implementations, gate electrode **220** includes multiple conductive layers, such as a W layer over a TiN layer. It is understood that gate electrode **220** and word line **204** may be a continuous conductive structure in some examples. In other words, gate electrode **220** may be viewed as part of word line **204** that forms gate structure **216**, or word line **204** may be viewed as the extension of gate electrode **220** to be coupled to peripheral circuits **202**.

(57) As shown in FIG. **2**, vertical transistor **210** can further include a pair of a source and a drain (S/D, dope regions, a.k.a., source electrode and drain electrode) formed at the two ends of semiconductor body **214** in the vertical direction (the z-direction), respectively. The source and drain can be doped with any suitable P-type dopants, such as boron (B) or Gallium (Ga), or any suitable N-type dopants, such as phosphorus (P) or arsenic (As). The source and drain can be separated by gate structure **216** in the vertical direction (the z-direction). In other words, gate structure **216** is formed vertically between the source and drain. As a result, one or more channels (not shown) of vertical transistor **210** can be formed in semiconductor body **214** vertically between the source and drain when a gate voltage applied to gate electrode **220** of gate structure **216** is above the threshold voltage of vertical transistor **210**. That is, each channel of vertical transistors **210** is also formed in the vertical direction along which semiconductor body **214** extends, according to some implementations.

(58) In some implementations, as shown in FIG. **2**, vertical transistor **210** is a multi-gate transistor. That is, gate structure **216** can be in contact with more than one side of semiconductor body **214** (e.g., four sides in FIG. **2**) to form more than one gate, such that more than one channel can be formed between the source and drain in operation. That is, different from the planar transistor that includes only a single planar gate (and resulting in a single planar channel), vertical transistor **210** shown in FIG. **2** can include multiple vertical gates on multiple sides of semiconductor body **214** due to the 3D structure of semiconductor body **214** and gate structure **216** that surrounds the multiple sides of semiconductor body **214**. As a result, compared with planar transistors, vertical transistor **210** shown in FIG. **2** can have a larger gate control area to achieve better channel control

with a smaller subthreshold swing. Since the channel is fully depleted, the leakage current (.sub.Ioff) of vertical transistor **210** can be significantly reduced a well. As described below in detail, the multi-gate vertical transistors can include double-gate vertical transistors (e.g., dual-side gate vertical transistors), tri-gate vertical transistors (e.g., tri-side gate vertical transistors), and GAA vertical transistors.

(59) It is understood that although vertical transistor **210** is shown as a multi-gate transistor in FIG. **2**, the vertical transistors disclosed herein may also include single-gate transistors as described below in detail. That is, gate structure **216** may be in contact with a single side of semiconductor body **214**, for example, for the purpose of increasing the transistor and memory cell density. It is also understood that although gate dielectric **218** is shown as being separate (a separate structure) from other gate dielectrics of adjacent vertical transistors (not shown), gate dielectric **218** may be part of a continuous dielectric layer having multiple gate dielectrics of vertical transistors.

(60) In planar transistors and some lateral multiple-gate transistors (e.g., FinFET), the active regions, such as semiconductor bodies (e.g., Fins), extend laterally (in the x-y plane), and the source and the drain are disposed at different locations in the same lateral plane (the x-y plane). In contrast, in vertical transistor **210**, semiconductor body **214** extends vertically (in the z-direction), and the source and the drain are disposed in the different lateral planes, according to some implementations. In some implementations, the source and the drain are formed at two ends of semiconductor body **214** in the vertical direction (the z-direction), respectively, thereby being overlapped in the plan view. As a result, the area (in the x-y plane) occupied by vertical transistor **210** can be reduced compared with planar transistor and lateral multiple-gate transistors. Also, the metal wiring coupled to vertical transistors **210** can be simplified as well since the interconnects can be routed in different planes. For example, bit lines **206** and storage units **212** may be formed on opposite sides of vertical transistor **210**. In one example, bit line **206** may be coupled to the source or the drain at the upper end of semiconductor body **214**, while storage unit **212** may be coupled to the other source or the drain at the lower end of semiconductor body **214**.

(61) As shown in FIG. **2**, storage unit **212** can be coupled to the source or the drain of vertical transistor **210**. Storage unit **212** can include any devices that are capable of storing binary data (e.g., 0 and 1), including but not limited to, capacitors for DRAM cells and FRAM cells, and PCM elements for PCM cells. In some implementations, vertical transistor **210** controls the selection and/or the state switch of the respective storage unit **212** coupled to vertical transistor **210**. In some implementations as shown in FIG. **3**, each memory cell **208** is a DRAM cell **302** including a transistor **304** (e.g., implementing using vertical transistors **210** in FIG. **2**) and a capacitor **306** (e.g., an example of storage unit **212** in FIG. **2**). The gate of transistor **304** (e.g., corresponding to gate electrode **220**) may be coupled to word line **204**, one of the source and the drain of transistor **304** may be coupled to bit line **206**, the other one of the source and the drain of transistor **304** may be coupled to one electrode of capacitor **306**, and the other electrode of capacitor **306** may be coupled to the ground. In some implementations as shown in FIG. **4**, each memory cell **208** is a PCM cell **402** including a transistor **404** (e.g., implementing using vertical transistors **210** in FIG. **2**) and a PCM element **406** (e.g., an example of storage unit **212** in FIG. **2**). The gate of transistor **404** (e.g., corresponding to gate electrode **220**) may be coupled to word line **204**, one of the source and the drain of transistor **404** may be coupled to the ground, the other one of the source and the drain of transistor **404** may be coupled to one electrode of PCM element **406**, and the other electrode of PCM element **406** may be coupled to bit line **206**.

(62) Peripheral circuits **202** can be coupled to memory cell array **201** through bit lines **206**, word lines **204**, and any other suitable metal wirings. As described above, peripheral circuits **202** can include any suitable circuits for facilitating the operations of memory cell array **201** by applying and sensing voltage signals and/or current signals through word lines **204** and bit lines **206** to and from each memory cell **208**. Peripheral circuits **202** can include various types of peripheral circuits formed using CMOS technologies.

(63) According to some aspects of the present disclosure, the vertical transistors of memory cells in a memory device (e.g., memory device **200**) are multi-gate transistors, and the gate dielectrics of vertical transistors in the word line direction are separate. For example, FIG. **5** illustrates a plan view of an array of memory cells **502** each including a vertical transistor in a memory device **500**, according to some aspects of the present disclosure. As shown in FIG. **5**, memory device **500** can include a plurality of word lines **504** each extending in a first lateral direction (the x-direction, referred to as the word line direction). Memory device **500** can also include a plurality of bit lines **506** each extending in a second lateral direction perpendicular to the first lateral direction (the y-direction, referred to as the bit line direction). It is understood that FIG. **5** does not illustrate a cross-section of memory device **500** in the same lateral plane, and word lines **504** and bit lines **506** may be formed in different lateral planes for ease of routing as described below in detail.
(64) Memory cells **502** can be formed at the intersections of word lines **504** and bit lines **506**. In some implementations, each memory cell **502** includes a vertical transistor (e.g., vertical transistor **210** in FIG. **2**) having a semiconductor body **508** and a gate structure **510**. Semiconductor body **508** can extend in the vertical direction (the z-direction, not shown) perpendicular to the first and second lateral directions. The vertical transistor can be a multi-gate transistor in which gate structure **510** is in contact with a plurality of sides (e.g., all **4** sides in FIG. **5**) of semiconductor body **508** (the active region in which channels are formed). As shown in FIG. **5**, the vertical transistor is a GAA transistor in which gate structure **510** fully circumscribes semiconductor body **508** in the plan view. That is, gate structure **510** circumscribes (e.g., surrounding and contacting) all four sides of semiconductor body **508** (having a rectangle or square-shaped cross-section) in the plan view, according to some implementations. Gate structure **510** can include a gate dielectric **512** fully circumscribes semiconductor body **508** in the plan view, and a gate electrode **514** fully circumscribes gate dielectric **512**. In some implementation, gate dielectric **512** is laterally between gate electrode **514** and semiconductor body **508** in the bit line direction and in the word line direction. As described above, gate electrode **514** may be part of word line **504**, and word line **504** may be an extension of gate electrode **514**. In some implementations, gate structures **510** of a row of the vertical transistors are continuous in the x-direction, as a shown in FIG. **5**.
(65) As shown in FIG. **5**, gate electrodes **514** of adjacent vertical transistors in the word line direction (the x-direction) are continuous, e.g., parts of a continuous conductive layer having gate electrodes **514** and **504**. In contrast, gate dielectrics **512** of adjacent vertical transistors in the word line direction are separate, e.g., not parts of a continuous dielectric layer having gate dielectrics **512**.
(66) FIG. **6**A illustrates a side view of a cross-section of a 3D memory device **600** including vertical transistors, according to some aspects of the present disclosure. 3D memory device **600** may be one example of memory device **500** including multi-gate vertical transistors in which gate structures fully circumscribes semiconductor bodies in the plan view, e.g., GAA vertical transistors. It is understood that FIG. **6**A is for illustrative purposes only and may not necessarily reflect the actual device structure (e.g., interconnections) in practice. As one example of 3D memory device **100** described above with respect to FIG. **1**A, 3D memory device **600** is a bonded chip including first semiconductor structure **102** and second semiconductor structure **104** stacked over first semiconductor structure **102**. First and second semiconductor structures **102** and **104** are jointed at bonding interface **106** therebetween, according to some implementations. As shown in FIG. **6**A, first semiconductor structure **102** can include a substrate **610**, which can include silicon (e.g., single crystalline silicon, c-Si), silicon germanium (SiGe), gallium arsenide (GaAs), germanium (Ge), silicon-on-insulator (SOI), or any other suitable materials.
(67) First semiconductor structure **102** can include peripheral circuits **612** on substrate **610**. In some implementations, peripheral circuits **612** includes a plurality of transistors **614** (e.g., planar transistors and/or 3D transistors). Trench isolations (e.g., shallow trench isolations (STIs)) and doped regions (e.g., wells, sources, and drains of transistors **614**) can be formed on or in substrate

610 as well.

(68) In some implementations, first semiconductor structure **102** further includes an interconnect layer **616** above peripheral circuits **612** to transfer electrical signals to and from peripheral circuits **612**. Interconnect layer **616** can include a plurality of interconnects (also referred to herein as "contacts"), including lateral interconnect lines and vertical interconnect access (VIA) contacts. As used herein, the term "interconnects" can broadly include any suitable types of interconnects, such as middle-end-of-line (MEOL) interconnects and back-end-of-line (BEOL) interconnects. Interconnect layer **616** can further include one or more interlayer dielectric (ILD) layers (also known as "intermetal dielectric (IMD) layers") in which the interconnect lines and via contacts can form. That is, interconnect layer **616** can include interconnect lines and via contacts in multiple ILD layers. In some implementations, peripheral circuits **612** are coupled to one another through the interconnects in interconnect layer **616**. The interconnects in interconnect layer **616** can include conductive materials including, but not limited to, W, Co, Cu, Al, doped silicon, silicides, or any combination thereof. The ILD layers can be formed with dielectric materials including, but not limited to, silicon oxide, silicon nitride, silicon oxynitride, low-k dielectrics, or any combination thereof.

(69) As shown in FIG. **6**A, first semiconductor structure **102** can further include a bonding layer **618** at bonding interface **106** and above interconnect layer **616** and peripheral circuits **612**. Bonding layer **618** can include a plurality of bonding contacts **619** and dielectrics electrically isolating bonding contacts **619**. Bonding contacts **619** can include conductive materials, such as Cu. The remaining area of bonding layer **618** can be formed with dielectric materials, such as silicon oxide. Bonding contacts **619** and surrounding dielectrics in bonding layer **618** can be used for hybrid bonding. Similarly, as shown in FIG. **6**A, second semiconductor structure **104** can also include a bonding layer **620** at bonding interface **106** and above bonding layer **618** of first semiconductor structure **102**. Bonding layer **620** can include a plurality of bonding contacts **621** and dielectrics electrically isolating bonding contacts **621**. Bonding contacts **621** can include conductive materials, such as Cu. The remaining area of bonding layer **620** can be formed with dielectric materials, such as silicon oxide. Bonding contacts **621** and surrounding dielectrics in bonding layer **620** can be used for hybrid bonding. Bonding contacts **621** are in contact with bonding contacts **619** at bonding interface **106**, according to some implementations. In some implementations, bonding layer **620** includes a dielectric layer opposing DRAM cells **624** with bit line **623** positioned between the dielectric layer and DRAM cells **624**, as shown in FIG. **6**A. The dielectric layer can include bonding interface **106** having bonding contacts **621**.

(70) Second semiconductor structure **104** can be bonded on top of first semiconductor structure **102** in a face-to-face manner at bonding interface **106**. In some implementations, bonding interface **106** is disposed between bonding layers **620** and **618** as a result of hybrid bonding (also known as "metal/dielectric hybrid bonding"), which is a direct bonding technology (e.g., forming bonding between surfaces without using intermediate layers, such as solder or adhesives) and can obtain metal-metal bonding and dielectric-dielectric bonding simultaneously. In some implementations, bonding interface **106** is the place at which bonding layers **620** and **618** are met and bonded. In practice, bonding interface **106** can be a layer with a certain thickness that includes the top surface of bonding layer **618** of first semiconductor structure **102** and the bottom surface of bonding layer **620** of second semiconductor structure **104**.

(71) In some implementations, second semiconductor structure **104** further includes an interconnect layer **622** including bit lines **623** above bonding layer **620** to transfer electrical signals. Interconnect layer **622** can include a plurality of interconnects, such as MEOL interconnects and BEOL interconnects. In some implementations, the interconnects in interconnect layer **622** also include local interconnects, such as bit lines **623** (e.g., an example of bit lines **506** in FIG. **5**), bit line contacts **625** (which may be omitted in some examples), and word line contacts **627**. Interconnect layer **622** can further include one or more ILD layers in which the interconnect lines and via

contacts can form. The interconnects in interconnect layer **622** can include conductive materials including, but not limited to, W, Co, Cu, Al, doped silicon, silicides, or any combination thereof. The ILD layers can be formed with dielectric materials including, but not limited to, silicon oxide, silicon nitride, silicon oxynitride, low-k dielectrics, or any combination thereof. In some implementations, peripheral circuits **612** includes a word line driver/row decoder coupled to word line contacts **627** in interconnect layer **622** through bonding contacts **621** and **619** in bonding layers **620** and **618** and interconnect layer **616**. In some implementations, peripheral circuits **612** includes a bit line driver/column decoder coupled to bit lines **623** and bit line contacts **625** in interconnect layer **622** through bonding contacts **621** and **619** in bonding layers **620** and **618** and interconnect layer **616**. In some implementations, bit line **623** is a metal bit line, as opposed to semiconductor bit lines (e.g., doped silicon bit lines). For example, bit line **623** may include W, Co, Cu, Al, or any other suitable metals having higher conductivities than doped silicon. In some implementations, bit line contact **625** is an ohmic contact, such as a metal silicide contact, as opposed to a Schottky contact. For example, bit line contact **625** may include metal silicides, such as WSi, CoSi, CuSi, AlSi, or any other suitable metal silicides having higher conductivities than doped silicon.
(72) In some implementations, second semiconductor structure **104** includes a DRAM device in which memory cells are provided in the form of an array of DRAM cells **624** (e.g., an example of memory cells **502** in FIG. **5**) above interconnect layer **622** and bonding layer **620**. That is, interconnect layer **622** including bit lines **623** can be disposed between bonding layer **620** and array of DRAM cells **624**. It is understood that the cross-section of 3D memory device **600** in FIG. **6**A may be made along the bit line direction (the y-direction), and one bit line **623** in interconnect layer **622** extending laterally in the y-direction may be coupled to a column of DRAM cells **624**.
(73) Each DRAM cell **624** can include a vertical transistor **626** (e.g., an example of vertical transistors **210** in FIG. **2**) and capacitor **628** (e.g., an example of storage unit **212** in FIG. **2**) coupled to the vertical transistor **626**. DRAM cell **624** can be a 1T1C cell consisting of one transistor and one capacitor. It is understood that DRAM cell **624** may be of any suitable configurations, such as 2T1C cell, 3T1C cell, etc.
(74) Vertical transistor **626** can be a MOSFET used to switch a respective DRAM cell **624**. In some implementations, vertical transistor **626** includes a semiconductor body **630** (the active region in which multiple channels can form) extending vertically (in the z-direction), and a gate structure **636** in contact with a plurality of sides of semiconductor body **630**. As described above, as in a GAA vertical transistor, semiconductor body **630** can have a cuboid shape or a cylinder shape, and gate structure **636** can fully circumscribe semiconductor body **630** in the plan view, for example, as shown in FIG. **5**. Gate structure **636** includes a gate electrode **634** and a gate dielectric **632** laterally between gate electrode **634** and semiconductor body **630**, according to some implementations. For example, for semiconductor body **630** having a cylinder shape, semiconductor body **630**, gate dielectric **632**, and gate electrode **634** may be disposed radially from the center of vertical transistor **626** in this order. In some implementations, gate dielectric **632** surrounds and contacts semiconductor body **630**, and gate electrode **634** surrounds and contacts gate dielectric **632**.
(75) As shown in FIG. **6**A, in some implementations, semiconductor body **630** has two ends (the upper end and lower end) in the vertical direction (the z-direction), and both ends extend beyond gate structure **636**, respectively, in the vertical direction (the z-direction) into ILD layers. That is, semiconductor body **630** can have a larger vertical dimension (e.g., the depth) than that of gate structure **636** (e.g., in the z-direction), and neither the upper end nor the lower end of semiconductor body **630** is flush with the respective end of gate structure **636**. Thus, short circuits between bit lines **623** and word lines/gate electrodes **634** or between word lines/gate electrodes **634** and capacitors **628** can be avoided. In some implementations, the two ILD layers into which semiconductor body **630** extends (e.g., the ILD layer vertically between bit line contacts **625** and word lines **634**, and the ILD layer vertically between word lines **634** and capacitors **628**) include the same dielectric material, such as silicon oxide. Vertical transistor **626** can further include a

source and a drain (both referred to as **638** as their locations may be interchangeable) disposed at the two ends (the upper end and lower end) of semiconductor body **630**, respectively, in the vertical direction (the z-direction). In some implementations, one of source and drain **638** (e.g., at the upper end in FIG. **6**A) is coupled to capacitor **628**, and the other one of source and drain **638** (e.g., at the lower end in FIG. **6**A) is coupled to bit line **623** (e.g., through bit line contact **625** or directly). That is, vertical transistor **626** can have a first terminal in the positive z-direction and a second terminal opposite the first terminal in the negative z-direction, as shown in FIG. **6**A. In some implementations, a metal bit line (e.g., bit line **623** made of a metal material) is coupled to the second terminal of vertical transistor **626** via an ohmic contact (e.g., bit line contact **625** made of a metal silicide material).

(76) In some implementations, semiconductor body **630** includes semiconductor materials, such as single crystalline silicon, polysilicon, amorphous silicon, Ge, any other semiconductor materials, or any combinations thereof. In one example, semiconductor body **630** may include single crystalline silicon. Source and drain **638** can be doped with N-type dopants (e.g., P or As) or P-type dopants (e.g., B or Ga) at a desired doping level. In some implementations, a silicide layer, such as a metal silicide layer, is formed between source/drain **638** of vertical transistor **626** and bit line **623** as bit line contact **625** or between source/drain **638** of vertical transistor **626** and the first electrode of capacitor **628** as capacitor contact **642** to reduce the contact resistance. In some implementations, gate dielectric **632** includes dielectric materials, such as silicon oxide, silicon nitride, or high-k dielectrics including, but not limited to, aluminum oxide (Al.sub.2O.sub.3), hafnium oxide (HfO.sub.2), tantalum oxide (Ta.sub.2O.sub.5), zirconium oxide (ZrO.sub.2), titanium oxide (TiO.sub.2), or any combination thereof. In some implementations, gate electrode **634** includes conductive materials including, but not limited to W, Co, Cu, Al, TiN, TaN, polysilicon, silicides, or any combination thereof. In some implementations, gate electrode **634** includes multiple conductive layers, such as a W layer over a TiN layer. In one example, gate structure **636** may be a “gate oxide/gate poly” gate in which gate dielectric **632** includes silicon oxide and gate electrode **634** includes doped polysilicon. In another example, gate structure **636** may be a high-k metal gate (HKMG) in which gate dielectric **632** includes a high-k dielectric and gate electrode **634** includes a metal.

(77) As described above, since gate electrode **634** may be part of a word line or extend in the word line direction (e.g., the x-direction in FIG. **5**) as a word line, although not directly shown in FIG. **6**A, second semiconductor structure **104** of 3D memory device **600** can also include a plurality of word lines (e.g., an example of word lines **504** in FIG. **5**, referred to as **634** as well) each extending in the word line direction (the x-direction). Each word line **634** can be coupled to a row of DRAM cells **624**. That is, bit line **623** and word line **634** can extend in two perpendicular lateral directions, and semiconductor body **630** of vertical transistor **626** can extend in the vertical direction perpendicular to the two lateral directions in which bit line **623** and word line **634** extend. Word lines **634** are in contact with word line contacts **627**, according to some implementations. In some implementations, word lines **634** include conductive materials including, but not limited to W, Co, Cu, Al, TiN, TaN, polysilicon, silicides, or any combination thereof. In some implementations, word line **634** includes multiple conductive layers, such as a W layer over a TiN layer.

(78) As shown in FIG. **6**A, vertical transistor **626** extends vertically through and contacts word lines **634**, and source or drain **638** of vertical transistor **626** at the lower end thereof is in contact with bit line contact **625** or in contact with bit line **623** directly, according to some implementations. Accordingly, word lines **634** and bit lines **623** can be disposed in different planes in the vertical direction due to the vertical arrangement of vertical transistor **626**, which simplifies the routing of word lines **634** and bit lines **623**. In some implementations, bit lines **623** are disposed vertically between bonding layer **620** and word lines **634**, and word lines **634** are disposed vertically between bit lines **623** and capacitors **628**. Word lines **634** can be coupled to peripheral circuits **612** in first semiconductor structure **102** through word line contacts **627** in interconnect

layer **622**, bonding contacts **621** and **619** in bonding layers **620** and **618**, and the interconnects in interconnect layer **616**. Similarly, bit lines **623** in interconnect layer **622** can be coupled to peripheral circuits **612** in first semiconductor structure **102** through bonding contacts **621** and **619** in bonding layers **620** and **618** and the interconnects in interconnect layer **616**.

(79) In some implementations, second semiconductor structure **104** further includes a plurality of air gaps **640** each disposed laterally between adjacent word lines **634**. Each air gap **640** can be a trench extending in the word line direction (e.g., the x-direction) in parallel with word lines **634** to separate adjacent rows of vertical transistors **626**. As described below with respect to the fabrication process, air gaps **640** may be formed due to the relatively small pitches of word lines **634** (and rows of DRAM cells **624**) in the bit line direction (e.g., the y-direction). On the other hand, the relatively large dielectric constant of air in air gaps **640** (e.g., about 4 times of the dielectric constant of silicon oxide) can improve the insulation effect between word lines **634** (and rows of DRAM cells **624**) compared with some dielectrics (e.g., silicon oxide).

(80) As shown in FIG. **6**A, in some implementations, capacitor **628** includes a first electrode above and coupled to source or drain **638** of vertical transistor **626**, e.g., the upper end of semiconductor body **630**, via a capacitor contact **642**. In some implementations, capacitor contact **642** is an ohmic contact, such as a metal silicide contact, as opposed to a Schottky contact. For example, capacitor contact **642** may include metal silicides, such as WSi, CoSi, CuSi, AlSi, or any other suitable metal silicides having higher conductivities than doped silicon. Capacitor **628** can also include a capacitor dielectric above and in contact with the first electrode, and a second electrode above and in contact with the capacitor dielectric. That is, capacitor **628** can be a vertical capacitor in which the electrodes and capacitor dielectric are stacked vertically (in the z-direction), and the capacitor dielectric can be sandwiched between the electrodes. In some implementations, each first electrode is coupled to source or drain **638** of a respective vertical transistor **626** in the same DRAM cell, while all second electrodes are coupled to a common plate **646** coupled to the ground, e.g., a common ground. Capacitor **628** can have a first end in the negative z-direction and a second end opposite the first end in the positive z-direction, as shown in FIG. **6**A. In some implementations, the first end of capacitor **628** is coupled to the first terminal of vertical transistor **626** via an ohmic contact (e.g., capacitor contact **642** made of a metal silicide material). It is understood that in some examples, the capacitor contacts described herein (e.g., **642**, **1024**, **1124**, **1224**, **1324**, **1742**, **2142**, and **2228**) may not be included in the memory devices. As shown in FIG. **6**A, second semiconductor structure **104** can further include a capacitor contact **647** (e.g., a conductor) in contact with common plate **646** for coupling capacitors **628** to peripheral circuits **612** or to the ground directly. In some implementations, capacitor contact **647** (e.g., a conductor) extends in the z-direction from the dielectric layer of bonding layer **620** to couple to the second end of capacitor **628** via common plate **646**, as shown in FIG. **6**A. In some implementation, the ILD layer in which capacitors **628** are formed has the same dielectric material as the two ILD layers into which semiconductor body **630** extends, such as silicon oxide.

(81) It is understood that the structure and configuration of capacitor **628** are not limited to the example in FIG. **6**A and may include any suitable structure and configuration, such as a planar capacitor, a stack capacitor, a multi-fins capacitor, a cylinder capacitor, a trench capacitor, or a substrate-plate capacitor. In some implementations, the capacitor dielectric includes dielectric materials, such as silicon oxide, silicon nitride, or high-k dielectrics including, but not limited to, Al.sub.2O.sub.3, HfO.sub.2, Ta.sub.2O.sub.5, ZrO.sub.2, TiO.sub.2, or any combination thereof. It is understood that in some examples, capacitor **628** may be a ferroelectric capacitor used in a FRAM cell, and the capacitor dielectric may be replaced by a ferroelectric layer having ferroelectric materials, such as lead zirconate titanate (PZT) or strontium bismuth tantalate (SBT). In some implementations, the electrodes include conductive materials including, but not limited to W, Co, Cu, Al, TiN, TaN, polysilicon, silicides, or any combination thereof.

(82) As shown in FIG. **6**A, vertical transistor **626** extends vertically through and contacts word

lines **634**, source or drain **638** of vertical transistor **626** at the lower end thereof is in contact with bit line contact **625**, and source or drain **638** of vertical transistor **626** at the upper end thereof is in contact with capacitor contact **642**, according to some implementations. That is, bit line **623** and capacitor **628** can be disposed in different planes in the vertical direction and coupled to opposite ends of vertical transistor **626** of DRAM cell **624** in the vertical direction due to the vertical arrangement of vertical transistor **626**. In some implementations, bit line **623** and capacitor **628** are disposed on opposite sides of vertical transistor **626** in the vertical direction, which simplifies the routing of bit lines **623** and reduces the coupling capacitance between bit lines **623** and capacitors **628** compared with DRAM cells in which the bit lines and capacitors are disposed on the same side of the planar transistors.

(83) As shown in FIG. **6**A, in some implementations, vertical transistors **626** are disposed vertically between capacitors **628** and bonding interface **106**. That is, vertical transistors **626** can be arranged closer to peripheral circuits **614** of first semiconductor structure **102** and bonding interface **106** than capacitors **628**. Since bit lines **623** and capacitors **628** are coupled to opposite ends of vertical transistors **626**, as described above, bit lines **623** (as part of interconnect layer **622**) are disposed vertically between vertical transistors **626** and bonding interface **106**, according to some implementations. As a result, interconnect layer **622** including bit lines **623** can be arranged close to bonding interface **106** to reduce the interconnect routing distance and complexity.

(84) In some implementations, second semiconductor structure **104** further includes a substrate **648** disposed above DRAM cells **624**. As described below with respect to the fabrication process, substrate **648** can be part of a carrier wafer. It is understood that in some examples, substrate **648** may not be included in second semiconductor structure **104**.

(85) As shown in FIG. **6**A, second semiconductor structure **104** can further include a pad-out interconnect layer **650** above substrate **648** and DRAM cells **624**. Pad-out interconnect layer **650** can include interconnects, e.g., contact pads **654**, in one or more ILD layers. Pad-out interconnect layer **650** and interconnect layer **622** can be formed on opposite sides of DRAM cells **624**. Capacitors **628** are disposed vertically between vertical transistors **626** and pad-out interconnect layer **650**, according to some implementations. In some implementations, the interconnects in pad-out interconnect layer **650** can transfer electrical signals between 3D memory device **600** and outside circuits, e.g., for pad-out purposes. In some implementations, second semiconductor structure **104** further includes one or more contacts **652** extending through substrate **648** and part of pad-out interconnect layer **650** to couple pad-out interconnect layer **650** to DRAM cells **624** and interconnect layer **622**. As a result, peripheral circuits **612** can be coupled to DRAM cells **624** through interconnect layers **616** and **622** as well as bonding layers **620** and **618**, and peripheral circuits **612** and DRAM cells **624** can be coupled to outside circuits through contacts **652** and pad-out interconnect layer **650**. Contact pads **654** and contacts **652** can include conductive materials including, but not limited to, W, Co, Cu, Al, silicides, or any combination thereof. In one example, contact pad **654** may include Al, and contact **652** may include W. In some implementations, contact **652** includes a via surrounded by a dielectric spacer (e.g., having silicon oxide) to electrically separate the via from substrate **648**. Depending on the thickness of substrate **648**, contact **652** can be an interlayer via (ILV) having a depth in the submicron-level (e.g., between 10 nm and 1 μm), or a through substrate via (TSV) having a depth in the micron- or tens micron-level (e.g., between 1 μm and 100 μm).

(86) It is understood that the pad-out of 3D memory devices is not limited to from second semiconductor structure **104** having DRAM cells **624** as shown in FIG. **6**A and may be from first semiconductor structure **102** having peripheral circuit **612**. For example, as shown in FIG. **6**B, a 3D memory device **601** may include pad-out interconnect layer **650** in first semiconductor structure **102**. Pad-out interconnect layer **650** can be disposed above and in contact with substrate **610** of first semiconductor structure **102** on which transistors **614** of peripheral circuit **612** are formed. In some implementations, first semiconductor structure **102** further includes one or more contacts **653**

extending vertically through substrate **610**. In some implementations, contact **653** couples the interconnects in interconnect layer **616** in first semiconductor structure **102** to contact pads **654** in pad-out interconnect layer **650** to make an electrical connection through substrate **610**. Contacts **653** can include conductive materials including, but not limited to, W, Co, Cu, Al, silicides, or any combination thereof. In one example, contact **653** may include W. In some implementations, contact **653** includes a via surrounded by a dielectric spacer (e.g., having silicon oxide) to electrically separate the via from substrate **610**. It is understood that in some examples, substrate **610** in FIG. **6**B may be a thinned substrate, e.g., compared with substrate **610** in FIG. **6**A. Depending on the thickness of substrate **610**, contact **653** can be an ILV having a depth in the submicron-level (e.g., between 10 nm and 1 μm), or a TSV having a depth in the micron- or tens micron-level (e.g., between 1 μm and 100 μm). It is understood that the details of the same components (e.g., materials, fabrication process, functions, etc.) in both 3D memory devices **600** and **601** are not repeated for ease of description. Pad-out from first semiconductor structure **102** including peripheral circuits **612** can reduce the interconnect distance between contact pad **654** and peripheral circuits **612**, thereby decreasing the parasitic capacitance from the interconnects and improving the electrical performance of 3D memory device **601**.

(87) It is also understood that the relative vertical positions between the semiconductor body and the respective gate structure and word line are not limited to the example shown in FIG. **6**A in which both the upper and lower ends of semiconductor body **630** extend beyond gate structure **636** (and word line **634**), respectively, depending on the various fabrication processes as described below in detail. For example, as shown in FIG. **6**C, a 3D memory device **603** may include vertical transistors **626** each having semiconductor body **630** and gate structure **636**, and one end of semiconductor body **630** in the vertical direction (the z-direction) may be flush with gate structure **636**. In some implementations, the upper or lower end of semiconductor body **630** that is in contact with capacitor contact **642** is flush with the respective end of gate structure **636** and word line **634**. That is, one of the upper and lower ends of semiconductor body **630** that is in contact with capacitor **628** does not extend beyond the respective end of gate structure **636** and word line **634**, according to some implementations. In some implementations, as shown in FIG. **6**C, the other end of semiconductor body **630** in the vertical direction that is in contact with bit line contact **625** still extends beyond the respective end of gate structure **636** and word line **634** into an ILD layer, which is vertically between bit line contacts **625** and word lines **634**. It is understood that the details of the same components (e.g., materials, fabrication process, functions, etc.) in both 3D memory devices **600** and **603** are not repeated for ease of description.

(88) It is further understood that the dielectric materials of the ILD layers into which the semiconductor bodies extend are not limited to the example shown in FIG. **6**A in which the ILD layers include silicon oxide, e.g., the same material as the ILD layer in which capacitors **628** are formed, depending on the various fabrication processes as described below in detail. For example, as shown in FIG. **6**D, a 3D memory device **605** may include two ILD layers **660** and **662** into which semiconductor body **630** extends. ILD layer **660** is vertically between bit line contacts **625** and word lines **634**, and ILD layer **662** is vertically between word lines **634** and capacitor contact **642**, according to some implementations. ILD layers **660** and **662** can include a dielectric material that is different from the dielectric material of the ILD layer in which capacitors **628** are formed. In some implementations, ILD layers **660** and **662** include silicon nitride, while the ILD layer of capacitors **628** includes silicon oxide. As shown in FIG. **6**D, in some implementations, one end of semiconductor body **630** in the vertical direction (the z-direction) that is in contact with capacitor **628** is flush with the respective end of ILD layer **662**. In some implementations, air gap **640** extends vertically through ILD layer **662** to separate ILD layer **662**, but does not extend further into ILD layer **660**, e.g., being stopped by ILD layer **660**. It is understood that the details of the same components (e.g., materials, fabrication process, functions, etc.) in both 3D memory devices **600** and **605** are not repeated for ease of description.

(89) It is further understood that the air gaps between word lines may be partially or fully filled with dielectrics. For example, as shown in FIG. **6**E, a memory device **607** may not include air gaps (e.g., air gaps **640** in FIG. **6**A) between adjacent word lines **634**. Instead, memory device **607** can include dielectric wall structures **641** (e.g., filled with dielectrics, such as silicon oxide) each formed between adjacent word lines **634**. It is understood that in some examples (not shown), air gaps **640** may still exist between word lines **634**, but with a smaller vertical dimension (depth) compared with word liens **634**. It is understood that the details of the same components (e.g., materials, fabrication process, functions, etc.) in both 3D memory devices **600** and **607** are not repeated for ease of description.

(90) It is further understood that more than one DRAM cell array may be stacked over one another to vertically scale up the number of DRAM cells. For example, as shown in FIG. **7**, a memory device **700** may further include a third semiconductor structure **702** having an array of DRAM cells **624** stacked over second and first semiconductor structures **104** and **102**. Third and second semiconductor structures **702** and **104** are jointed at another bonding interface **704** therebetween, according to some implementations. Third and second semiconductor structures **702** and **104** may have the same arrays of DRAM cells **624** and interconnect layers **622** and thus, the details of DRAM cells **624** and interconnect layer **622** in third semiconductor structure **702** are not repeated for ease of description.

(91) Third and second semiconductor structures **702** and **104** can be bonded in a face-to-face manner, such that at least some components (e.g., DRAM cells **624**) in third and second semiconductor structures **702** and **104** can be symmetric with respect bonding interface **704**, according to some implementations. Bonding interface **704** can be formed vertically between DRAM cells **624** in third semiconductor structure **702** and DRAM cells **624** in second semiconductor structure **104**. As shown in FIG. **7**, in some implementations, capacitors **628** in second semiconductor structure **104** are disposed vertically between bonding interface **704** and vertical transistors **626** in second semiconductor structure **104**, and capacitors **628** in third semiconductor structure **702** are disposed vertically between bonding interface **704** and vertical transistors **626** in third semiconductor structure **702**. That is, capacitors **628** in second semiconductor structure **104** and capacitors **628** in third semiconductor structure **702** can be disposed on opposite sides of bonding interface **704**. In some implementations, the second electrodes of capacitors **628** in third semiconductor structure **702** are in contact with common plate **646** at bonding interface **704**.

(92) In some implementations, 3D memory device **700** includes additional interconnects that couple DRAM cells **624** in third semiconductor structure **702** to peripheral circuits **612** across bonding interfaces **704** and **106**, such as word line contacts **734** coupling word lines **634** in third semiconductor structure **702** and peripheral circuits **612** in first semiconductor structure **102**. As shown in FIG. **7**, third semiconductor structure **702**, as opposed to first or second semiconductor structure **102** or **104**, can include pad-out interconnect layer **650**. In some implementations, vertical transistors **626** in third semiconductor structure **702** are disposed vertically between capacitors **628** in third semiconductor structure **702** and pad-out interconnect layer **650**. It is understood that the details of the same components (e.g., materials, fabrication process, functions, etc.) in both 3D memory devices **600** and **700** are not repeated for ease of description.

(93) It is understood that the architecture of multiple memory cell arrays shown in FIG. **7** is not limited to the design of DRAM cells **624** and may be applied to any suitable memory cells disclosed herein. It is also understood that various designs of memory cells disclosed herein may be mixed in the architecture of multiple memory cell arrays shown in FIG. **7**. For example, second and third semiconductor structures **104** and **702** may include different designs of memory cells disclosed herein.

(94) It is further understood that the memory cell array is not limited to the example shown in FIGS. **5**, **6**A-**6**D, and **7** in which the vertical transistors are GAA transistors and may be any other

suitable multi-gate vertical transistors. For example, FIG. **8** illustrates a plan view of another array of memory cells **802** each including a vertical transistor in a memory device **800**, according to some aspects of the present disclosure. As shown in FIG. **8**, memory device **800** can include a plurality of word lines **804** each extending in a first lateral direction (the x-direction, referred to as the word line direction). Memory device **800** can also include a plurality of bit lines **806** each extending in a second lateral direction perpendicular to the first lateral direction (the y-direction, referred to as the bit line direction). It is understood that FIG. **8** does not illustrate a cross-section of memory device **800** in the same lateral plane, and word lines **804** and bit lines **806** may be formed in different lateral planes for ease of routing as described below in detail.

(95) Memory cells **802** can be formed at the intersections of word lines **804** and bit lines **806**. In some implementations, each memory cell **802** includes a vertical transistor (e.g., vertical transistor **210** in FIG. **2**) having a semiconductor body **808** and a gate structure **810**. The vertical transistor of memory cell **802** in FIG. **8** may be an example of a tri-gate vertical transistor (e.g., tri-side gate vertical transistor). Semiconductor body **808** can extend in the vertical direction (the z-direction, not shown) perpendicular to the first and second lateral directions. Gate structure **810** can be in contact with a plurality of sides (e.g., three of all four sides in FIG. **8**) of semiconductor body **808** (the active region in which channels are formed). That is, different from the GAA vertical transistor in memory cell **502** in FIG. **5**, gate structure **810** of the vertical transistor in memory cell **802** partially circumscribes semiconductor body **808** in the plan view. That is, gate structure **810** circumscribes (e.g., surrounding and contacting) three of all four sides of semiconductor body **808** (having a rectangle or square-shaped cross-section) in the plan view, according to some implementations. Gate structure **810** does not surround and contact at least one side of semiconductor body **808**, according to some implementations. Gate structure **810** can include a gate dielectric **812** partially or fully circumscribes semiconductor body **808** in the plan view, and a gate electrode **814** partially circumscribes gate dielectric **812**. Thus, the vertical transistor having gate structure **810** may be viewed as a “tri-gate” vertical transistor in which gate structure **810** is in contact with two opposite sides of semiconductor body **808** in the word line direction (the x-direction) and one side of semiconductor body **808** in the bit line direction (the y-direction). As described above, gate electrode **814** may be part of word line **804**, and word line **804** may be an extension of gate electrode **814**. For example, as shown in FIG. **8**, one edge of each word line **804** may be formed aligned with the same side of each semiconductor body **808**, such that gate electrode **814** may not extend to the side of semiconductor body **808** to form a GAA transistor. By arranging semiconductor bodies **808** of memory cells **802** to be aligned with one side of word lines **804**, the pitch of word lines **804** and/or the pitch of memory cells **802** in the bit line direction (the y-direction) can be further increased to reduce the fabrication complexity and increase the production yield. In some implementations, gate structures **810** of a row of the vertical transistors are continuous in the x-direction, as a shown in FIG. **8**.

(96) Similar to memory device **500** in FIG. **5**, as shown in FIG. **8**, gate electrodes **814** of adjacent vertical transistors in the word line direction (the x-direction) are continuous, e.g., parts of a continuous conductive layer having gate electrodes **814** and **804**. In contrast, gate dielectrics **812** of adjacent vertical transistors in the word line direction are separate, e.g., not parts of a continuous dielectric layer having gate dielectrics **812**.

(97) FIG. **9** illustrates a side view of a cross-section of a 3D memory device **900** including vertical transistors, according to some aspects of the present disclosure. 3D memory device **900** may be one example of memory device **800** including multi-gate vertical transistors in which gate structures partially circumscribes semiconductor bodies in the plan view. 3D memory device **900** is similar to 3D memory device **600** in FIG. **6**A except for the different structures of multi-gate vertical transistors in DRAM cells **624**. It is understood that the details of the same components (e.g., materials, fabrication process, functions, etc.) in both 3D memory devices **600** and **900** are not repeated for ease of description. Similar to FIG. **6**A, the cross-section of 3D memory device **900** in

FIG. **9** may be made along the bit line direction (the y-direction).
(98) Vertical transistor **926** can be a MOSFET used to switch a respective DRAM cell **624**. In some implementations, vertical transistor **926** includes semiconductor body **630** (the active region in which multiple channels can form) extending vertically (in the z-direction), and a gate structure **936** in contact with a plurality of sides of semiconductor body **630**. Semiconductor body **630** can have a cuboid shape or a cylinder shape, and gate structure **936** can partially circumscribe semiconductor body **630** in the plan view, for example, as shown in FIG. **8**. As shown in FIG. **9**, gate structure **936** does not extend to at least one side of semiconductor body **630**, according to some implementations. Gate structure **936** includes a gate electrode **934** and a gate dielectric **932** laterally between gate electrode **934** and semiconductor body **630**, according to some implementations. As shown in FIG. **9**, gate electrode **934** does not extend to at least one side of semiconductor body **630**, according to some implementation. Due to the increased pitch of word lines **934** and/or the pitch of DRAM cells **624** in the bit line direction (they-direction), the air gaps between word lines **934** may be partially or fully filled with dielectrics.
(99) It is also understood that the number of gates in multi-gate transistors may vary, e.g., not limited by the GAA vertical transistor example in FIG. **5** and the tri-gate vertical transistor example in FIG. **8**. For example, multi-gate vertical transistors may also include double-gate vertical transistors (a.k.a. dual-side gate vertical transistors) in which the gate structure is in contact with two sides of the semiconductor body, such as the two opposite sides in the bit line direction or in the word line direction.
(100) It is further understood that although storage units are described as capacitors **628** in FIGS. **6**A-**6**D, **7**, and **9**, storage units may include any other suitable devices, such as PCM elements, as described above with respect to FIG. **4** in some examples. For example, the capacitor dielectric of capacitor **628** may be replaced with a phase-change material layer, such as chalcogenide alloys, vertically sandwiched between the electrodes to form a PCM element. Also, instead of coupling source or drain **638** of vertical transistor **626** or **926** to bit line **623**, the electrode of the PCM element may be coupled to bit line **623**, while source or drain **638** of vertical transistor **626** or **926** may be coupled to the ground, e.g., a common ground plate.
(101) According to some aspects of the present disclosure, the vertical transistors of memory cells in a memory device (e.g., memory device **200**) are single-gate transistors, and the gate dielectrics of vertical transistors in the word line direction are continuous. For example, FIG. **16** illustrates a plan view of still another array of memory cells **1602** each including a vertical transistor in a memory device **1600**, according to some aspects of the present disclosure. As shown in FIG. **16**, memory device **1600** can include a plurality of word lines **1604** each extending in a first lateral direction (the x-direction, referred to as the word line direction). Memory device **1600** can also include a plurality of bit lines **1606** each extending in a second lateral direction perpendicular to the first lateral direction (the y-direction, referred to as the bit line direction). It is understood that FIG. **16** does not illustrate a cross-section of memory device **1600** in the same lateral plane, and word lines **1604** and bit lines **1606** may be formed in different lateral planes for ease of routing as described below in detail.
(102) Memory cells **1602** can be formed at the intersections of word lines **1604** and bit lines **1606**. In some implementations, each memory cell **1602** includes a vertical transistor (e.g., vertical transistor **210** in FIG. **2**) having a semiconductor body **1608** and a gate structure **1610**. Semiconductor body **1608** can extend in a substrate in the vertical direction (the z-direction, not shown) perpendicular to the first and second lateral directions. The vertical transistor can be a single-gate transistor in which gate structure **1610** is in contact with a single side (e.g., one of four sides in FIG. **16**) of semiconductor body **1608** (the active region in which channels are formed). As shown in FIG. **16**, the vertical transistor is a single-gate transistor in which gate structure **1610** abuts one side of semiconductor body **1608** (having a rectangle or square-shaped cross-section) in the bit line direction (the y-direction) in the plan view. Gate structure **1610** does not surround and

contact other three sides of semiconductor body **1608**, according to some implementations. Gate structure **1610** can include a gate dielectric **1612** abuts one side of semiconductor body **1608** in the plan view, and a gate electrode **1614** in contact with gate dielectric **1612**. In some implementations, gate dielectric **1612** is laterally between gate electrode **1614** and semiconductor body **1608** in the bit line direction (the y-direction). As described above, gate electrode **1614** may be part of word line **1604**, and word line **1604** may be an extension of gate electrode **1614**. That is, gate electrodes **1614** of adjacent vertical transistors in the word line direction (the x-direction) are continuous, e.g., parts of a continuous conductive layer having gate electrodes **1614** and **1604**. In some implementations, gate structures **1610** of a row of the vertical transistors are continuous in the x-direction, as a shown in FIG. **16**.

(103) Different from separate gate dielectrics **512** and **812** in FIGS. **5** and **8**, as shown in FIG. **16**, gate dielectrics **1612** of adjacent vertical transistors in the word line direction are continuous, e.g., parts of a continuous dielectric layer having gate dielectrics **1612** and extending in the word line direction to abut vertical transistors in the same row on the same side. Gate structures **1610** can be thus viewed as parts of a continuous structure extending in the word line direction at which the continuous structure abut vertical transistors in the same row on the same side.

(104) As shown in FIG. **16**, two adjacent vertical transistors of memory cells (e.g., **1602**A and **1602**B) in the bit line direction (the y-direction) are mirror-symmetric to one another, according to some implementations. As described below with respect to the fabrication process, semiconductor bodies **1608** of each pair of two adjacent vertical transistors of memory cells (e.g., **1602**A and **1602**B) in the bit line direction (the y-direction) can be formed by separating a semiconductor pillar into two pieces using a trench isolation **1616** extending in the word line direction (the x-direction) and in parallel with word lines **1604**. Trench isolations **1616** and word lines **1604** can be disposed in an interleaved manner in the bit line direction. In some implementations, trench isolation **1616** is formed in the middle of the semiconductor pillars (not shown) such that the resulting pair of semiconductor bodies **1608** are mirror-symmetric to one another with respect to trench isolation **1616**, so are the pair of vertical transistors having semiconductor bodies **1608** when the respective gate structures **1610** are mirror-symmetric to one another with respect to trench isolation **1616** as well.

(105) It is understood that in some examples, trench isolations **1616** extending in the word line directions may not be formed such that two adjacent semiconductor bodies **1608** separated by a respective trench isolation **1616** may be merged as a single semiconductor body having two opposite sides in the bit line direction in contact with gate structure **1610**. That is, without trench isolations **1616**, the adjacent single-gate vertical transistors may be merged to form a double-gate vertical transistor with increased gate control area and lower leakage current. The gate structure of the double-gate vertical transistor may include two mirror-symmetric gate structures **1610** in FIG. **16**, such that both sides of the merged semiconductor body **1608** in the bit line direction may be in contact with the gate structure in the double-gate vertical transistor. On the other hand, by splitting the double-gate vertical transistors into single-gate vertical transistors using trench isolations **1616**, the number of memory cells **1602** (and the cell density) in the bit line direction can be doubled compared to double-gate vertical transistors without unduly complexing the fabrication process (e.g., compared with using SADP process).

(106) FIG. **17** illustrates a side view of a cross-section of yet another 3D memory device **1700** including vertical transistors, according to some aspects of the present disclosure. 3D memory device **1700** may be one example of memory device **1600** including single-gate vertical transistors in which gate structures abut a single side of semiconductor bodies in the plan view. It is understood that FIG. **17** is for illustrative purposes only and may not necessarily reflect the actual device structure (e.g., interconnections) in practice. As one example of 3D memory device **100** described above with respect to FIG. **1**A, 3D memory device **1700** is a bonded chip including first semiconductor structure **102** and second semiconductor structure **104** stacked over first

semiconductor structure **102**. First and second semiconductor structures **102** and **104** are jointed at bonding interface **106** therebetween, according to some implementations. As shown in FIG. **17**, first semiconductor structure **102** can include a substrate **1710**, which can include silicon (e.g., single crystalline silicon, c-Si), SiGe, GaAs, Ge, SOI, or any other suitable materials.

(107) First semiconductor structure **102** can include peripheral circuits **1712** on substrate **1710**. In some implementations, peripheral circuits **1712** include a plurality of transistors **1714** (e.g., planar transistors and/or 3D transistors). Trench isolations (e.g., shallow trench isolations (STIs)) and doped regions (e.g., wells, sources, and drains of transistors **1714**) can be formed on or in substrate **1710** as well.

(108) In some implementations, first semiconductor structure **102** further includes an interconnect layer **1716** above peripheral circuits **1712** to transfer electrical signals to and from peripheral circuits **1712**. Interconnect layer **1716** can include a plurality of interconnects (also referred to herein as "contacts"), including lateral interconnect lines and VIA contacts. Interconnect layer **1716** can further include one or more ILD layers in which the interconnect lines and via contacts can form. That is, interconnect layer **1716** can include interconnect lines and via contacts in multiple ILD layers. In some implementations, peripheral circuits **1712** are coupled to one another through the interconnects in interconnect layer **1716**. The interconnects in interconnect layer **1716** can include conductive materials including, but not limited to, W, Co, Cu, Al, doped silicon, silicides, or any combination thereof. The ILD layers can be formed with dielectric materials including, but not limited to, silicon oxide, silicon nitride, silicon oxynitride, low-k dielectrics, or any combination thereof.

(109) As shown in FIG. **17**, first semiconductor structure **102** can further include a bonding layer **1718** at bonding interface **106** and above interconnect layer **1716** and peripheral circuits **1712**. Bonding layer **1718** can include a plurality of bonding contacts **1719** and dielectrics electrically isolating bonding contacts **1719**. Bonding contacts **1719** can include conductive materials, such as Cu. The remaining area of bonding layer **1718** can be formed with dielectric materials, such as silicon oxide. Bonding contacts **1719** and surrounding dielectrics in bonding layer **1718** can be used for hybrid bonding. Similarly, as shown in FIG. **17**, second semiconductor structure **104** can also include a bonding layer **1720** at bonding interface **106** and above bonding layer **1718** of first semiconductor structure **102**. Bonding layer **1720** can include a plurality of bonding contacts **1721** and dielectrics electrically isolating bonding contacts **1721**. Bonding contacts **1721** can include conductive materials, such as Cu. The remaining area of bonding layer **1720** can be formed with dielectric materials, such as silicon oxide. Bonding contacts **1721** and surrounding dielectrics in bonding layer **1720** can be used for hybrid bonding. Bonding contacts **1721** are in contact with bonding contacts **1719** at bonding interface **106**, according to some implementations. In some implementations, bonding layer **1720** includes a dielectric layer opposing DRAM cells **1724** with bit line **1723** positioned between the dielectric layer and DRAM cells **1724**, as shown in FIG. **17**. The dielectric layer can include bonding interface **106** having bonding contacts **1721**.

(110) Second semiconductor structure **104** can be bonded on top of first semiconductor structure **102** in a face-to-face manner at bonding interface **106**. In some implementations, bonding interface **106** is disposed between bonding layers **1720** and **1718** as a result of hybrid bonding (also known as "metal/dielectric hybrid bonding"), which is a direct bonding technology (e.g., forming bonding between surfaces without using intermediate layers, such as solder or adhesives) and can obtain metal-metal bonding and dielectric-dielectric bonding simultaneously. In some implementations, bonding interface **106** is the place at which bonding layers **1720** and **1718** are met and bonded. In practice, bonding interface **106** can be a layer with a certain thickness that includes the top surface of bonding layer **1718** of first semiconductor structure **102** and the bottom surface of bonding layer **1720** of second semiconductor structure **104**.

(111) In some implementations, second semiconductor structure **104** further includes an interconnect layer **1722** including bit lines **1723** above bonding layer **1720** to transfer electrical

signals. Interconnect layer **1722** can include a plurality of interconnects, such as MEOL interconnects and BEOL interconnects. In some implementations, the interconnects in interconnect layer **1722** also include local interconnects, such as bit lines **1723** (e.g., an example of bit lines **1606** in FIG. **16**) and word line contacts (not shown). Interconnect layer **1722** can further include one or more ILD layers in which the interconnect lines and via contacts can form. The interconnects in interconnect layer **1722** can include conductive materials including, but not limited to, W, Co, Cu, Al, doped silicon, silicides, or any combination thereof. The ILD layers can be formed with dielectric materials including, but not limited to, silicon oxide, silicon nitride, silicon oxynitride, low-k dielectrics, or any combination thereof. In some implementations, peripheral circuits **1712** include a word line driver/row decoder coupled to the word line contacts in interconnect layer **1722** through bonding contacts **1721** and **1719** in bonding layers **1720** and **1718** and interconnect layer **1716**. In some implementations, peripheral circuits **1712** include a bit line driver/column decoder coupled to bit lines **1723** and bit line contacts (not shown in FIG. **17**, e.g., bit line contacts **625** in FIGS. **6**A-**6**E) in interconnect layer **1722** through bonding contacts **1721** and **1719** in bonding layers **1720** and **1718** and interconnect layer **1716**. In some implementations, bit line **1723** is a metal bit line, as opposed to semiconductor bit lines (e.g., doped silicon bit lines). For example, bit line **1723** may include W, Co, Cu, Al, or any other suitable metals having higher conductivities than doped silicon. In some implementations, the bit line contact (not shown in FIG. **17**, e.g., bit line contacts **625** in FIGS. **6**A-**6**E) is an ohmic contact, such as a metal silicide contact, as opposed to a Schottky contact. For example, the bit line contact may include metal silicides, such as WSi, CoSi, CuSi, AlSi, or any other suitable metal silicides having higher conductivities than doped silicon.

(112) In some implementations, second semiconductor structure **104** includes a DRAM device in which memory cells are provided in the form of an array of DRAM cells **1724** (e.g., an example of memory cells **1602** in FIG. **16**) above interconnect layer **1722** and bonding layer **1720**. That is, interconnect layer **1722** including bit lines **1723** can be disposed between bonding layer **1720** and array of DRAM cells **1724**. It is understood that the cross-section of 3D memory device **1700** in FIG. **17** may be made along the bit line direction (the y-direction), and one bit line **1723** in interconnect layer **1722** extending laterally in the y-direction may be coupled to a column of DRAM cells **1724**.

(113) Each DRAM cell **1724** can include a vertical transistor **1726** (e.g., an example of vertical transistors **210** in FIG. **2**) and capacitor **1728** (e.g., an example of storage unit **212** in FIG. **2**) coupled to the vertical transistor **1726**. DRAM cell **1724** can be a 1T1C cell consisting of one transistor and one capacitor. It is understood that DRAM cell **1724** may be of any suitable configurations, such as 2T1C cell, 3T1C cell, etc. To better illustrate vertical transistor **1726**, FIG. **18** illustrates a perspective view of an array of vertical transistor **1726**, according to some aspects of the present disclosure. FIGS. **17** and **18** will be described together when describing vertical transistors **1726**.

(114) Vertical transistor **1726** can be a MOSFET used to switch a respective DRAM cell **1724**. In some implementations, vertical transistor **1726** includes a semiconductor body **1730** (the active region in which a channel can form) extending vertically (in the z-direction), and a gate structure **1736** in contact with one side of semiconductor body **1730** in the bit line direction (the y-direction). As described above, as in a single-gate vertical transistor, semiconductor body **1730** can have a cuboid shape or a cylinder shape, and gate structure **1736** can abut a single side of semiconductor body **1730** in the plan view, for example, as shown in FIGS. **17** and **18**. Gate structure **1736** includes a gate electrode **1734** and a gate dielectric **1732** laterally between gate electrode **1734** and semiconductor body **1730** in the bit line direction, according to some implementations. In some implementations, gate dielectric **1732** abuts one side of semiconductor body **1730**, and gate electrode **1734** abuts gate dielectric **1732**.

(115) As shown in FIGS. **17** and **18**, in some implementations, semiconductor body **1730** has two

ends (the upper end and lower end) in the vertical direction (the z-direction), and at least one end (e.g., the lower end in FIGS. **17** and **18**) extends beyond gate dielectric **1732** in the vertical direction (the z-direction) into ILD layers. In some implementations, one end (e.g., the upper end in FIGS. **17** and **18**) of semiconductor body **1730** is flush with the respective end (e.g., the upper end in FIGS. **17** and **18**) of gate dielectric **1732**. In some implementations, both ends (the upper end and lower end) of semiconductor body **1730** extend beyond gate electrode **1734**, respectively, in the vertical direction (the z-direction) into ILD layers. That is, semiconductor body **1730** can have a larger vertical dimension (e.g., the depth) than that of gate electrode **1734** (e.g., in the z-direction), and neither the upper end nor the lower end of semiconductor body **1730** is flush with the respective end of gate electrode **1734**. Thus, short circuits between bit lines **1723** and word lines/gate electrodes **1734** or between word lines/gate electrodes **1734** and capacitors **1728** can be avoided. Vertical transistor **1726** can further include a source and a drain (both referred to as **1738** as their locations may be interchangeable) disposed at the two ends (the upper end and lower end) of semiconductor body **1730**, respectively, in the vertical direction (the z-direction). In some implementations, one of source and drain **1738** (e.g., at the upper end in FIGS. **17** and **18**) is coupled to capacitor **1728**, and the other one of source and drain **1738** (e.g., at the lower end in FIGS. **17** and **18**) is coupled to bit line **1723**. That is, vertical transistor **1726** can have a first terminal in the positive z-direction and a second terminal opposite the first terminal in the negative z-direction, as shown in FIG. **17**. In some implementations, a metal bit line (e.g., bit line **1723** made of a metal material) is coupled to the second terminal of vertical transistor **1726** via an ohmic contact (e.g., a bit line contact made of a metal silicide material).

(116) In some implementations, semiconductor body **1730** includes semiconductor materials, such as single crystalline silicon, polysilicon, amorphous silicon, Ge, any other semiconductor materials, or any combinations thereof. In one example, semiconductor body **1730** may include single crystalline silicon. Source and drain **1738** can be doped with N-type dopants (e.g., P or As) or P-type dopants (e.g., B or Ga) at a desired doping level. In some implementations, a silicide layer, such as a metal silicide layer, is formed between source/drain **1738** of vertical transistor **1726** and bit line **1723** as the bit line contact or between source/drain **1738** of vertical transistor **1726** and the first electrode of capacitor **1728** as capacitor contact **1742** to reduce the contact resistance. In some implementations, gate dielectric **1732** includes dielectric materials, such as silicon oxide, silicon nitride, or high-k dielectrics including, but not limited to, Al.sub.2O.sub.3, HfO.sub.2, Ta.sub.2O.sub.5, ZrO.sub.2, TiO.sub.2, or any combination thereof. In some implementations, gate electrode **1734** includes conductive materials including, but not limited to W, Co, Cu, Al, TiN, TaN, polysilicon, silicides, or any combination thereof. In some implementations, gate electrode **1734** includes multiple conductive layers, such as a W layer over a TiN layer, as shown in FIGS. **17** and **18**. In one example, gate structure **1736** may be a “gate oxide/gate poly” gate in which gate dielectric **1732** includes silicon oxide and gate electrode **1734** includes doped polysilicon. In another example, gate structure **1736** may be an HKMG in which gate dielectric **1732** includes a high-k dielectric and gate electrode **1734** includes a metal.

(117) As described above, since gate electrode **1734** may be part of a word line or extend in the word line direction (e.g., the x-direction in FIG. **18**) as a word line, as shown in FIG. **18**, second semiconductor structure **104** of 3D memory device **1700** can also include a plurality of word lines (e.g., an example of word lines **1604** in FIG. **16**, referred to as **1734** as well) each extending in the word line direction (the x-direction). Each word line **1734** can be coupled to a row of DRAM cells **1724**. That is, bit line **1723** and word line **1734** can extend in two perpendicular lateral directions, and semiconductor body **1730** of vertical transistor **1726** can extend in the vertical direction perpendicular to the two lateral directions in which bit line **1723** and word line **1734** extend. Word lines **1734** are in contact with word line contacts (not shown), according to some implementations. In some implementations, word lines **1734** include conductive materials including, but not limited to W, Co, Cu, Al, TiN, TaN, polysilicon, silicides, or any combination thereof. In some

implementations, word line **1734** includes multiple conductive layers, such as a W layer over a TiN layer, as shown in FIG. **17**.

(118) As shown in FIGS. **17** and **18**, vertical transistor **1726** extends vertically through and contacts word lines **1734**, and source or drain **1738** of vertical transistor **1726** at the lower end thereof is in contact with bit line **1723** (or bit line contact if any), according to some implementations. Accordingly, word lines **1734** and bit lines **1723** can be disposed in different planes in the vertical direction due to the vertical arrangement of vertical transistor **1726**, which simplifies the routing of word lines **1734** and bit lines **1723**. In some implementations, bit lines **1723** are disposed vertically between bonding layer **1720** and word lines **1734**, and word lines **1734** are disposed vertically between bit lines **1723** and capacitors **1728**. Word lines **1734** can be coupled to peripheral circuits **1712** in first semiconductor structure **102** through word line contacts (not shown) in interconnect layer **1722**, bonding contacts **1721** and **1719** in bonding layers **1720** and **1718**, and the interconnects in interconnect layer **1716**. Similarly, bit lines **1723** in interconnect layer **1722** can be coupled to peripheral circuits **1712** in first semiconductor structure **102** through bonding contacts **1721** and **1719** in bonding layers **1720** and **1718** and the interconnects in interconnect layer **1716**.

(119) As described above with respect to FIG. **16**, vertical transistors **1726** can be arranged in a mirror-symmetric manner to increase the density of DRAM cells **1724** in the bit line direction (the y-direction). As shown in FIG. **17**, two adjacent vertical transistors **1726** in the bit line direction are mirror-symmetric to one another with respect to a trench isolation **1760** (e.g., corresponding to trench isolation **1616** in FIG. **16**), according to some implementations. That is, second semiconductor structure **104** can include a plurality of trench isolations **1760** each extending in the word line direction (the x-direction) in parallel with word lines **1734** and disposed between semiconductor bodies **1730** of two adjacent rows of vertical transistors **1726**. In some implementations, the rows of vertical transistors **1726** separated by trench isolation **1760** are mirror-symmetric to one another with respect to trench isolation **1760**. Trench isolation **1760** can be formed with dielectric materials including, but not limited to, silicon oxide, silicon nitride, silicon oxynitride, low-k dielectrics, or any combination thereof. It is understood that trench isolation **1760** may include an air gap each disposed laterally between adjacent semiconductor bodies **1730**. As described below with respect to the fabrication process, air gaps may be formed due to the relatively small pitches of vertical transistors **1726** in the bit line direction (e.g., the y-direction). On the other hand, the relatively large dielectric constant of air in air gaps (e.g., about 4 times of the dielectric constant of silicon oxide) can improve the insulation effect between vertical transistors **1726** (and rows of DRAM cells **1724**) compared with some dielectrics (e.g., silicon oxide). Similarly, in some implementations, air gaps are formed laterally between word lines/gate electrodes **1734** in the bit line direction as well, depending on the pitches of word lines/gate electrodes **1734** in the bit line direction.

(120) As shown in FIG. **17**, in some implementations, capacitor **1728** includes a first electrode above and coupled to source or drain **1738** of vertical transistor **1726**, e.g., the upper end of semiconductor body **1730**, via a capacitor contact **1742**. In some implementations, capacitor contact **1742** is an ohmic contact, such as a metal silicide contact, as opposed to a Schottky contact. For example, capacitor contact **1742** may include metal silicides, such as WSi, CoSi, CuSi, AlSi, or any other suitable metal silicides having higher conductivities than doped silicon. Capacitor **1728** can also include a capacitor dielectric above and in contact with the first electrode, and a second electrode above and in contact with the capacitor dielectric. That is, capacitor **1728** can be a vertical capacitor in which the electrodes and capacitor dielectric are stacked vertically (in the z-direction), and the capacitor dielectric can be sandwiched between the electrodes. In some implementations, each first electrode is coupled to source or drain **1738** of a respective vertical transistor **1726** in the same DRAM cell, while all second electrodes are coupled to a common plate **1746** coupled to the ground, e.g., a common ground. Capacitor **1728** can have a first end in the negative z-direction and a second end opposite the first end in the positive z-direction, as shown in

FIG. **17**. In some implementations, the first end of capacitor **1728** is coupled to the first terminal of vertical transistor **1726** via an ohmic contact (e.g., capacitor contact **1742** made of a metal silicide material). As shown in FIG. **17**, second semiconductor structure **104** can further include a capacitor contact **1747** (e.g., a conductor) in contact with common plate **1746** for coupling capacitors **1728** to peripheral circuits **1712** or to the ground directly. In some implementations, capacitor contact **1747** (e.g., a conductor) extends in the z-direction from the dielectric layer of bonding layer **1720** to couple to the second end of capacitor **1728** via common plate **1746**, as shown in FIG. **17**. In some implementation, the ILD layer in which capacitors **1728** are formed has the same dielectric material as the two ILD layers into which semiconductor body **1730** extends, such as silicon oxide.
(121) It is understood that the structure and configuration of capacitor **1728** are not limited to the example in FIG. **17** and may include any suitable structure and configuration, such as a planar capacitor, a stack capacitor, a multi-fins capacitor, a cylinder capacitor, a trench capacitor, or a substrate-plate capacitor. In some implementations, the capacitor dielectric includes dielectric materials, such as silicon oxide, silicon nitride, or high-k dielectrics including, but not limited to, Al.sub.2O.sub.3, HfO.sub.2, Ta.sub.2O.sub.5, ZrO.sub.2, TiO.sub.2, or any combination thereof. It is understood that in some examples, capacitor **1728** may be a ferroelectric capacitor used in a FRAM cell, and the capacitor dielectric may be replaced by a ferroelectric layer having ferroelectric materials, such as PZT or SBT. In some implementations, the electrodes include conductive materials including, but not limited to W, Co, Cu, Al, TiN, TaN, polysilicon, silicides, or any combination thereof.
(122) As shown in FIG. **17**, vertical transistor **1726** extends vertically through and contacts word lines **1734**, source or drain **1738** of vertical transistor **1726** at the lower end thereof is in contact with bit line **1723**, and source or drain **1738** of vertical transistor **1726** at the upper end thereof is coupled to capacitor **1728**, according to some implementations. That is, bit line **1723** and capacitor **1728** can be disposed in different planes in the vertical direction and coupled to opposite ends of vertical transistor **1726** of DRAM cell **1724** in the vertical direction due to the vertical arrangement of vertical transistor **1726**. In some implementations, bit line **1723** and capacitor **1728** are disposed on opposite sides of vertical transistor **1726** in the vertical direction, which simplifies the routing of bit lines **1723** and reduces the coupling capacitance between bit lines **1723** and capacitors **1728** compared with DRAM cells in which the bit lines and capacitors are disposed on the same side of the planar transistors.
(123) As shown in FIG. **17**, in some implementations, vertical transistors **1726** are disposed vertically between capacitors **1728** and bonding interface **106**. That is, vertical transistors **1726** can be arranged closer to peripheral circuits **1714** of first semiconductor structure **102** and bonding interface **106** than capacitors **1728**. Since bit lines **1723** and capacitors **1728** are coupled to opposite ends of vertical transistors **1726**, as described above, bit lines **1723** (as part of interconnect layer **1722**) are disposed vertically between vertical transistors **1726** and bonding interface **106**, according to some implementations. As a result, interconnect layer **1722** including bit lines **1723** can be arranged close to bonding interface **106** to reduce the interconnect routing distance and complexity.
(124) In some implementations, second semiconductor structure **104** further includes a substrate **1748** disposed above DRAM cells **1724**. As described below with respect to the fabrication process, substrate **1748** can be part of a carrier wafer. It is understood that in some examples, substrate **1748** may not be included in second semiconductor structure **104**.
(125) As shown in FIG. **17**, second semiconductor structure **104** can further include a pad-out interconnect layer **1750** above substrate **1748** and DRAM cells **1724**. Pad-out interconnect layer **1750** can include interconnects, e.g., contact pads **1754**, in one or more ILD layers. Pad-out interconnect layer **1750** and interconnect layer **1722** can be formed on opposite sides of DRAM cells **1724**. Capacitors **1728** are disposed vertically between vertical transistors **1726** and pad-out interconnect layer **1750**, according to some implementations. In some implementations, the

interconnects in pad-out interconnect layer **1750** can transfer electrical signals between 3D memory device **1700** and outside circuits, e.g., for pad-out purposes. In some implementations, second semiconductor structure **104** further includes one or more contacts **1752** extending through substrate **1748** and part of pad-out interconnect layer **1750** to couple pad-out interconnect layer **1750** to DRAM cells **1724** and interconnect layer **1722**. As a result, peripheral circuits **1712** can be coupled to DRAM cells **1724** through interconnect layers **1716** and **1722** as well as bonding layers **1720** and **1718**, and peripheral circuits **1712** and DRAM cells **1724** can be coupled to outside circuits through contacts **1752** and pad-out interconnect layer **1750**. Contact pads **1754** and contacts **1752** can include conductive materials including, but not limited to, W, Co, Cu, Al, silicides, or any combination thereof. In one example, contact pad **1754** may include Al, and contact **1752** may include W. In some implementations, contact **1752** includes a via surrounded by a dielectric spacer (e.g., having silicon oxide) to electrically separate the via from substrate **1748**. Depending on the thickness of substrate **1748**, contact **1752** can be an ILV having a depth in the submicron-level (e.g., between 10 nm and 1 μm), or a TSV having a depth in the micron- or tens micron-level (e.g., between 1 μm and 100 μm).

(126) Although not shown, it is understood that the pad-out of 3D memory devices is not limited to from second semiconductor structure **104** having DRAM cells **1724** as shown in FIG. **17** and may be from first semiconductor structure **102** having peripheral circuit **1712** in the similar manner as described above with respect to FIG. **6**B. Although not shown, it is also understood that the air gaps between word lines **1734** and/or between semiconductor bodies **1730** may be partially or fully filled with dielectrics in the similar manner as described above with respect to FIG. **6**E. Although not shown, it is further understood that more than one array of DRAM cells **1724** may be stacked over one another to vertically scale up the number of DRAM cells **1724** in the similar manner as described above with respect to FIG. **7**.

(127) As described above, in some examples, trench isolations **1616** extending in the word line directions in FIG. **16** may not be formed such that two adjacent semiconductor bodies **1608** separated by a respective trench isolation **1616** may be merged as a single semiconductor body having two opposite sides in the bit line direction in contact with gate structure **1610**. That is, without trench isolations **1616**, the adjacent single-gate vertical transistors may be merged to form a double-gate vertical transistor (e.g., dual-side gate vertical transistor) with increased gate control area and lower leakage current. For example, FIG. **20** illustrates a plan view of yet another array of memory cells **2002** each including a vertical transistor in a memory device **2000**, according to some aspects of the present disclosure. As shown in FIG. **20**, memory device **2000** can include a plurality of word lines **2004** each extending in a first lateral direction (the x-direction, referred to as the word line direction). Memory device **2000** can also include a plurality of bit lines **2006** each extending in a second lateral direction perpendicular to the first lateral direction (the y-direction, referred to as the bit line direction). It is understood that FIG. **20** does not illustrate a cross-section of memory device **2000** in the same lateral plane, and word lines **2004** and bit lines **2006** may be formed in different lateral planes for ease of routing as described below in detail.

(128) Memory cells **2002** can be formed at the intersections of word lines **2004** and bit lines **2006**. In some implementations, each memory cell **2002** includes a vertical transistor (e.g., vertical transistor **210** in FIG. **2**) having a semiconductor body **2008** and a gate structure **2010**. Semiconductor body **2008** can extend in a substrate in the vertical direction (the z-direction, not shown) perpendicular to the first and second lateral directions. The vertical transistor can be a double-gate transistor in which gate structure **2010** is in contact with two sides (e.g., two of four sides in FIG. **20**) of semiconductor body **2008** (the active region in which channels are formed). As shown in FIG. **20**, the vertical transistor is a double-gate transistor in which gate structure **2010** abuts two opposite sides of semiconductor body **1608** (having a rectangle or square-shaped cross-section) in the bit line direction (the y-direction) in the plan view. Gate structure **2010** does not surround and contact the other two sides of semiconductor body **2008** in the word line direction

(the x-direction), according to some implementations. That is, gate structure **2010** can partially circumscribes semiconductor body **2008** in the plan view. Gate structure **2010** can include a gate dielectric **2012** abuts two opposite sides of semiconductor body **2008** in the plan view, and a gate electrode **2014** in contact with gate dielectric **2012**. In some implementations, gate dielectric **2012** is laterally between gate electrode **2014** and semiconductor body **2008** in the bit line direction (the y-direction). As described above, gate electrode **2014** may be part of word line **2004**, and word line **2004** may be an extension of gate electrode **2014**. That is, gate electrodes **1614** of adjacent vertical transistors in the word line direction (the x-direction) are continuous, e.g., parts of a continuous conductive layer having gate electrodes **1614** and **1604**. In some implementations, gate structures **2010** of a row of the vertical transistors are continuous in the x-direction, as a shown in FIG. **20**.

(129) Different from separate gate dielectrics **512** and **812** in FIGS. **5** and **8**, as shown in FIG. **20**, gate dielectrics **2012** of adjacent vertical transistors in the word line direction are continuous, e.g., parts of a continuous dielectric layer having gate dielectrics **2012** and extending in the word line direction. Gate structures **2010** can be thus viewed as parts of a continuous structure extending in the word line direction at which the continuous structure intersects vertical transistors in the same row.

(130) FIG. **21** illustrates a side view of a cross-section of yet another 3D memory device **2100** including vertical transistors, according to some aspects of the present disclosure. 3D memory device **2100** may be one example of memory device **2000** including double-gate vertical transistors in which gate structures abut two sides of semiconductor bodies in the plan view. It is understood that FIG. **21** is for illustrative purposes only and may not necessarily reflect the actual device structure (e.g., interconnections) in practice. As one example of 3D memory device **100** described above with respect to FIG. **1**A, 3D memory device **2100** is a bonded chip including first semiconductor structure **102** and second semiconductor structure **104** stacked over first semiconductor structure **102**. First and second semiconductor structures **102** and **104** are jointed at bonding interface **106** therebetween, according to some implementations. As shown in FIG. **21**, first semiconductor structure **102** can include a substrate **2110**, which can include silicon (e.g., single crystalline silicon, c-Si), SiGe, GaAs, Ge, SOI, or any other suitable materials.

(131) First semiconductor structure **102** can include peripheral circuits **2112** on substrate **2110**. In some implementations, peripheral circuits **2112** include a plurality of transistors **2114** (e.g., planar transistors and/or 3D transistors). Trench isolations (e.g., shallow trench isolations (STIs)) and doped regions (e.g., wells, sources, and drains of transistors **2114**) can be formed on or in substrate **2110** as well.

(132) In some implementations, first semiconductor structure **102** further includes an interconnect layer **2116** above peripheral circuits **2112** to transfer electrical signals to and from peripheral circuits **2112**. Interconnect layer **2116** can include a plurality of interconnects (also referred to herein as "contacts"), including lateral interconnect lines and VIA contacts. Interconnect layer **1716** can further include one or more ILD layers in which the interconnect lines and via contacts can form. That is, interconnect layer **2116** can include interconnect lines and via contacts in multiple ILD layers. In some implementations, peripheral circuits **2112** are coupled to one another through the interconnects in interconnect layer **2116**. The interconnects in interconnect layer **2116** can include conductive materials including, but not limited to, W, Co, Cu, Al, doped silicon, silicides, or any combination thereof. The ILD layers can be formed with dielectric materials including, but not limited to, silicon oxide, silicon nitride, silicon oxynitride, low-k dielectrics, or any combination thereof.

(133) As shown in FIG. **21**, first semiconductor structure **102** can further include a bonding layer **2118** at bonding interface **106** and above interconnect layer **2116** and peripheral circuits **2112**. Bonding layer **2118** can include a plurality of bonding contacts **2119** and dielectrics electrically isolating bonding contacts **2119**. Bonding contacts **2119** can include conductive materials, such as Cu. The remaining area of bonding layer **2118** can be formed with dielectric materials, such as

silicon oxide. Bonding contacts **2119** and surrounding dielectrics in bonding layer **2118** can be used for hybrid bonding. Similarly, as shown in FIG. **21**, second semiconductor structure **104** can also include a bonding layer **2120** at bonding interface **106** and above bonding layer **2118** of first semiconductor structure **102**. Bonding layer **2120** can include a plurality of bonding contacts **2121** and dielectrics electrically isolating bonding contacts **2121**. Bonding contacts **2121** can include conductive materials, such as Cu. The remaining area of bonding layer **2120** can be formed with dielectric materials, such as silicon oxide. Bonding contacts **2121** and surrounding dielectrics in bonding layer **2120** can be used for hybrid bonding. Bonding contacts **2121** are in contact with bonding contacts **2119** at bonding interface **106**, according to some implementations. In some implementations, bonding layer **2120** includes a dielectric layer opposing DRAM cells **2124** with bit line **2123** positioned between the dielectric layer and DRAM cells **2124**, as shown in FIG. **21**. The dielectric layer can include bonding interface **106** having bonding contacts **2121**.

(134) Second semiconductor structure **104** can be bonded on top of first semiconductor structure **102** in a face-to-face manner at bonding interface **106**. In some implementations, bonding interface **106** is disposed between bonding layers **2120** and **2118** as a result of hybrid bonding (also known as "metal/dielectric hybrid bonding"), which is a direct bonding technology (e.g., forming bonding between surfaces without using intermediate layers, such as solder or adhesives) and can obtain metal-metal bonding and dielectric-dielectric bonding simultaneously. In some implementations, bonding interface **106** is the place at which bonding layers **2120** and **2118** are met and bonded. In practice, bonding interface **106** can be a layer with a certain thickness that includes the top surface of bonding layer **2118** of first semiconductor structure **102** and the bottom surface of bonding layer **2120** of second semiconductor structure **104**.

(135) In some implementations, second semiconductor structure **104** further includes an interconnect layer **2122** including bit lines **2123** above bonding layer **2120** to transfer electrical signals. Interconnect layer **2122** can include a plurality of interconnects, such as MEOL interconnects and BEOL interconnects. In some implementations, the interconnects in interconnect layer **2122** also include local interconnects, such as bit lines **2123** (e.g., an example of bit lines **2006** in FIG. **20**) and word line contacts (not shown). Interconnect layer **2122** can further include one or more ILD layers in which the interconnect lines and via contacts can form. The interconnects in interconnect layer **2122** can include conductive materials including, but not limited to, W, Co, Cu, Al, doped silicon, silicides, or any combination thereof. The ILD layers can be formed with dielectric materials including, but not limited to, silicon oxide, silicon nitride, silicon oxynitride, low-k dielectrics, or any combination thereof. In some implementations, peripheral circuits **2112** include a word line driver/row decoder coupled to the word line contacts in interconnect layer **2122** through bonding contacts **2121** and **2119** in bonding layers **2120** and **2118** and interconnect layer **2116**. In some implementations, peripheral circuits **2112** include a bit line driver/column decoder coupled to bit lines **2123** and the bit line contacts in interconnect layer **2122** through bonding contacts **2121** and **2119** in bonding layers **2120** and **2118** and interconnect layer **2116**. In some implementations, bit line **2123** is a metal bit line, as opposed to semiconductor bit lines (e.g., doped silicon bit lines). For example, bit line **2123** may include W, Co, Cu, Al, or any other suitable metals having higher conductivities than doped silicon. In some implementations, the bit line contact is an ohmic contact, such as a metal silicide contact, as opposed to a Schottky contact. For example, the bit line contact may include metal silicides, such as WSi, CoSi, CuSi, AlSi, or any other suitable metal silicides having higher conductivities than doped silicon.

(136) In some implementations, second semiconductor structure **104** includes a DRAM device in which memory cells are provided in the form of an array of DRAM cells **2124** (e.g., an example of memory cells **2002** in FIG. **20**) above interconnect layer **2122** and bonding layer **2120**. That is, interconnect layer **2122** including bit lines **2123** can be disposed between bonding layer **2120** and array of DRAM cells **2124**. It is understood that the cross-section of 3D memory device **2100** in FIG. **21** may be made along the bit line direction (the y-direction), and one bit line **2123** in

interconnect layer **2122** extending laterally in the y-direction may be coupled to a column of DRAM cells **2124**.

(137) Each DRAM cell **2124** can include a vertical transistor **2126** (e.g., an example of vertical transistors **210** in FIG. **2**) and capacitor **2128** (e.g., an example of storage unit **212** in FIG. **2**) coupled to the vertical transistor **2126**. DRAM cell **2124** can be a 1T1C cell consisting of one transistor and one capacitor. It is understood that DRAM cell **2124** may be of any suitable configurations, such as 2T1C cell, 3T1C cell, etc.

(138) Vertical transistor **2126** can be a MOSFET used to switch a respective DRAM cell **2124**. In some implementations, vertical transistor **2126** includes a semiconductor body **2130** (the active region in which channels can form) extending vertically (in the z-direction), and a gate structure **2136** in contact with two opposite sides of semiconductor body **2130** in the bit line direction (the y-direction). As described above, as in a double-gate vertical transistor, semiconductor body **2130** can have a cuboid shape or a cylinder shape, and gate structure **2136** can abut two sides of semiconductor body **2130** in the plan view, for example, as shown in FIG. **21**. Gate structure **2136** includes a gate electrode **2134** and a gate dielectric **2132** laterally between gate electrode **2134** and semiconductor body **2130** in the bit line direction, according to some implementations. In some implementations, gate dielectric **2132** abuts two sides of semiconductor body **2130**, and gate electrode **2134** abuts gate dielectric **2132**.

(139) As shown in FIG. **21**, in some implementations, semiconductor body **2130** has two ends (the upper end and lower end) in the vertical direction (the z-direction), and at least one end (e.g., the lower end in FIG. **21**) extends beyond gate dielectric **2132** in the vertical direction (the z-direction) into ILD layers. In some implementations, one end (e.g., the upper end in FIG. **21**) of semiconductor body **2130** is flush with the respective end (e.g., the upper end in FIG. **21**) of gate dielectric **2132**. In some implementations, both ends (the upper end and lower end) of semiconductor body **2130** extend beyond gate electrode **2134**, respectively, in the vertical direction (the z-direction) into ILD layers. That is, semiconductor body **2130** can have a larger vertical dimension (e.g., the depth) than that of gate electrode **2134** (e.g., in the z-direction), and neither the upper end nor the lower end of semiconductor body **2130** is flush with the respective end of gate electrode **2134**. Thus, short circuits between bit lines **2123** and word lines/gate electrodes **2134** or between word lines/gate electrodes **2134** and capacitors **2128** can be avoided. Vertical transistor **2126** can further include a source and a drain (both referred to as **2138** as their locations may be interchangeable) disposed at the two ends (the upper end and lower end) of semiconductor body **2130**, respectively, in the vertical direction (the z-direction). In some implementations, one of source and drain **2138** (e.g., at the upper end in FIG. **21**) is coupled to capacitor **2128**, and the other one of source and drain **2138** (e.g., at the lower end in FIG. **21**) is coupled to bit line **2123**. That is, vertical transistor **2126** can have a first terminal in the positive z-direction and a second terminal opposite the first terminal in the negative z-direction, as shown in FIG. **21**. In some implementations, a metal bit line (e.g., bit line **2123** made of a metal material) is coupled to the second terminal of vertical transistor **2126** via an ohmic contact (e.g., a bit line contact made of a metal silicide material).

(140) In some implementations, semiconductor body **2130** includes semiconductor materials, such as single crystalline silicon, polysilicon, amorphous silicon, Ge, any other semiconductor materials, or any combinations thereof. In one example, semiconductor body **2130** may include single crystalline silicon. Source and drain **2138** can be doped with N-type dopants (e.g., P or As) or P-type dopants (e.g., B or Ga) at a desired doping level. In some implementations, a silicide layer, such as a metal silicide layer, is formed between source/drain **2138** of vertical transistor **2126** and bit line **2123** as the bit line contact or between source/drain **1738** of vertical transistor **2126** and the first electrode of capacitor **2128** as capacitor contact **2142** to reduce the contact resistance. In some implementations, gate dielectric **2132** includes dielectric materials, such as silicon oxide, silicon nitride, or high-k dielectrics including, but not limited to, Al.sub.2O.sub.3, HfO.sub.2,

Ta.sub.2O.sub.5, ZrO.sub.2, TiO.sub.2, or any combination thereof. In some implementations, gate electrode **2134** includes conductive materials including, but not limited to W, Co, Cu, Al, TiN, TaN, polysilicon, silicides, or any combination thereof. In some implementations, gate electrode **2134** includes multiple conductive layers, such as a W layer over a TiN layer. In one example, gate structure **2136** may be a "gate oxide/gate poly" gate in which gate dielectric **2132** includes silicon oxide and gate electrode **2134** includes doped polysilicon. In another example, gate structure **2136** may be an HKMG in which gate dielectric **2132** includes a high-k dielectric and gate electrode **2134** includes a metal.
(141) As described above, since gate electrode **2134** may be part of a word line or extend in the word line direction as a word line, second semiconductor structure **104** of 3D memory device **2100** can also include a plurality of word lines (e.g., an example of word lines **2004** in FIG. **20**, referred to as **2134** as well) each extending in the word line direction. Each word line **2134** can be coupled to a row of DRAM cells **2124**. That is, bit line **2123** and word line **2134** can extend in two perpendicular lateral directions, and semiconductor body **2130** of vertical transistor **2126** can extend in the vertical direction perpendicular to the two lateral directions in which bit line **2123** and word line **2134** extend. Word lines **2134** are in contact with word line contacts (not shown), according to some implementations. In some implementations, word lines **2134** include conductive materials including, but not limited to W, Co, Cu, Al, TiN, TaN, polysilicon, silicides, or any combination thereof. In some implementations, word line **2134** includes multiple conductive layers, such as a W layer over a TiN layer.
(142) As shown in FIG. **21**, vertical transistor **2126** extends vertically through and contacts word lines **2134**, and source or drain **2138** of vertical transistor **2126** at the lower end thereof is in contact with bit line **2123** (or bit line contact if any), according to some implementations. Accordingly, word lines **2134** and bit lines **2123** can be disposed in different planes in the vertical direction due to the vertical arrangement of vertical transistor **2126**, which simplifies the routing of word lines **2134** and bit lines **2123**. In some implementations, bit lines **2123** are disposed vertically between bonding layer **2120** and word lines **2134**, and word lines **2134** are disposed vertically between bit lines **2123** and capacitors **2128**. Word lines **2134** can be coupled to peripheral circuits **2112** in first semiconductor structure **102** through word line contacts in interconnect layer **2122**, bonding contacts **2121** and **2119** in bonding layers **2120** and **2118**, and the interconnects in interconnect layer **2116**. Similarly, bit lines **2123** in interconnect layer **2122** can be coupled to peripheral circuits **2112** in first semiconductor structure **102** through bonding contacts **2121** and **2119** in bonding layers **2120** and **2118** and the interconnects in interconnect layer **2116**.
(143) In some implementations, second semiconductor structure **104** further includes a plurality of air gaps **2140** each disposed laterally between adjacent word lines **2134**. Each air gap **2140** can be a trench extending in the word line direction (e.g., the x-direction) in parallel with word lines **2134** to separate adjacent rows of vertical transistors **2126**. As described below with respect to the fabrication process, air gaps **2140** may be formed due to the relatively small pitches of word lines **2134** (and rows of DRAM cells **2124**) in the bit line direction (e.g., the y-direction). On the other hand, the relatively large dielectric constant of air in air gaps **2140** (e.g., about 4 times of the dielectric constant of silicon oxide) can improve the insulation effect between word lines **2134** (and rows of DRAM cells **2124**) compared with some dielectrics (e.g., silicon oxide).
(144) As shown in FIG. **21**, in some implementations, capacitor **2128** includes a first electrode above and coupled to source or drain **2138** of vertical transistor **2126**, e.g., the upper end of semiconductor body **2130**, via a capacitor contact **2142**. In some implementations, capacitor contact **2142** is an ohmic contact, such as a metal silicide contact, as opposed to a Schottky contact. For example, capacitor contact **2142** may include metal silicides, such as WSi, CoSi, CuSi, AlSi, or any other suitable metal silicides having higher conductivities than doped silicon. Capacitor **2128** can also include a capacitor dielectric above and in contact with the first electrode, and a second electrode above and in contact with the capacitor dielectric. That is, capacitor **2128** can be a

vertical capacitor in which the electrodes and capacitor dielectric are stacked vertically (in the z-direction), and the capacitor dielectric can be sandwiched between the electrodes. In some implementations, each first electrode is coupled to source or drain **2138** of a respective vertical transistor **2126** in the same DRAM cell, while all second electrodes are coupled to a common plate **2146** coupled to the ground, e.g., a common ground. Capacitor **2128** can have a first end in the negative z-direction and a second end opposite the first end in the positive z-direction, as shown in FIG. **21**. In some implementations, the first end of capacitor **2128** is coupled to the first terminal of vertical transistor **2126** via an ohmic contact (e.g., capacitor contact **2142** made of a metal silicide material). As shown in FIG. **21**, second semiconductor structure **104** can further include a capacitor contact **2147** (e.g., a conductor) in contact with common plate **2146** for coupling capacitors **2128** to peripheral circuits **2112** or to the ground directly. In some implementations, capacitor contact **2147** (e.g., a conductor) extends in the z-direction from the dielectric layer of bonding layer **2120** to couple to the second end of capacitor **2128** via common plate **2146**, as shown in FIG. **21**. In some implementation, the ILD layer in which capacitors **2128** are formed has the same dielectric material as the two ILD layers into which semiconductor body **2130** extends, such as silicon oxide.

(145) It is understood that the structure and configuration of capacitor **2128** are not limited to the example in FIG. **21** and may include any suitable structure and configuration, such as a planar capacitor, a stack capacitor, a multi-fins capacitor, a cylinder capacitor, a trench capacitor, or a substrate-plate capacitor. In some implementations, the capacitor dielectric includes dielectric materials, such as silicon oxide, silicon nitride, or high-k dielectrics including, but not limited to, Al.sub.2O.sub.3, HfO.sub.2, Ta.sub.2O.sub.5, ZrO.sub.2, TiO.sub.2, or any combination thereof. It is understood that in some examples, capacitor **2128** may be a ferroelectric capacitor used in a FRAM cell, and the capacitor dielectric may be replaced by a ferroelectric layer having ferroelectric materials, such as PZT or SBT. In some implementations, the electrodes include conductive materials including, but not limited to W, Co, Cu, Al, TiN, TaN, polysilicon, silicides, or any combination thereof.

(146) As shown in FIG. **21**, vertical transistor **2126** extends vertically through and contacts word lines **2134**, source or drain **2138** of vertical transistor **2126** at the lower end thereof is in contact with bit line **2123** directly or through the bit line contact, and source or drain **2138** of vertical transistor **2126** at the upper end thereof is coupled to capacitor **2128**, according to some implementations. That is, bit line **2123** and capacitor **2128** can be disposed in different planes in the vertical direction and coupled to opposite ends of vertical transistor **2126** of DRAM cell **2124** in the vertical direction due to the vertical arrangement of vertical transistor **2126**. In some implementations, bit line **2123** and capacitor **2128** are disposed on opposite sides of vertical transistor **2126** in the vertical direction, which simplifies the routing of bit lines **2123** and reduces the coupling capacitance between bit lines **2123** and capacitors **2128** compared with DRAM cells in which the bit lines and capacitors are disposed on the same side of the planar transistors.

(147) As shown in FIG. **21**, in some implementations, vertical transistors **2126** are disposed vertically between capacitors **2128** and bonding interface **106**. That is, vertical transistors **2126** can be arranged closer to peripheral circuits **2114** of first semiconductor structure **102** and bonding interface **106** than capacitors **2128**. Since bit lines **2123** and capacitors **2128** are coupled to opposite ends of vertical transistors **2126**, as described above, bit lines **2123** (as part of interconnect layer **2122**) are disposed vertically between vertical transistors **2126** and bonding interface **106**, according to some implementations. As a result, interconnect layer **2122** including bit lines **2123** can be arranged close to bonding interface **106** to reduce the interconnect routing distance and complexity.

(148) In some implementations, second semiconductor structure **104** further includes a substrate **2148** disposed above DRAM cells **2124**. As described below with respect to the fabrication process, substrate **2148** can be part of a carrier wafer. It is understood that in some examples, substrate **2148** may not be included in second semiconductor structure **104**.

(149) As shown in FIG. **21**, second semiconductor structure **104** can further include a pad-out interconnect layer **2150** above substrate **2148** and DRAM cells **2124**. Pad-out interconnect layer **2150** can include interconnects, e.g., contact pads **2154**, in one or more ILD layers. Pad-out interconnect layer **2150** and interconnect layer **2122** can be formed on opposite sides of DRAM cells **2124**. Capacitors **2128** are disposed vertically between vertical transistors **2126** and pad-out interconnect layer **2150**, according to some implementations. In some implementations, the interconnects in pad-out interconnect layer **2150** can transfer electrical signals between 3D memory device **2100** and outside circuits, e.g., for pad-out purposes. In some implementations, second semiconductor structure **104** further includes one or more contacts **2152** extending through substrate **2148** and part of pad-out interconnect layer **2150** to couple pad-out interconnect layer **2150** to DRAM cells **2124** and interconnect layer **2122**. As a result, peripheral circuits **2112** can be coupled to DRAM cells **2124** through interconnect layers **2116** and **2122** as well as bonding layers **2120** and **2118**, and peripheral circuits **2112** and DRAM cells **2124** can be coupled to outside circuits through contacts **2152** and pad-out interconnect layer **2150**. Contact pads **2154** and contacts **2152** can include conductive materials including, but not limited to, W, Co, Cu, Al, silicides, or any combination thereof. In one example, contact pad **2154** may include Al, and contact **2152** may include W. In some implementations, contact **2152** includes a via surrounded by a dielectric spacer (e.g., having silicon oxide) to electrically separate the via from substrate **2148**. Depending on the thickness of substrate **2148**, contact **2152** can be an ILV having a depth in the submicron-level (e.g., between 10 nm and 1 μm), or a TSV having a depth in the micron- or tens micron-level (e.g., between 1 μm and 100 μm).
(150) Although not shown, it is understood that the pad-out of 3D memory devices is not limited to from second semiconductor structure **104** having DRAM cells **2124** as shown in FIG. **21** and may be from first semiconductor structure **102** having peripheral circuit **2112** in the similar manner as described above with respect to FIG. **6**B. Although not shown, it is also understood that the air gaps between word lines **2134** may be partially or fully filled with dielectrics in the similar manner as described above with respect to FIG. **6**E. Although not shown, it is further understood that more than one array of DRAM cells **2124** may be stacked over one another to vertically scale up the number of DRAM cells **2124** in the similar manner as described above with respect to FIG. **7**.
(151) FIG. **27** illustrates a block diagram of a system **2700** having a memory device, according to some aspects of the present disclosure. System **2700** can be a mobile phone, a desktop computer, a laptop computer, a tablet, a vehicle computer, a gaming console, a printer, a positioning device, a wearable electronic device, a smart sensor, a virtual reality (VR) device, an argument reality (AR) device, or any other suitable electronic devices having storage therein. As shown in FIG. **27**, system **2700** can include a host **2708** and a memory system **2702** having one or more memory devices **2704** and a memory controller **2706**. Host **2708** can be a processor of an electronic device, such as a central processing unit (CPU), or a system-on-chip (SoC), such as an application processor (AP). Host **2708** can be configured to send or receive the data to or from memory devices **2704**.
(152) Memory device **2704** can be any memory devices disclosed herein, such as 3D memory devices **100** and **101**, memory devices **200**, **500**, **800**, **1600**, and **2000**, and 3D memory devices **600**, **601**, **603**, **605**, **607**, **700**, **900**, **1700**, and **2100**. In some implementations, memory device **2704** includes an array of memory cells each including a vertical transistor, as described above in detail.
(153) Memory controller **2706** is coupled to memory device **2704** and host **2708** and is configured to control memory device **2704**, according to some implementations. Memory controller **2706** can manage the data stored in memory device **2704** and communicate with host **2708**. Memory controller **2706** can be configured to control operations of memory device **2704**, such as read, write, and refresh operations. Memory controller **2706** can also be configured to manage various functions with respect to the data stored or to be stored in memory device **2704** including, but not limited to refresh and timing control, command/request translation, buffer and schedule, and power

management. In some implementations, memory controller **2706** is further configured to determines the maximum memory capacity that the computer system can use, the number of memory banks, memory type and speed, memory particle data depth and data width, and other important parameters. Any other suitable functions may be performed by memory controller **2706** as well. Memory controller **2706** can communicate with an external device (e.g., host **2708**) according to a particular communication protocol. For example, memory controller **2706** may communicate with the external device through at least one of various interface protocols, such as a USB protocol, an MMC protocol, a peripheral component interconnection (PCI) protocol, a PCI-express (PCI-E) protocol, an advanced technology attachment (ATA) protocol, a serial-ATA protocol, a parallel-ATA protocol, a small computer small interface (SCSI) protocol, an enhanced small disk interface (ESDI) protocol, an integrated drive electronics (IDE) protocol, a Firewire protocol, etc.

(154) FIGS. **10**A-**10**M illustrate a fabrication process for forming a 3D memory device including vertical transistors, according to some aspects of the present disclosure. FIGS. **11**A-**11**I illustrate a fabrication process for forming another 3D memory device including vertical transistors, according to some aspects of the present disclosure. FIGS. **12**A-**12**H illustrate a fabrication process for forming still another 3D memory device including vertical transistors, according to some aspects of the present disclosure. FIGS. **13**A-**13**H illustrate a fabrication process for forming yet another 3D memory device including vertical transistors, according to some aspects of the present disclosure. FIGS. **14**A-**14**E illustrate a fabrication process for forming yet another 3D memory device including vertical transistors, according to some aspects of the present disclosure. FIGS. **15**A-**15**D illustrate a fabrication process for forming yet another 3D memory device including vertical transistors, according to some aspects of the present disclosure. FIGS. **19**A-**19**M illustrate a fabrication process for forming yet another 3D memory device including vertical transistors, according to some aspects of the present disclosure. FIGS. **22**A-**22**M illustrate a fabrication process for forming yet another 3D memory device including vertical transistors, according to some aspects of the present disclosure. FIG. **23** illustrates a flowchart of a method **2300** for forming a 3D memory device including vertical transistors, according to some aspects of the present disclosure. Examples of the 3D memory devices depicted in FIGS. **10**A-**10**M include 3D memory devices **600** and **601** depicted in FIGS. **6**A and **6**B. Examples of the 3D memory devices depicted in FIGS. **11**A-**11**I include 3D memory device **900** depicted in FIG. **9**. Examples of the 3D memory devices depicted in FIGS. **12**A-**12**H include 3D memory device **603** depicted in FIG. **6**C. Examples of the 3D memory devices depicted in FIGS. **13**A-**13**H include 3D memory device **605** depicted in FIG. **6**D. Examples of the 3D memory devices depicted in FIGS. **14**A-**14**E and **15**A-**15**D include 3D memory device **700** depicted in FIG. **7**. Examples of the 3D memory devices depicted in FIGS. **19**A-**19**M include 3D memory device **1700** depicted in FIG. **17**. Examples of the 3D memory devices depicted in FIGS. **22**A-**22**M include 3D memory device **2100** depicted in FIG. **21**. FIGS. **10**A-**10**M, **11**A-**11**I, **12**A-**12**H, **13**A-**13**H, **14**A-**14**E, **15**A-**15**D, **19**A-**19**M, **22**A-**22**M, and **23** will be described together. It is understood that the operations shown in method **2300** are not exhaustive and that other operations can be performed as well before, after, or between any of the illustrated operations. Further, some of the operations may be performed simultaneously, or in a different order than shown in FIG. **2300**.

(155) In some implementations, a first semiconductor structure including a peripheral circuit is formed. As depicted in FIG. **10**L or **19**L, a first semiconductor structure including peripheral circuits is formed. In some implementations, a second semiconductor structure including a first array of memory cells and a plurality of bit lines coupled to the memory cells is formed. Each of the memory cells can include a vertical transistor, and a storage unit coupled to the vertical transistor. A respective one of the bit lines and a respective storage unit are coupled to opposite ends of each one of the memory cells vertically. As depicted in FIG. **10**L, **11**I, **12**H, **13**H, **19**L, or **22**L, a second semiconductor structure including an array of DRAM cells, each of which includes a

vertical transistor, and a capacitor coupled to the vertical transistor, is formed. The second semiconductor structure also includes a plurality of bit lines coupled to the memory cells, and a respective one of the bit lines and a respective storage unit are coupled to opposite ends of each one of the memory cells vertically. In some implementations, the first semiconductor structure and the second semiconductor structure are bonded in a face-to-face manner, such that the first array of memory cells is coupled to the peripheral circuit across a bonding interface. As depicted in FIGS. **10**L and **10**M, **19**L, or **22**L, the first and second semiconductor structures are bonded in a face-to-face manner, such that the array of DRAM cells is coupled to the peripheral circuit across a bonding interface.

(156) Referring to FIG. **23**, method **2300** starts at operation **2302**, in which a peripheral circuit is formed on a first substrate. The first substrate can include a silicon substrate. In some implementations, an interconnect layer is formed above the peripheral circuit. The interconnect layer can include a plurality of interconnects in one or more ILD layers.

(157) As illustrated in FIG. **10**L, a plurality of transistors **1042** are formed on a silicon substrate **1038**. Transistors **1042** can be formed by a plurality of processes including, but not limited to, photolithography, dry/wet etch, thin film deposition, thermal growth, implantation, chemical mechanical polishing (CMP), and any other suitable processes. In some implementations, doped regions are formed in silicon substrate **1038** by ion implantation and/or thermal diffusion, which function, for example, as the source and drain of transistors **1042**. In some implementations, isolation regions (e.g., STIs) are also formed in silicon substrate **1038** by wet/dry etch and thin film deposition. Transistors **1042** can form peripheral circuits **1040** on silicon substrate **1038**.

(158) As illustrated in FIG. **10**L, an interconnect layer **1044** can be formed above peripheral circuits **1040** having transistors **1042**. Interconnect layer **1044** can include interconnects of MEOL and/or BEOL in a plurality of ILD layers to make electrical connections with peripheral circuits **1040**. In some implementations, interconnect layer **1044** includes multiple ILD layers and interconnects therein formed in multiple processes. For example, the interconnects in interconnect layers **1044** can include conductive materials deposited by one or more thin film deposition processes including, but not limited to, chemical vapor deposition (CVD), physical vapor deposition (PVD), atomic layer deposition (ALD), electroplating, electroless plating, or any combination thereof. Fabrication processes to form the interconnects can also include photolithography, CMP, wet/dry etch, or any other suitable processes. The ILD layers can include dielectric materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. The ILD layers and interconnects illustrated in FIG. **10**L can be collectively referred to as interconnect layer **1044**.

(159) Method **2300** proceeds to operation **2304**, as illustrated in FIG. **23**, in which a first bonding layer is formed above the peripheral circuit (and the interconnect layer). The first bonding layer can include a first bonding contact. As illustrated in FIG. **10**L, a bonding layer **1046** is formed above interconnect layer **1044** and peripheral circuits **1040**. Bonding layer **1046** can include a plurality of bonding contacts **1047** surrounded by dielectrics. In some implementations, a dielectric layer (e.g., ILD layer) is deposited on the top surface of interconnect layer **1044** by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. Bonding contacts **1047** then can be formed through the dielectric layer and in contact with the interconnects in interconnect layer **1044** by first patterning contact holes through the dielectric layer using patterning process (e.g., photolithography and dry/wet etch of dielectric materials in the dielectric layer). The contact holes can be filled with a conductor (e.g., Cu). In some implementations, filling the contact holes includes depositing a barrier layer, an adhesion layer, and/or a seed layer before depositing the conductor.

(160) Method **2300** proceeds to operation **2306**, as illustrated in FIG. **23**, in which an array of memory cells each including a vertical transistor and a storage unit is formed on a second substrate. The second substrate can include a carrier substrate. The storage unit can include a capacitor or a

PCM element. In some implementations, a capacitor is formed to be coupled to the vertical transistor in the respective memory cell.

(161) For example, FIG. **24** illustrates a flowchart of a method **2400** for forming an array of memory cells each including a vertical transistor, according to some aspects of the present disclosure. At operation **2402** in FIG. **24**, a stack of dielectric layers is formed on a substrate. In some implementations, to form the stack of dielectric layers, three layers having a first dielectric, a second dielectric, and the first dielectric, respectively, are subsequently deposited on the substrate. The first dielectric can include silicon oxide, and the second dielectric can include silicon nitride. The layer having the second dielectric can act as a sacrificial layer sandwiched vertically between two layers having the first dielectric. The sacrificial layer can be removed by selectively etching against the two layers having the first dielectric and replaced with a conductive layer in the later processes.

(162) As illustrated in FIG. **10**A, a stack of a silicon oxide layer **1004**, a silicon nitride layer **1006**, and a silicon oxide layer **1008** is formed on a silicon substrate **1002**. To form the dielectric stack, silicon oxide, silicon nitride, and silicon oxide are subsequently deposited onto silicon substrate **1002** using one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. In some implementations, silicon oxide layer **1004** is formed by oxidizing the top portion of silicon substrate **1002** using dry oxidation and/or we oxidation, such as in situ steam generation (ISSG) oxidation process. In some implementations, the thickness of silicon oxide layer **1004** (e.g., ISSG silicon oxide) is smaller than the thickness of silicon oxide layer **1008** (e.g., CVD silicon oxide). FIG. **10**A illustrates both the side view (in the top portion of FIG. **10**A) of a cross-section along the y-direction (the bit line direction, e.g., in the CC plane) and the plan view (in the bottom portion of FIG. **10**A) of a cross-section in the x-y plane (e.g., in the AA plane through silicon nitride layer **1006**). The same drawing layout is arranged in FIGS. **10**B-**10**G as well.

(163) At operation **2404** in FIG. **24**, a semiconductor body extending vertically from the substrate through the stack of dielectric layers is formed. In some implementations, to form the semiconductor body, an opening extending through the stack of dielectric layers is etched to expose part of the substrate, and the semiconductor body is epitaxially grown from the exposed part of the substrate in the opening.

(164) As illustrated in FIG. **10**B, an array of openings **1010** is formed, each of which extends vertically (in the z-direction) through the stack of silicon oxide layer **1008**, silicon nitride layer **1006**, and silicon oxide layer **1004** to silicon substrate **1002**. As a result, parts of silicon substrate **1002** can be exposed from openings **1010**. In some implementations, a lithography process is performed to pattern the array of openings **1010** using an etch mask (e.g., a photoresist mask), for example, based on the design of word lines and bit lines, and one or more dry etching and/or wet etching processes, such as reactive ion etch (RIE), are performed to etch openings **1010** through silicon oxide layer **1008**, silicon nitride layer **1006**, and silicon oxide layer **1004** until being stopped by silicon substrate **1002**.

(165) As illustrated in FIG. **10**C, an array of semiconductor bodies **1012** are formed in openings **1010**. Semiconductor body **1012** can be epitaxially grown from the respective exposed part of silicon substrate **1002** in the respective opening **1010**. The fabrication processes for epitaxially growing semiconductor body **1012** can include, but not limited to, vapor-phase epitaxy (VPE), liquid-phase epitaxy (LPE), molecular-beam epitaxy (MPE), or any combinations thereof. The epitaxy can occur upward (toward the positive z-direction) from the exposed parts of silicon substrate **1002** in openings **1010**. Semiconductor body **1012** thus can have the same material as silicon substrate **1002**, in the form of single crystalline silicon. Depending on the shape of opening **1010**, semiconductor body **1012** can have the same shape as opening **1010**, such as a cuboid shape or a cylinder shape. In some implementations, a planarization process, such as CMP, is performed to remove excess parts of semiconductor bodies **1012** beyond the top surface of silicon oxide layer

1008. As a result, an array of semiconductor bodies **1012** (e.g., single crystalline silicon bodies) extending vertically (in the z-direction) from silicon substrate **1002** through the stack of silicon oxide layer **1008**, silicon nitride layer **1006**, and silicon oxide layer **1004** is formed thereby, according to some implementations.

(166) At operation **2406** in FIG. **24**, one of the stack of dielectric layers is removed to expose part of the semiconductor body. In some implementations, to remove the one of the stack of dielectric layers, a trench is etched through at least part of the stack of dielectric layers to expose the layer having the second dielectric, and the layer having the second dielectric (e.g., the sacrificial layer) is etched away via the trench. In some implementations, the trench is etched between adjacent rows of semiconductor bodies without touching any sides of the semiconductor bodies.

(167) As illustrated in FIG. **10**D, a plurality of trenches **1014** (slits openings) each extending laterally along the word line direction (the x-direction) and extending vertically through at least silicon oxide layer **1008** and silicon nitride layer **1006** are formed to expose silicon nitride layer **1006**. As a result, parts of silicon nitride layer **1006** can be exposed from trenches **1014**. In some implementations, a lithography process is performed to pattern trenches **1014** using an etch mask (e.g., a photoresist mask), for example, based on the design of word lines (word line trenches). As shown in FIG. **10**D, trench **1014** is patterned to be formed between adjacent rows of semiconductor bodies **1012** without touching any sides of semiconductor bodies **1012**, such that semiconductor bodies **1012** are not exposed from any side thereof. In one example, trench **1014** is patterned to be formed in the middle between adjacent rows of semiconductor bodies **1012**, e.g., having the same distance two adjacent rows of semiconductor bodies **1012**. In some implementations, one or more dry etching and/or wet etching processes, such as RIE, are performed to etch trenches **1014** through silicon oxide layer **1008**, silicon nitride layer **1006**, and silicon oxide layer **1004** until being stopped by silicon substrate **1002**. It is understood that in some examples, the etching of trenches **1014** may not go all the way to silicon substrate **1002** and may be stopped at silicon oxide layer **1004** so long as silicon nitride layer **1006** is exposed from trenches **1014**.

(168) As illustrated in FIG. **10**E, silicon nitride layer **1006** (shown in FIG. **10**D) is removed to expose parts of semiconductor bodies **1012** abutting silicon nitride layer **1006**. In some implementations, silicon nitride layer **1006** is etched away via trenches **1014**. For example, a wet etchant including phosphoric acid may be applied through trenches **1014** to selectively wet etch silicon nitride layer **1006** without etching silicon oxide layers **1004** and **1008** as well as semiconductor bodies **1012** and silicon substrate **1002**. As a result, lateral recesses **1016** can be formed vertically between silicon oxide layers **1004** and **1008** thereby, which expose parts of semiconductor bodies **1012**. As shown in the plan view, all sides of each semiconductor body **1012** can be exposed from lateral recesses **1016**.

(169) At operation **2408** in FIG. **24**, a gate structure in contact with a plurality of sides of the exposed part of the semiconductor body is formed. In some implementations, to form the gate structure, a gate dielectric is formed over the exposed part of the semiconductor body, a conductive layer is deposited over the gate dielectric, and the conductive layer is patterned to form a gate electrode over the gate dielectric.

(170) As illustrated in FIG. **10**F, a gate dielectric **1018** is formed over the exposed part of each semiconductor body **1012**, e.g., surrounding and contacting all the sides of the exposed part of semiconductor body **1012**. As shown in the plan view, gate dielectric **1018** can fully circumscribe a respective semiconductor body **1012**. In some implementations, a wet oxidation and/or a dry oxidation process, such as ISSG, is performed to form native oxide (e.g., silicon oxide) on semiconductor body **1012** (e.g., single crystalline silicon) as gate dielectric **1018**. In some implementations, gate dielectric **1018** is formed by depositing a layer of dielectric, such as silicon oxide, over the exposed part of semiconductor body **1012** through trenches **1014** and lateral recesses **1016** using one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof, without filling lateral recesses **1016** and trenches **1014**.

(171) As illustrated in FIG. **10**G, a conductive layer **1020** is formed over gate dielectrics **1018** in lateral recesses **1016** (shown in FIG. **10**F) through trenches **1014**. In some implementations, conductive layer **1020** is formed by depositing conductive materials, such as metal or metal compounds (e.g., TiN), over gate dielectrics **1018** using one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof, through trenches **1014** to fill lateral recesses **1016**. In one example, the deposition of conductive layer **1020** is controlled not to fill trenches **1014**. It is understood that in some examples, the deposition of conductive layer **1020** may fill trenches **1014** as well. Thus, a planarization process, e.g., CMP, may be performed to remove the excess conductive layer **1020** over the top surface of silicon oxide layer **1008**, and conductive layer **1020** may be patterned to form a gate electrode over a respective gate dielectric. For example, trenches **1014** filled with conductive layer **1020** may be patterned and etched again to separate conductive layers **1020** between adjacent rows of semiconductor bodies **1012** and gate dielectrics **1018**. As described above, a lithography process can be performed to pattern trenches **1014** again using an etch mask (e.g., a photoresist mask), for example, based on the design of word lines (word line trenches).

(172) As a result, patterned conductive layers **1020** can become word lines each extending in the word line direction (the x-direction) and being separated by adjacent trenches **1014**, and parts of patterned conductive layers **1020** that are over gate dielectrics **1018** (e.g., fully circumscribes a respective gate dielectric **1018** in the plan view) can become gate electrodes. Gate structures each including a respective gate dielectric **1018** over the exposed part of semiconductor body **1012** and a respective gate electrode (part of conductive layer **1020**) over gate dielectric **1018** can be formed thereby. Since conductive layer **1020** remains on all sides of semiconductor body **1012** (and gate dielectric **1018** thereover) when patterning conductive layer **1020** (etching trenches **1014**), the gate structure is in contact with all sides of semiconductor body **1012**, according to some implementations, as shown in FIG. **10**G. As shown in the plan view, the gate structure (having gate dielectric **1018** and the gate electrode) can fully circumscribe a respective semiconductor body **1012**, and all sides of each semiconductor body **1012** can be surrounded and contacted by the respective gate structure. Comparing FIG. **10**G with FIG. **10**A, silicon nitride layer **1006** (sacrificial layer) in FIG. **10**A is eventually replaced with conductive layer **1020** in FIG. **10**G, according to some implementations.

(173) At operation **2410** in FIG. **24**, a first end of the semiconductor body away from the substrate is doped. As illustrated in FIG. **10**G, the exposed upper end of each semiconductor body **1012**, e.g., one of the two ends of semiconductor body **1012** in the vertical direction (the z-direction) that is away from silicon substrate **1002**, is doped to form a source/drain **1021** (e.g., a source terminal of a vertical transistor). In some implementations, an implantation process and/or thermal diffusion process are performed to dope P-type dopants or N-type dopants to exposed upper ends of semiconductor bodies **1012** to form sources/drains **1021**. In some implementations, a silicide layer is formed on source/drain **1021** by performing a silicidation process at the exposed upper ends of semiconductor bodies **1012**.

(174) At operation **2412** in FIG. **24**, a storage unit in contact with the semiconductor body, e.g., the doped first end thereof, is formed. The storage unit can include a capacitor or a PCM element. In some implementations, to form a storage unit that is a capacitor, a first electrode is formed on the doped first end of the semiconductor body, a capacitor dielectric is formed on the first electrode, and a second electrode is formed on the capacitor dielectric.

(175) As illustrated in FIG. **10**H, one or more ILD layers are formed over the top surface of silicon oxide layer **1008**, for example, by depositing dielectrics using one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. It is understood that in some examples in which the ILD layers include silicon oxide, the same material as silicon oxide layer **1008**, the boundary and interface between the ILD layer and silicon oxide layer **1008** may become indistinguishable after the deposition. Depending on the lateral dimensions

of trenches **1014** (shown in FIG. **10**G), trenches **1014** may not be fully filled with dielectrics (e.g., silicon oxide) when forming the ILD layers and thus, become air gaps **1022** between adjacent word lines (patterned conductive layers **1020**). It is understood that in some examples, when the lateral dimensions of trenches **1014** are sufficiently large, dielectrics may fully fill trenches **1014** during the formation of the ILD layers, thereby eliminating air gaps **1022**.

(176) As illustrated in FIG. **10**H, capacitor contacts **1024**, first electrodes, capacitor dielectrics, and second electrodes of capacitors **1026**, and common plate **1028** are subsequently formed in the ILD layers to be coupled to semiconductor bodies **1012**. In some implementations, each capacitor contact **1024** is formed on a respective source/drain **1021**, e.g., the doped upper end of a respective semiconductor body **1012** by patterning and etching an electrode hole aligned with respective source/drain **1021** using lithography and etching processes and depositing conductive materials to fill the electrode hole using thin film deposition processes. In some implementations, common plate **1028** is formed on the second electrodes of capacitors **1026** by patterning and etching an electrode trench aligned with capacitors **1026** using lithography and etching processes and depositing conductive materials to fill the electrode trench using thin film deposition processes.

(177) At operation **2414** in FIG. **24**, the substrate is removed to expose a second end of the semiconductor body opposite to the first end. As illustrated in FIG. **10**I, a carrier substrate **1030** (a.k.a. a handle substrate) is bonded onto the front side of silicon substrate **1002** on which devices are formed using any suitable bonding processes, such as anodic bonding, fusion bonding, transfer bonding, adhesive bonding, and eutectic bonding. The bonded structure can then be flipped upside down, such that silicon substrate **1002** become above carrier substrate **1030**.

(178) As illustrated in FIG. **10**J, silicon substrate **1002** (shown in FIG. **10**I) is removed to expose the undoped upper ends of semiconductor bodies **1012** (used to be the lower ends before flipping over). In some implementations, planarization processes (e.g., CMP) and/or etching processes are performed to remove silicon substrate **1002** until being stopped by silicon oxide layer **1004** and the upper ends of semiconductor bodies **1012**.

(179) At operation **2416** in FIG. **24**, the exposed second end of the semiconductor body is doped. As illustrated in FIG. **10**J, the exposed upper end of each semiconductor body **1012**, e.g., one of the two ends of semiconductor body **1012** in the vertical direction (the z-direction) that is away from carrier substrate **1030**, is doped to form another source/drain **1023** (e.g., a source terminal of the vertical transistor). In some implementations, an implantation process and/or thermal diffusion process are performed to dope P-type dopants or N-type dopants to exposed upper ends of semiconductor bodies **1012** to form sources/drains **1023**. In some implementations, a silicide layer is formed on source/drain **1023** by performing a silicidation process at the exposed upper ends of semiconductor bodies **1012**. As a result, vertical transistors having semiconductor body **1012**, sources/drains **1021** and **1023**, gate dielectric **1018**, and the gate electrode (part of conductive layer **1020**) are formed thereby, as shown in FIG. **10**J, according to some implementations. As described above, capacitors **1026** each having the first and second electrodes and the capacitor dielectric are thereby formed as well, and DRAM cells **1080** each having a multi-gate vertical transistor and a capacitor coupled to the multi-gate vertical transistor are thereby formed, as shown in FIG. **10**J, according to some implementations.

(180) Referring back to FIG. **23**, method **2300** proceeds to operation **2308**, as illustrated in FIG. **23**, in which an interconnect layer including bit lines is formed above the array of memory cells. As illustrated in FIG. **10**K, an interconnect layer **1032** can be formed above DRAM cells **1080**. Interconnect layer **1032** can include interconnects of MEOL and/or BEOL in a plurality of ILD layers to make electrical connections with DRAM cells **1080**. In some implementations, interconnect layer **1032** includes multiple ILD layers and interconnects therein formed in multiple processes. For example, the interconnects in interconnect layers **1032** can include conductive materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, electroplating, electroless plating, or any combination thereof. Fabrication

processes to form the interconnects can also include photolithography, CMP, wet/dry etch, or any other suitable processes. The ILD layers can include dielectric materials deposited on silicon oxide layer **1004** by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. The ILD layers and interconnects illustrated in FIG. **10**K can be collectively referred to as interconnect layer **1032**.

(181) As shown in FIG. **24**, at operation **2418**, to form the interconnect layer, a bit line is formed on the doped second end. As illustrated in FIG. **10**K, bit line **1034** can be formed on sources/drains **1023** by patterning and etching a trench aligned with respective source/drain **1023** using lithography and etching processes and depositing conductive materials to fill the trench using thin film deposition processes. In some implementations, forming bit line **1034** includes depositing a metal layer onto the exposed end of semiconductor body **1012**. As a result, bit line **1034** and capacitor **1026** can be formed on opposite sides of semiconductor body **1012** and coupled to opposite ends of semiconductor body **1012**. It is understood that additional local interconnects, such as word line contacts **1039**, capacitor contacts **1083** (e.g., a conductor), and bit line contacts **1041** (e.g., a metal silicide contact) may be similarly formed as well. In some implementations, bit line contact **1041** (e.g., a metal silicide contact) is formed on the exposed end of semiconductor body **1012**, and bit line **1034** is formed on bit line contact **1041**.

(182) Method **2300** proceeds to operation **2310**, as illustrated in FIG. **23**, in which a second bonding layer is formed above the array of memory cells and the interconnect layer. The second bonding layer can include a second bonding contact. As illustrated in FIG. **10**K, a bonding layer **1036** is formed above interconnect layer **1032** and DRAM cells **1080**. Bonding layer **1036** can include a plurality of bonding contacts **1037** surrounded by dielectrics. In some implementations, a dielectric layer (e.g., ILD layer) is deposited on the top surface of interconnect layer **1032** by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. Bonding contacts **1037** can then be formed through the dielectric layer and in contact with the interconnects in interconnect layer **1032** by first patterning contact holes through the dielectric layer using patterning process (e.g., photolithography and dry/wet etch of dielectric materials in the dielectric layer). The contact holes can be filled with a conductor (e.g., Cu). In some implementations, filling the contact holes includes depositing a barrier layer, an adhesion layer, and/or a seed layer before depositing the conductor.

(183) Method **2300** proceeds to operation **2312**, as illustrated in FIG. **23**, in which the first semiconductor structure and the second semiconductor structure are bonded in a face-to-face manner, such that the first array of memory cells is coupled to the peripheral circuit across a bonding interface. The bonding can include hybrid bonding. In some implementations, the first bonding contact is in contact with the second bonding contact at the bonding interface after the bonding. In some implementations, the second semiconductor structure is above the first semiconductor structure after the bonding. In some implementations, the first semiconductor structure is above the second semiconductor structure after the bonding.

(184) As illustrated in FIG. **10**L, carrier substrate **1030** and components formed thereon (e.g., DRAM cells **1080**) are flipped upside down. As illustrated in FIG. **10**M, bonding layer **1036** facing down is bonded with bonding layer **1046** facing up, e.g., in a face-to-face manner, thereby forming a bonding interface **1050**. In some implementations, a treatment process, e.g., a plasma treatment, a wet treatment, and/or a thermal treatment, is applied to the bonding surfaces prior to the bonding. Although not shown in FIGS. **10**L and **10**M, silicon substrate **1038** and components formed thereon (e.g., peripheral circuits **1040**) can be flipped upside down, and bonding layer **1046** facing down can be bonded with bonding layer **1036** facing up, e.g., in a face-to-face manner, thereby forming bonding interface **1050**. After the bonding, bonding contacts **1037** in bonding layer **1036** and bonding contacts **1047** in bonding layer **1046** are aligned and in contact with one another, such that DRAM cells **1080** can be electrically connected to peripheral circuits **1040** across bonding interface **1050**. It is understood that in the bonded chip, DRAM cells **1080** may be either above or

below peripheral circuits **1040**. Nevertheless, bonding interface **1050** can be formed vertically between peripheral circuits **1040** and DRAM cells **1080** after the bonding.

(185) Method **2300** proceeds to operation **2314**, as illustrated in FIG. **23**, in which a pad-out interconnect layer is formed on the backside of the first semiconductor structure or the second semiconductor structure. As illustrated in FIG. **10**M, a pad-out interconnect layer **1052** is formed above on the backside of carrier substrate **1030**. Pad-out interconnect layer **1052** can include interconnects, such as pad contacts **1054**, formed in one or more ILD layers. Pad contacts **1054** can include conductive materials including, but not limited to, W, Co, Cu, Al, doped silicon, silicides, or any combination thereof. The ILD layers can include dielectric materials including, but not limited to, silicon oxide, silicon nitride, silicon oxynitride, low-k dielectrics, or any combination thereof. In some implementations, after the bonding, contacts **1056** are formed extending vertically through carrier substrate **1030**, for example, by wet/dry etching processes, followed by depositing conductive materials. Contacts **1056** can be in contact with the interconnects in pad-out interconnect layer **1052**. It is understood that in some examples, carrier substrate **1030** may be thinned or removed after bonding and prior to forming pad-out interconnect layer **1052** and contacts **1056**, for example, using planarization processes and/or etching processes.

(186) Although not shown, it is understood that in some examples, pad-out interconnect layer **1052** may be formed above on the backside of silicon substrate **1038**, and contacts **1056** may be formed extending vertically through silicon substrate **1038**. Silicon substrate **1038** may be thinned prior to forming pad-out interconnect layer **1052** and contacts **1056**, for example, using planarization processes and/or etching processes.

(187) As described above, FIGS. **10**A-**10**M illustrates a fabrication process of forming a DRAM array having a vertical transistor in which the gate structure is in contact with all sides of the semiconductor body, in the form of a GAA transistor. In some implementations as shown in FIG. **11**A-**11**I, by changing the layout of word line trenches, a DRAM array having a vertical transistor in which the gate structure is in contact with some sides (e.g., three of the four sides) of the semiconductor body are formed with a relatively larger pitch of word lines and reduced fabrication complexity.

(188) As illustrated in FIG. **11**A, a stack of a silicon oxide layer **1104**, a silicon nitride layer **1106**, and a silicon oxide layer **1108** is formed on a silicon substrate **1102**. To form the dielectric stack, silicon oxide, silicon nitride, and silicon oxide are subsequently deposited onto silicon substrate **1102** using one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. In some implementations, silicon oxide layer **1104** is formed by oxidizing the top portion of silicon substrate **1102** using dry oxidation and/or we oxidation, such as ISSG oxidation process. In some implementations, the thickness of silicon oxide layer **1104** (e.g., ISSG silicon oxide) is smaller than the thickness of silicon oxide layer **1108** (e.g., CVD silicon oxide). Besides the side view of the cross-section along the y-direction (e.g., the bit line direction) shown in the top portion of FIG. **11**A, the plan view of the cross-section in the x-y plane through silicon nitride layer **1106** is also shown in the bottom portion of FIG. **11**A. The same drawing layout is arranged in FIGS. **11**B-**11**E as well.

(189) As illustrated in FIG. **11**B, an array of semiconductor bodies **1112** each extending vertically through the stack of silicon oxide layer **1108**, silicon nitride layer **1106**, and silicon oxide layer **1104** are formed. Semiconductor body **1112** can be epitaxially grown from the respective exposed part of silicon substrate **1102** in a respective opening (not shown). The fabrication processes for epitaxially growing semiconductor body **1112** can include, but not limited to, VPE, LPE, MPE, or any combinations thereof. The epitaxy can occur upward (toward the positive z-direction) from the exposed parts of silicon substrate **1102** in the openings. Semiconductor body **1112** thus can have the same material as silicon substrate **1102**, in the form of single crystalline silicon. In some implementations, a planarization process, such as CMP, is performed to remove excess parts of semiconductor bodies **1112** beyond the top surface of silicon oxide layer **1108**. As a result, an array

of semiconductor bodies **1112** (e.g., single crystalline silicon bodies) extending vertically (in the z-direction) from silicon substrate **1102** through the stack of silicon oxide layer **1108**, silicon nitride layer **1106**, and silicon oxide layer **1104** is formed thereby, according to some implementation.

(190) In some implementations, at operation **2406** in FIG. **24**, one of the stack of dielectric layers is removed to expose part of the semiconductor body. In some implementations, to remove the one of the stack of dielectric layers, a trench is etched through at least part of the stack of dielectric layers to expose the layer having the second dielectric, and the layer having the second dielectric (e.g., the sacrificial layer) is etched away via the trench. In some implementations, the trench is etched aligned with one side of the semiconductor body to expose the semiconductor body from the side.

(191) As illustrated in FIG. **11**C, a plurality of trenches **1114** (slits openings) each extending laterally along the word line direction (the x-direction) and extending vertically through at least silicon oxide layer **1108** and silicon nitride layer **1106** are formed to expose silicon nitride layer **1106**. As a result, parts of silicon nitride layer **1106** can be exposed from trenches **1114**. In some implementations, a lithography process is performed to pattern trenches **1114** using an etch mask (e.g., a photoresist mask), for example, based on the design of word lines (word line trenches). As shown in FIG. **11**C, trench **1114** is patterned to be formed between adjacent rows of semiconductor bodies **1112** and aligned with one side of semiconductor bodies **1112** to expose the semiconductor bodies **1112** from the side, according to some implementations. That is, trench **1114** can be patterned to touch one side of semiconductor bodies **1112**, such that semiconductor bodies **1112** are exposed from the side. In some implementations, one or more dry etching and/or wet etching processes, such as RIE, are performed to etch trenches **1114** through silicon oxide layer **1108**, silicon nitride layer **1106**, and silicon oxide layer **1104** until being stopped by silicon substrate **1102**. It is understood that in some examples, the etching of trenches **1114** may not go all the way to silicon substrate **1102** and may be stopped at silicon oxide layer **1104** so long as silicon nitride layer **1106** is exposed from trenches **1114**.

(192) As illustrated in FIG. **11**D, silicon nitride layer **1106** (shown in FIG. **11**C) is removed to expose parts of semiconductor bodies **1112** abutting silicon nitride layer **1106**. In some implementations, silicon nitride layer **1106** is etched away via trenches **1114**. For example, a wet etchant including phosphoric acid may be applied through trenches **1114** to selectively wet etch silicon nitride layer **1106** without etching silicon oxide layers **1104** and **1108** as well as semiconductor bodies **1112** and silicon substrate **1102**. As a result, lateral recesses **1116** can be formed vertically between silicon oxide layers **1104** and **1008** thereby, which expose parts of semiconductor bodies **1112**.

(193) As illustrated in FIG. **11**D, a gate dielectric **1118** is formed over the exposed part of each semiconductor body **1112**, e.g., surrounding and contacting all the sides of the exposed part of semiconductor body **1112**. As shown in the plan view, gate dielectric **1118** can fully circumscribe a respective semiconductor body **1112**. In some implementations, a wet oxidation and/or a dry oxidation process, such as ISSG, is performed to form native oxide (e.g., silicon oxide) on semiconductor body **1112** (e.g., single crystalline silicon) as gate dielectric **1118**. In some implementations, gate dielectric **1118** is formed by depositing a layer of dielectric, such as silicon oxide, over the exposed part of semiconductor body **1112** through trenches **1114** and lateral recesses **1116** using one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof, without filling lateral recesses **1116** and trenches **1114**.

(194) Different from FIG. **10**F in which gate dielectric **1118** has a uniform vertical dimension (thickness in the z-direction) because all sides of semiconductor body **1112** is surrounded by lateral recess **1016** having the same vertical dimension, in FIG. **11**D, because one side of semiconductor body **1112** is aligned with and touches trench **1114** having a greater vertical dimension than that of lateral recess **1116**, part of gate dielectric **1118** formed on that side of semiconductor body **1112** (referred to as elongated gate dielectric part **1119**) can have a greater vertical dimension than the remainder of gate dielectric **1118** formed on other sides of semiconductor body **1112** touching

lateral recess **1116**, as shown in the side view of FIG. **11**D.

(195) As illustrated in FIG. **11**E, a conductive layer **1120** is formed over gate dielectric **1118** in lateral recesses **1116** (shown in FIG. **11**D) through trenches **1114**, but not over elongated gate dielectric part **1119**. In some implementations, conductive layer **1120** is formed by depositing conductive materials, such as metal or metal compounds (e.g., TiN), over gate dielectrics **1118** using one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof, through trenches **1114** to fill lateral recesses **1116**. In one example, the deposition of conductive layer **1120** is controlled not to fill trenches **1114** (and not over elongated gate dielectric part **1119**). It is understood that in some examples, the deposition of conductive layer **1120** may fill trenches **1114** as well. Thus, a planarization process, e.g., CMP, may be performed to remove the excess conductive layer **1120** over the top surface of silicon oxide layer **1108**, and conductive layer **1120** may be patterned to form a gate electrode over only gate dielectric **1118**, but not elongated gate dielectric part **1119**. For example, trenches **1114** filled with conductive layer **1120** may be patterned and etched again to separate conductive layers **1120** between adjacent rows of semiconductor bodies **1112** and gate dielectrics **1118**. As described above, a lithography process can be performed to pattern trenches **1114** again using an etch mask (e.g., a photoresist mask), for example, based on the design of word lines (word line trenches).

(196) As a result, patterned conductive layers **1120** can become word lines each extending in the word line direction (the x-direction) and being separated by adjacent trenches **1114**, and parts of patterned conductive layers **1120** that are over gate dielectrics **1118**, but not elongated gate dielectric part **1119**, can become gate electrodes. Gate structures each including a respective gate dielectric **1118** over the exposed part semiconductor body **1112** and a respective gate electrode (part of conductive layer **1120**) over gate dielectric **1118** can be formed thereby. Since conductive layer **1120** remains on only some sides of semiconductor body **1012** (and gate dielectric **1018** thereover) when patterning conductive layer **1120** (etching trenches **1114**), the gate structure is in contact with some, but not all, sides of semiconductor body **1012**, according to some implementations, as shown in FIG. **11**E. As shown in the plan view, the gate structure (having gate dielectric **1118** and the gate electrode) can partially circumscribe a respective semiconductor body **1112**, and not all sides of each semiconductor body **1112** can be surrounded and contacted by the respective gate structure. Compared with the pitch of word lines **1020** in the example in FIG. **10**G, the pitch of word lines **1120** in FIG. **11**E may be increased to reduce the fabrication complexity.

(197) As illustrated in FIG. **11**E, the exposed upper end of each semiconductor body **1112**, e.g., one of the two ends of semiconductor body **1112** in the vertical direction (the z-direction) that is away from silicon substrate **1102**, is doped to form a source/drain **1121** (e.g., a source terminal of a vertical transistor). In some implementations, an implantation process and/or thermal diffusion process are performed to dope P-type dopants or N-type dopants to exposed upper ends of semiconductor bodies **1112** to form sources/drains **1021**.

(198) As illustrated in FIG. **11**F, one or more ILD layers are formed over the top surface of silicon oxide layer **1108**, for example, by depositing dielectrics using one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. It is understood that in some examples in which the ILD layers include silicon oxide, the same material as silicon oxide layer **1108**, the boundary and interface between the ILD layer and silicon oxide layer **1108** may become indistinguishable after the deposition. Due to the relatively larger lateral dimensions of trenches **1114** (shown in FIG. **11**E) compared with trenches **1014** (as a result of a larger pitch of word lines **1120**), trenches **1114** may be fully filled or at least partially with dielectrics (e.g., silicon oxide) when forming the ILD layers and thus, eliminating air gaps **1022** or at least reducing air gaps **1022** between adjacent word lines (patterned conductive layers **1120**).

(199) As illustrated in FIG. **11**F, capacitor contacts **1124**, first electrodes, capacitor dielectrics, and second electrodes of capacitors **1126**, and common plate **1128** are subsequently formed in the ILD layers to be coupled to semiconductor bodies **1112**. In some implementations, each capacitor

contact **1124** is formed on a respective source/drain **1121**, e.g., the doped upper end of a respective semiconductor body **1112** by patterning and etching an electrode hole aligned with respective source/drain **1121** using lithography and etching processes and depositing conductive materials to fill the electrode hole using thin film deposition processes. In some implementations, common plate **1128** is formed on the second electrodes of capacitors **1126** by patterning and etching an electrode trench aligned with capacitors **1126** using lithography and etching processes and depositing conductive materials to fill the electrode trench using thin film deposition processes.

(200) As illustrated in FIG. **11**G, a carrier substrate **1130** (a.k.a. a handle substrate) is bonded onto the front side of silicon substrate **1102** on which devices are formed using any suitable bonding processes, such as anodic bonding, fusion bonding, transfer bonding, adhesive bonding, and eutectic bonding. The bonded structure can then be flipped upside down, such that silicon substrate **1102** become above carrier substrate **1130**.

(201) As illustrated in FIG. **11**H, silicon substrate **1102** (shown in FIG. **11**G) is removed to expose the undoped upper ends of semiconductor bodies **1112** (used to be the lower ends before flipping over). In some implementations, planarization processes (e.g., CMP) and/or etching processes are performed to remove silicon substrate **1102** until being stopped by silicon oxide layer **1104** and the upper ends of semiconductor bodies **1112**.

(202) As illustrated in FIG. **11**H, the exposed upper end of each semiconductor body **1112**, e.g., one of the two ends of semiconductor body **1112** in the vertical direction (the z-direction) that is away from carrier substrate **1130**, is doped to form another source/drain **1123** (e.g., a drain terminal of the vertical transistor). In some implementations, an implantation process and/or thermal diffusion process are performed to dope P-type dopants or N-type dopants to exposed upper ends of semiconductor bodies **1112** to form sources/drains **1123**. As a result, multi-gate vertical transistors having semiconductor body **1112**, sources/drains **1121** and **1123**, gate dielectric **1118** (not including elongated gate dielectric part **1119**), and the gate electrode (part of conductive layer **1120**) are formed thereby, as shown in FIG. **11**H, according to some implementations. As described above, capacitors **1126** are thereby formed as well, and DRAM cells **1180** each having a multi-gate vertical transistor and a capacitor coupled to the multi-gate vertical transistor are thereby formed, as shown in FIG. **11**H, according to some implementations.

(203) As illustrated in FIG. **11**I, an interconnect layer **1132** can be formed above DRAM cells **1180**. Interconnect layer **1132** can include interconnects of MEOL and/or BEOL in a plurality of ILD layers to make electrical connections with DRAM cells **1180**. In some implementations, interconnect layer **1132** includes multiple ILD layers and interconnects therein formed in multiple processes. For example, the interconnects in interconnect layers **1132** can include conductive materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, electroplating, electroless plating, or any combination thereof. Fabrication processes to form the interconnects can also include photolithography, CMP, wet/dry etch, or any other suitable processes. The ILD layers can include dielectric materials deposited on silicon oxide layer **1104** by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. The ILD layers and interconnects illustrated in FIG. **11**I can be collectively referred to as interconnect layer **1132**. As illustrated in FIG. **11**I, bit line **1134** can be formed on sources/drains **1123** by patterning and etching a trench aligned with respective source/drain **1123** using lithography and etching processes and depositing conductive materials to fill the trench using thin film deposition processes. In some implementations, forming bit line **1134** includes depositing a metal layer onto the exposed end of semiconductor body **1112**.

(204) As illustrated in FIG. **11**I, a bonding layer **1136** is formed above interconnect layer **1132** and DRAM cells **1180**. Bonding layer **1136** can include a plurality of bonding contacts **1137** surrounded by dielectrics. In some implementations, a dielectric layer (e.g., ILD layer) is deposited on the top surface of interconnect layer **1132** by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. Bonding contacts **1137** can then be

formed through the dielectric layer and in contact with the interconnects in interconnect layer **1132** by first patterning contact holes through the dielectric layer using patterning process (e.g., photolithography and dry/wet etch of dielectric materials in the dielectric layer). The contact holes can be filled with a conductor (e.g., Cu). In some implementations, filling the contact holes includes depositing a barrier layer, an adhesion layer, and/or a seed layer before depositing the conductor.

(205) As described above, FIGS. **10**A-**10**M illustrates a fabrication process of forming a DRAM cell array from a three-layer dielectric stack having a sacrificial layer (e.g., silicon nitride layer **1006**) sandwiched between two dielectric layers (e.g., silicon oxide layers **1004** and **1008**). It is understood that the configuration of the dielectric stack from which the DRAM cell array is formed can vary in other examples, resulting DRAM cells with different structures, such as in 3D memory devices **603** and **605** in FIGS. **6**C and **6**D. In some implementations as shown in FIGS. **12**A-**12**H, a DRAM cell array is formed from a two-layer dielectric stack having a sacrificial layer on a dielectric layer.

(206) At operation **2402** in FIG. **24**, a stack of dielectric layers is formed on a substrate. In some implementations, to form the stack of dielectric layers, two layers having a first dielectric and a second dielectric, respectively, are subsequently deposited on the substrate. The first dielectric can include silicon oxide, and the second dielectric can include silicon nitride. The layer having the second dielectric can act as a sacrificial layer on the layer having the first dielectric. The sacrificial layer can be removed by selectively etching against the other layer having the first dielectric and replaced with a conductive layer in the later processes.

(207) As illustrated in FIG. **12**A, a stack of a silicon oxide layer **1204** and a silicon nitride layer **1206** is formed on a silicon substrate **1202**. To form the dielectric stack, silicon oxide and silicon nitride are subsequently deposited onto silicon substrate **1202** using one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. In some implementations, silicon oxide layer **1204** is formed by oxidizing the top portion of silicon substrate **1202** using dry oxidation and/or we oxidation, such as in situ steam generation (ISSG) oxidation process.

(208) As illustrated in FIG. **12**B, an array of semiconductor bodies **1212** each extending vertically through the stack of silicon nitride layer **1206** and silicon oxide layer **1204** are formed. Semiconductor body **1212** can be epitaxially grown from the respective exposed part of silicon substrate **1202** in a respective opening (not shown). The fabrication processes for epitaxially growing semiconductor body **1212** can include, but not limited to, VPE, LPE, MPE, or any combinations thereof. The epitaxy can occur upward (toward the positive z-direction) from the exposed parts of silicon substrate **1202** in the openings. Semiconductor body **1212** thus can have the same material as silicon substrate **1202**, in the form of single crystalline silicon. In some implementations, a planarization process, such as CMP, is performed to remove excess parts of semiconductor bodies **1212** beyond the top surface of silicon nitride layer **1206**. As a result, an array of semiconductor bodies **1212** (e.g., single crystalline silicon bodies) extending vertically (in the z-direction) from silicon substrate **1202** through the stack of silicon nitride layer **1206** and silicon oxide layer **1204** is formed thereby, according to some implementation.

(209) As illustrated in FIG. **12**C, a plurality of trenches **1214** (slits openings) each extending laterally along the word line direction (the x-direction) and extending vertically through at least silicon nitride layer **1206** are formed to expose silicon nitride layer **1206**. As a result, parts of silicon nitride layer **1206** can be exposed from trenches **1214**. In some implementations, a lithography process is performed to pattern trenches **1214** using an etch mask (e.g., a photoresist mask), for example, based on the design of word lines (word line trenches). In some implementations, one or more dry etching and/or wet etching processes, such as RIE, are performed to etch trenches **1214** through silicon nitride layer **1206** and silicon oxide layer **1204** until being stopped by silicon substrate **1202**. It is understood that in some examples, the etching of

trenches **1214** may not go all the way to silicon substrate **1202** and may be stopped at silicon oxide layer **1204** so long as silicon nitride layer **1206** is exposed from trenches **1214**.

(210) As illustrated in FIG. **12**D, silicon nitride layer **1206** (shown in FIG. **12**C) is removed to expose parts of semiconductor bodies **1212** abutting silicon nitride layer **1206**. In some implementations, silicon nitride layer **1206** is etched away via trenches **1214**. For example, a wet etchant including phosphoric acid may be applied through trenches **1214** to selectively wet etch silicon nitride layer **1206** without etching silicon oxide layer **1204** as well as semiconductor bodies **1212** and silicon substrate **1202**. As a result, lateral recesses **1216** can be formed thereby, which expose parts of semiconductor bodies **1212**. It is understood that in some examples, the top surface of silicon nitride layer **1206** may be exposed, such that trenches **1214** may not be needed to remove silicon nitride layer **1206**. Dry etching and/or wet etching processes may be applied directly on silicon nitride layer **1206** to remove silicon nitride layer **1206** (from FIG. **12**B to FIG. **12**D directly without going through FIG. **12**C).

(211) As illustrated in FIG. **12**D, a gate dielectric **1218** is formed over the exposed part of each semiconductor body **1212**, e.g., surrounding and contacting all the sides of the exposed part of semiconductor body **1212**. In some implementations, a wet oxidation and/or a dry oxidation process, such as ISSG, is performed to form native oxide (e.g., silicon oxide) on semiconductor body **1212** (e.g., single crystalline silicon) as gate dielectric **1218**. In some implementations, gate dielectric **1218** is formed by depositing a layer of dielectric, such as silicon oxide, over the exposed part of semiconductor body **1212** through trenches **1214** and lateral recesses **1216** using one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof, without filling lateral recesses **1216** and trenches **1214**. Due to the omission of silicon oxide layer **1008**, the upper end of gate dielectric **1218** can be flush with the upper end of semiconductor body **1212** as shown in FIG. **12**D, while the upper end of gate dielectric **1018** is below the upper end of semiconductor body **1012** in FIG. **10**F.

(212) As illustrated in FIG. **12**E, a conductive layer **1220** is formed over gate dielectrics **1218** in lateral recesses **1216** (shown in FIG. **12**D) through trenches **1214**. In some implementations, conductive layer **1220** is formed by depositing conductive materials, such as metal or metal compounds (e.g., TiN), over gate dielectrics **1218** using one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof, through trenches **1214** to fill lateral recesses **1216**. In one example, the deposition of conductive layer **1220** is controlled not to fill trenches **1214**. It is understood that in some examples, the deposition of conductive layer **1220** may fill trenches **1214** as well. Thus, a planarization process, e.g., CMP, may be performed to remove the excess conductive layer **1220** to expose the upper ends of semiconductor bodies **1212**, and conductive layer **1220** may be patterned to form a gate electrode over a respective gate dielectric. For example, trenches **1214** filled with conductive layer **1220** may be patterned and etched again to separate conductive layers **1220** between adjacent rows of semiconductor bodies **1212** and gate dielectrics **1218**. As described above, a lithography process can be performed to pattern trenches **1214** again using an etch mask (e.g., a photoresist mask), for example, based on the design of word lines (word line trenches). Due to the omission of silicon oxide layer **1008**, the top surface of conductive layer **1220** (including gate electrodes and word line) can be flush with the upper end of semiconductor body **1212** as shown in FIG. **12**E, while the top surface of word line **1020** is below the upper end of semiconductor body **1012** in FIG. **10**G.

(213) As a result, patterned conductive layers **1220** can become word lines each extending in the word line direction (the x-direction) and being separated by adjacent trenches **1214**, and parts of patterned conductive layers **1220** that are over gate dielectrics **1218** (e.g., fully circumscribes a respective gate dielectric **1218** in the plan view) can become gate electrodes. Gate structures each including a respective gate dielectric **1218** over the exposed part semiconductor body **1212** and a respective gate electrode (part of conductive layer **1220**) over gate dielectric **1218** can be formed thereby. Comparing FIG. **12**E with FIG. **12**A, silicon nitride layer **1206** (sacrificial layer) in FIG.

12A is eventually replaced with conductive layer **1220** in FIG. **12**E, according to some implementations.
(214) As illustrated in FIG. **12**E, the exposed upper end of each semiconductor body **1212**, e.g., one of the two ends of semiconductor body **1212** in the vertical direction (the z-direction) that is away from silicon substrate **1202**, is doped to form a source/drain **1221**. In some implementations, an implantation process and/or thermal diffusion process are performed to dope P-type dopants or N-type dopants to exposed upper ends of semiconductor bodies **1212** to form sources/drains **1221**.
(215) As illustrated in FIG. **12**F, one or more ILD layers are formed over the top surface of conductive layer **1220**, for example, by depositing dielectrics using one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. Depending on the lateral dimensions of trenches **1214** (shown in FIG. **12**E), trenches **1214** may not be fully filled with dielectrics (e.g., silicon oxide) when forming the ILD layers and thus, become air gaps **1222** between adjacent word lines (patterned conductive layers **1220**). It is understood that in some examples, when the lateral dimensions of trenches **1214** are sufficiently large, dielectrics may fully fill trenches **1214** during the formation of the ILD layers, thereby eliminating air gaps **1222**.
(216) As illustrated in FIG. **12**F, capacitor contacts **1224**, first electrodes, capacitor dielectrics, and second electrodes of capacitors **1226**, and a common plate **1228** are subsequently formed in the ILD layers to be coupled to semiconductor bodies **1212**. In some implementations, each capacitor contact **1224** is formed on a respective source/drain **1221**, e.g., the doped upper end of a respective semiconductor body **1212** by patterning and etching an electrode hole aligned with respective source/drain **1221** using lithography and etching processes and depositing conductive materials to fill the electrode hole using thin film deposition processes. In some implementations, common plate **1228** is formed on the second electrodes of capacitors **1226** by patterning and etching an electrode trench aligned with capacitors **1226** using lithography and etching processes and depositing conductive materials to fill the electrode trench using thin film deposition processes.
(217) As illustrated in FIG. **12**G, a carrier substrate (a.k.a. a handle substrate) **1230** is bonded onto the front side of silicon substrate **1202** on which devices are formed using any suitable bonding processes, such as anodic bonding, fusion bonding, transfer bonding, adhesive bonding, and eutectic bonding. The bonded structure can then be flipped upside down, such that silicon substrate **1202** become above carrier substrate **1230**.
(218) As illustrated in FIG. **12**G, silicon substrate **1202** (shown in FIG. **12**F) is removed to expose the undoped upper ends of semiconductor bodies **1212** (used to be the lower ends before flipping over). In some implementations, planarization processes (e.g., CMP) and/or etching processes are performed to remove silicon substrate **1202** until being stopped by silicon oxide layer **1204** and the upper ends of semiconductor bodies **1212**.
(219) As illustrated in FIG. **12**G, the exposed upper end of each semiconductor body **1212**, e.g., one of the two ends of semiconductor body **1212** in the vertical direction (the z-direction) that is away from carrier substrate **1230**, is doped to form another source/drain **1223**. In some implementations, an implantation process and/or thermal diffusion process are performed to dope P-type dopants or N-type dopants to exposed upper ends of semiconductor bodies **1212** to form sources/drains **1223**. As a result, multi-gate vertical transistors having semiconductor body **1212**, sources/drains **1221** and **1223**, gate dielectric **1218**, and the gate electrode (part of conductive layer **1220**) are formed thereby, as shown in FIG. **12**G, according to some implementations. As described above, capacitors each having first and second electrodes **1224** and **1228** and capacitor dielectric **1226** are thereby formed as well, and DRAM cells **1280** each having a multi-gate vertical transistor and a capacitor coupled to the multi-gate vertical transistor are thereby formed, as shown in FIG. **12**G, according to some implementations.
(220) As illustrated in FIG. **12**H, an interconnect layer **1232** can be formed above DRAM cells **1280**. Interconnect layer **1232** can include interconnects of MEOL and/or BEOL in a plurality of

ILD layers to make electrical connections with DRAM cells **1280**. In some implementations, interconnect layer **1232** includes multiple ILD layers and interconnects therein formed in multiple processes. For example, the interconnects in interconnect layers **1232** can include conductive materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, electroplating, electroless plating, or any combination thereof. Fabrication processes to form the interconnects can also include photolithography, CMP, wet/dry etch, or any other suitable processes. The ILD layers can include dielectric materials deposited on silicon oxide layer **1204** by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. The ILD layers and interconnects illustrated in FIG. **12**H can be collectively referred to as interconnect layer **1232**. As illustrated in FIG. **12**H, bit line **1234** can be formed on sources/drains **1223** by patterning and etching a trench aligned with respective source/drain **1223** using lithography and etching processes and depositing conductive materials to fill the trench using thin film deposition processes. In some implementations, forming bit line **1234** includes depositing a metal layer onto the exposed end of semiconductor body **1212**.

(221) As illustrated in FIG. **12**H, a bonding layer **1236** is formed above interconnect layer **1232** and DRAM cells **1280**. Bonding layer **1236** can include a plurality of bonding contacts **1237** surrounded by dielectrics. In some implementations, a dielectric layer (e.g., ILD layer) is deposited on the top surface of interconnect layer **1232** by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. Bonding contacts **1237** can then be formed through the dielectric layer and in contact with the interconnects in interconnect layer **1232** by first patterning contact holes through the dielectric layer using patterning process (e.g., photolithography and dry/wet etch of dielectric materials in the dielectric layer). The contact holes can be filled with a conductor (e.g., Cu). In some implementations, filling the contact holes includes depositing a barrier layer, an adhesion layer, and/or a seed layer before depositing the conductor.

(222) In some implementations as shown in FIGS. **13**A-**13**H, a DRAM cell array is formed from a four-layer dielectric stack having a sacrificial layer (e.g., a silicon oxide layer) sandwiched between two dielectric layers (e.g., silicon nitride layers) on a pad layer (e.g., a silicon oxide layer).

(223) At operation **2402** in FIG. **24**, a stack of dielectric layers is formed on a substrate. In some implementations, to form the stack of dielectric layers, four layers having a first dielectric, a second dielectric, a third dielectric, and the second dielectric, respectively, are subsequently deposited on the substrate. The second dielectric can include silicon nitride, and the third dielectric can include silicon oxide. The layer having the third dielectric can act as a sacrificial layer vertically sandwiched between the two layers having the second dielectric. The sacrificial layer can be removed by selectively etching against the other layer having the second dielectric and replaced with a conductive layer in the later processes.

(224) As illustrated in FIG. **13**A, a stack of a silicon oxide layer **1304**, a silicon nitride layer **1306**, a silicon oxide layer **1308**, and a silicon nitride layer **1309** is formed on a silicon substrate **1302**. To form the dielectric stack, silicon oxide and silicon nitride are subsequently and alternatively deposited onto silicon substrate **1202** using one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. In some implementations, silicon oxide layer **1304** (a pad layer) is formed by oxidizing the top portion of silicon substrate **1302** using dry oxidation and/or we oxidation, such as ISSG oxidation process. In some implementations, the thickness of silicon oxide layer **1304** (e.g., ISSG silicon oxide) is smaller than the thickness of silicon oxide layer **1308** (e.g., CVD silicon oxide).

(225) As illustrated in FIG. **13**B, an array of semiconductor bodies **1312** each extending vertically through the stack of silicon oxide layer **1304**, silicon nitride layer **1306**, silicon oxide layer **1308**, and silicon nitride layer **1309** are formed. Semiconductor body **1312** can be epitaxially grown from the respective exposed part of silicon substrate **1302** in a respective opening (not shown). The fabrication processes for epitaxially growing semiconductor body **1312** can include, but not limited

to, VPE, LPE, MPE, or any combinations thereof. The epitaxy can occur upward (toward the positive z-direction) from the exposed parts of silicon substrate **1302** in the openings. Semiconductor body **1312** thus can have the same material as silicon substrate **1302**, in the form of single crystalline silicon. In some implementations, a planarization process, such as CMP, is performed to remove excess parts of semiconductor bodies **1312** beyond the top surface of silicon nitride layer **1309**. As a result, an array of semiconductor bodies **1312** (e.g., single crystalline silicon bodies) extending vertically (in the z-direction) from silicon substrate **1302** through the stack of silicon oxide layer **1304**, silicon nitride layer **1306**, silicon oxide layer **1308**, and silicon nitride layer **1309** is formed thereby, according to some implementation.

(226) At operation **2406** in FIG. **24**, one of the stack of dielectric layers is removed to expose part of the semiconductor body. In some implementations, to remove the one of the stack of dielectric layers, a trench is etched through at least part of the stack of dielectric layers to expose the layer having the third dielectric, and the layer having the third dielectric (e.g., the sacrificial layer) is etched away via the trench.

(227) As illustrated in FIG. **13**C, a plurality of trenches **1314** (slits openings) each extending laterally along the word line direction (the x-direction) and extending vertically through at least silicon nitride layer **1309** and silicon oxide layer **1308** are formed to expose silicon oxide layer **1308**. As a result, parts of silicon oxide layer **1308** can be exposed from trenches **1314**. In some implementations, a lithography process is performed to pattern trenches **1314** using an etch mask (e.g., a photoresist mask), for example, based on the design of word lines (word line trenches). In some implementations, one or more dry etching and/or wet etching processes, such as RIE, are performed to etch trenches **1314** through silicon nitride layer **1309** and silicon oxide layer **1308** until being stopped by silicon nitride layer **1306**. It is understood that in some examples, the etching of trenches **1314** may go further into silicon nitride layer **1306**, but not into silicon oxide layer **1304**.

(228) As illustrated in FIG. **13**D, silicon oxide layer **1308** (shown in FIG. **13**C) is removed to expose parts of semiconductor bodies **1312** abutting silicon oxide layer **1308**. In some implementations, silicon oxide layer **1308** is etched away via trenches **1214**. For example, a wet etchant including hydrofluoric acid may be applied through trenches **1314** to selectively wet etch silicon oxide layer **1308** without etching silicon nitride layers **1309** and **1306** as well as semiconductor bodies **1312**. As a result, lateral recesses **1316** can be formed thereby, which expose parts of semiconductor bodies **1312**.

(229) As illustrated in FIG. **13**E, a gate dielectric **1318** is formed over the exposed part of each semiconductor body **1312**, e.g., surrounding and contacting all the sides of the exposed part of semiconductor body **1312**. In some implementations, a wet oxidation and/or a dry oxidation process, such as ISSG, is performed to form native oxide (e.g., silicon oxide) on semiconductor body **1312** (e.g., single crystalline silicon) as gate dielectric **1318**. In some implementations, gate dielectric **1318** is formed by depositing a layer of dielectric, such as silicon oxide, over the exposed part of semiconductor body **1312** through trenches **1314** and lateral recesses **1316** using one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof, without filling lateral recesses **1316** and trenches **1314**. Due to the existence of silicon nitride layer **1309**, the upper end of gate dielectric **1318** can be below the upper end of semiconductor body **1312** in FIG. **13**E.

(230) As illustrated in FIG. **13**F, a conductive layer **1320** is formed over gate dielectrics **1318** in lateral recesses **1316** (shown in FIG. **13**E) through trenches **1314**. In some implementations, conductive layer **1320** is formed by depositing conductive materials, such as metal or metal compounds (e.g., TiN), over gate dielectrics **1318** using one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof, through trenches **1314** to fill lateral recesses **1316**. In one example, the deposition of conductive layer **1320** is controlled not to fill trenches **1314**. It is understood that in some examples, the deposition of conductive layer

1320 may fill trenches **1314** as well. Thus, a planarization process, e.g., CMP, may be performed to remove the excess conductive layer **1320** to expose the upper ends of semiconductor bodies **1312**, and conductive layer **1320** may be patterned to form a gate electrode over a respective gate dielectric. For example, trenches **1314** filled with conductive layer **1320** may be patterned and etched again to separate conductive layers **1320** between adjacent rows of semiconductor bodies **1312** and gate dielectrics **1318**. As described above, a lithography process can be performed to pattern trenches **1314** again using an etch mask (e.g., a photoresist mask), for example, based on the design of word lines (word line trenches). Due to the existence of silicon nitride layer **1309**, the top surface of conductive layer **1320** (including gate electrodes and word line) can be below the upper end of semiconductor body **1312** in FIG. **13**F.

(231) As a result, patterned conductive layers **1320** can become word lines each extending in the word line direction (the x-direction) and being separated by adjacent trenches **1314**, and parts of patterned conductive layers **1320** that are over gate dielectrics **1318** (e.g., fully circumscribes a respective gate dielectric **1318** in the plan view) can become gate electrodes. Gate structures each including a respective gate dielectric **1318** over the exposed part semiconductor body **1312** and a respective gate electrode (part of conductive layer **1320**) over gate dielectric **1318** can be formed thereby. Comparing FIG. **13**F with FIG. **13**A, silicon oxide layer **1308** (sacrificial layer) in FIG. **13**A is eventually replaced with conductive layer **1320** in FIG. **13**F, according to some implementations.

(232) As illustrated in FIG. **13**F, the exposed upper end of each semiconductor body **1312**, e.g., one of the two ends of semiconductor body **1312** in the vertical direction (the z-direction) that is away from silicon substrate **1302**, is doped to form a source/drain **1321** (e.g., a source terminal of a vertical transistor). In some implementations, an implantation process and/or thermal diffusion process are performed to dope P-type dopants or N-type dopants to exposed upper ends of semiconductor bodies **1312** to form sources/drains **1321**.

(233) As illustrated in FIG. **13**G, one or more ILD layers are formed over the top surface of silicon nitride layer **1309**, for example, by depositing dielectrics using one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. Depending on the lateral dimensions of trenches **1314** (shown in FIG. **13**F), trenches **1314** may not be fully filled with dielectrics (e.g., silicon oxide) when forming the ILD layers and thus, become air gaps **1322** between adjacent word lines (patterned conductive layers **1320**). It is understood that in some examples, when the lateral dimensions of trenches **1314** are sufficiently large, dielectrics may fully fill trenches **1314** during the formation of the ILD layers, thereby eliminating air gaps **1322**.

(234) As illustrated in FIG. **13**G, capacitor contacts **1324**, first electrodes, capacitor dielectrics, and second electrodes of capacitors **1326**, and a common plate **1328** are subsequently formed in the ILD layers to be coupled to semiconductor bodies **1312**. In some implementations, each capacitor contact **1324** is formed on a respective source/drain **1321**, e.g., the doped upper end of a respective semiconductor body **1312** by patterning and etching an electrode hole aligned with respective source/drain **1321** using lithography and etching processes and depositing conductive materials to fill the electrode hole using thin film deposition processes. In some implementations, common plate **1328** is formed on capacitors **1326** by patterning and etching an electrode trench aligned with capacitors **1326** using lithography and etching processes and depositing conductive materials to fill the electrode trench using thin film deposition processes.

(235) As illustrated in FIG. **13**H, a carrier substrate **1330** (a.k.a. a handle substrate) is bonded onto the front side of silicon substrate **1302** on which devices are formed using any suitable bonding processes, such as anodic bonding, fusion bonding, transfer bonding, adhesive bonding, and eutectic bonding. The bonded structure can then be flipped upside down, such that silicon substrate **1302** become above carrier substrate **1330**.

(236) As illustrated in FIG. **13**H, silicon substrate **1302** (shown in FIG. **13**G) is removed to expose the undoped upper ends of semiconductor bodies **1312** (used to be the lower ends before flipping

over). In some implementations, planarization processes (e.g., CMP) and/or etching processes are performed to remove silicon substrate **1302** until being stopped by silicon oxide layer **1304** and the upper ends of semiconductor bodies **1312**.
(237) As illustrated in FIG. **13**H, the exposed upper end of each semiconductor body **1312**, e.g., one of the two ends of semiconductor body **1312** in the vertical direction (the z-direction) that is away from carrier substrate **1330**, is doped to form another source/drain **1323** (e.g., a drain terminal of the vertical transistor). In some implementations, an implantation process and/or thermal diffusion process are performed to dope P-type dopants or N-type dopants to exposed upper ends of semiconductor bodies **1312** to form sources/drains **1323**. As a result, vertical transistors having semiconductor body **1312**, sources/drains **1321** and **1323**, gate dielectric **1318**, and the gate electrode (part of conductive layer **1320**) are formed thereby, as shown in FIG. **13**H, according to some implementations. As described above, capacitors each having first and second electrodes **1324** and **1328** and capacitor dielectric **1326** are thereby formed as well, and DRAM cells **1380** each having a multi-gate vertical transistor and a capacitor coupled to the multi-gate vertical transistor are thereby formed, as shown in FIG. **13**H, according to some implementations.
(238) As illustrated in FIG. **13**H, an interconnect layer **1332** can be formed above DRAM cells **1380**. Interconnect layer **1332** can include interconnects of MEOL and/or BEOL in a plurality of ILD layers to make electrical connections with DRAM cells **1380**. In some implementations, interconnect layer **1332** includes multiple ILD layers and interconnects therein formed in multiple processes. For example, the interconnects in interconnect layers **1332** can include conductive materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, electroplating, electroless plating, or any combination thereof. Fabrication processes to form the interconnects can also include photolithography, CMP, wet/dry etch, or any other suitable processes. The ILD layers can include dielectric materials deposited on silicon oxide layer **1304** by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. The ILD layers and interconnects illustrated in FIG. **13**H can be collectively referred to as interconnect layer **1332**. As illustrated in FIG. **13**H, bit line **1334** can be formed on sources/drains **1323** by patterning and etching a trench aligned with respective source/drain **1323** using lithography and etching processes and depositing conductive materials to fill the trench using thin film deposition processes. In some implementations, forming bit line **1334** includes depositing a metal layer onto the exposed end of semiconductor body **1312**.
(239) As illustrated in FIG. **13**H, a bonding layer **1336** is formed above interconnect layer **1332** and DRAM cells **1380**. Bonding layer **1336** can include a plurality of bonding contacts **1337** surrounded by dielectrics. In some implementations, a dielectric layer (e.g., ILD layer) is deposited on the top surface of interconnect layer **1332** by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. Bonding contacts **1337** can then be formed through the dielectric layer and in contact with the interconnects in interconnect layer **1332** by first patterning contact holes through the dielectric layer using patterning process (e.g., photolithography and dry/wet etch of dielectric materials in the dielectric layer). The contact holes can be filled with a conductor (e.g., Cu). In some implementations, filling the contact holes includes depositing a barrier layer, an adhesion layer, and/or a seed layer before depositing the conductor.
(240) A third semiconductor structure including a second array of memory cells can be formed. Each of the memory cells can also include a vertical transistor, and a storage unit coupled to the vertical transistor. The second semiconductor structure and the third semiconductor structure can be bonded in a face-to-face manner. In some implementations, the second and third semiconductor structures are bonded prior to bonding the first and second semiconductor structures. For example, as shown in FIG. **23**, the second and third semiconductor structures may be bonded prior to operation **2312**, e.g., between operation **2306** and operation **2308**.
(241) As illustrated in FIG. **14**A, two semiconductor structures **1000** and **1400** are formed

separately (e.g., in parallel) using any suitable fabrication processes disclosed herein (e.g., in FIGS. **10**A-**10**H). For ease of description, the fabrication process of forming semiconductor structure **1400** is not repeated and is the same as that of forming semiconductor structure **1000**. Thus, two semiconductor structures **1000** and **1400** may have the same devices therein.

(242) As illustrated in FIG. **14**A, semiconductor structure **1400** is flipped upside down. As illustrated in FIG. **14**B, semiconductor structure **1400** facing down is bonded with semiconductor structure **1000** facing up, e.g., in a face-to-face manner, thereby forming a bonding interface **1402**, using any suitable substrate/wafer bonding processes including, for example, hybrid bonding (as described above in detail), anodic bonding, and fusion (direct) bonding. In one example, fusion bonding may be performed between layers of silicon and silicon, silicon and silicon oxide, or silicon oxide and silicon oxide with pressure and heat. In another example, anodic bonding may be performed between layers of silicon oxide (in an ionic glass) and silicon with voltage, pressure, and heat. It is understood that depending on the bonding process, dielectric layers (e.g., silicon oxide layers) may be formed on one or both sides of bonding interface **1402**. For example, silicon oxide layers may be formed on the top surfaces of semiconductor structures **1000** and **1400** to allow SiO.sub.2—SiO.sub.2 bonding using fusion bonding. In some implementations, common plate **1028** of semiconductor structure **1400** are in contact with common plate of semiconductor structure **1000** at bonding interface **1402** and thus, may be viewed as a common electrode (e.g., common ground plate) of both semiconductor structures **1000** and **1400**.

(243) As illustrated in FIG. **14**B, silicon substrate **1002** (shown in FIG. **14**A) of semiconductor structure **1400** (on top of semiconductor structure **1000** after bonding) is removed to expose the undoped upper ends of semiconductor bodies **1012** (used to be the lower ends before flipping over). In some implementations, planarization processes (e.g., CMP) and/or etching processes are performed to remove silicon substrate **1002** of semiconductor structure **1400** until being stopped by silicon oxide layer **1004** and the upper ends of semiconductor bodies **1012** of semiconductor structure **1400**.

(244) As illustrated in FIG. **14**B, the exposed upper end of each semiconductor body **1012** of semiconductor structure **1400**, e.g., one of the two ends of semiconductor body **1012** in the vertical direction (the z-direction) that is away from semiconductor structure **1000**, is doped to form another source/drain **1023**. In some implementations, an implantation process and/or thermal diffusion process are performed to dope P-type dopants or N-type dopants to exposed upper ends of semiconductor bodies **1012** of semiconductor structure **1400** to form sources/drains **1023**. As a result, multi-gate vertical transistors having semiconductor body **1012**, sources/drains **1021** and **1023**, gate dielectric **1018**, and the gate electrode (part of conductive layer **1020**) are formed thereby in semiconductor structure **1400**, as shown in FIG. **14**B, according to some implementations. As described above, capacitors **1026** are thereby formed as well, and DRAM cells **1080** each having a multi-gate vertical transistor and a capacitor coupled to the multi-gate vertical transistor are thereby formed of semiconductor structure **1400**, as shown in FIG. **14**B, according to some implementations.

(245) As illustrated in FIG. **14**C, an interconnect layer **1032** can be formed above DRAM cells **1080**. Interconnect layer **1032** can include interconnects of MEOL and/or BEOL in a plurality of ILD layers to make electrical connections with DRAM cells **1080**. In some implementations, interconnect layer **1032** includes multiple ILD layers and interconnects therein formed in multiple processes. For example, the interconnects in interconnect layers **1032** can include conductive materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, electroplating, electroless plating, or any combination thereof. Fabrication processes to form the interconnects can also include photolithography, CMP, wet/dry etch, or any other suitable processes. The ILD layers can include dielectric materials deposited on silicon oxide layer **1004** by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. The ILD layers and interconnects illustrated in FIG. **14**C can be

collectively referred to as interconnect layer **1032**.

(246) As illustrated in FIG. **14**D, a carrier substrate **1030** (a.k.a. a handle substrate) is bonded onto the front side of semiconductor structure **1400** on which devices are formed using any suitable bonding processes, such as anodic bonding, fusion bonding, transfer bonding, adhesive bonding, and eutectic bonding. The bonded structure can then be flipped upside down, such that semiconductor structure **1000** become above carrier substrate **1030** (not shown in FIG. **14**D).

(247) As illustrated in FIG. **14**D, silicon substrate **1002** of semiconductor structure **1000** (shown in FIG. **14**C) is removed to expose the undoped upper ends of semiconductor bodies **1012** of semiconductor structure **1000** (used to be the lower ends before flipping over). In some implementations, planarization processes (e.g., CMP) and/or etching processes are performed to remove silicon substrate **1002** of semiconductor structure **1000** until being stopped by silicon oxide layer **1004** and the upper ends of semiconductor bodies **1012** of semiconductor structure **1000**.

(248) As illustrated in FIG. **14**D, the exposed upper end of each semiconductor body **1012** of semiconductor structure **1000**, e.g., one of the two ends of semiconductor body **1012** in the vertical direction (the z-direction) that is away from semiconductor structure **1400**, is doped to form another source/drain **1023**. In some implementations, an implantation process and/or thermal diffusion process are performed to dope P-type dopants or N-type dopants to exposed upper ends of semiconductor bodies **1012** of semiconductor structure **1000** to form sources/drains **1023**. As a result, vertical transistors having semiconductor body **1012**, sources/drains **1021** and **1023**, gate dielectric **1018**, and the gate electrode (part of conductive layer **1020**) are formed thereby in semiconductor structure **1000**, as shown in FIG. **14**D, according to some implementations. As described above, capacitors **1026** are thereby formed as well, and DRAM cells **1080** each having a multi-gate vertical transistor and a capacitor coupled to the multi-gate vertical transistor are thereby formed of semiconductor structure **1000**, as shown in FIG. **14**D, according to some implementations.

(249) As illustrated in FIG. **14**E, an interconnect layer **1032** can be formed above DRAM cells **1080** in semiconductor structure **1000**. Interconnect layer **1032** can include interconnects of MEOL and/or BEOL in a plurality of ILD layers to make electrical connections with DRAM cells **1080** in semiconductor structure **1000**. In some implementations, interconnect layer **1032** includes multiple ILD layers and interconnects therein formed in multiple processes. For example, the interconnects in interconnect layers **1032** can include conductive materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, electroplating, electroless plating, or any combination thereof. Fabrication processes to form the interconnects can also include photolithography, CMP, wet/dry etch, or any other suitable processes. The ILD layers can include dielectric materials deposited on silicon oxide layer **1004** by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. The ILD layers and interconnects illustrated in FIG. **14**E can be collectively referred to as interconnect layer **1032**.

(250) As illustrated in FIG. **14**E, a bonding layer **1036** is formed above interconnect layer **1032** and DRAM cells **1080** in semiconductor structure **1000**. Bonding layer **1036** can include a plurality of bonding contacts **1037** surrounded by dielectrics. In some implementations, a dielectric layer (e.g., ILD layer) is deposited on the top surface of interconnect layer **1032** by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. Bonding contacts **1037** can then be formed through the dielectric layer and in contact with the interconnects in interconnect layer **1032** by first patterning contact holes through the dielectric layer using patterning process (e.g., photolithography and dry/wet etch of dielectric materials in the dielectric layer). The contact holes can be filled with a conductor (e.g., Cu). In some implementations, filling the contact holes includes depositing a barrier layer, an adhesion layer, and/or a seed layer before depositing the conductor.

(251) The bonded structure shown in FIG. **14**E then can be bonded with a semiconductor structure

including peripheral circuits in a face-to-face manner, as described above in detail with respect to operation **2312** in FIG. **23** and FIGS. **10**L and **10**M.
(252) In some implementations, the second and third semiconductor structures are bonded after bonding the first and second semiconductor structures. For example, as shown in FIG. **23**, the second and third semiconductor structures may be bonded after operation **2312**, e.g., between operation **2312** and operation **2314**.
(253) As illustrated in FIG. **15**A, a bonded semiconductor structure **1500** is formed after the fabrication process shown in FIG. **10**L by removing carrier substrate **1030**. A semiconductor structure **1000** is formed separately (e.g., in parallel) using any suitable fabrication processes disclosed herein (e.g., in FIGS. **10**A-**10**H). For ease of description, the fabrication processes of forming semiconductor structures **1000** and **1500** are not repeated.
(254) As illustrated in FIG. **15**A, semiconductor structure **1000** is flipped upside down. As illustrated in FIG. **15**B, semiconductor structure **1000** facing down is bonded with semiconductor structure **1500** facing up, e.g., in a face-to-face manner, thereby forming a bonding interface **1502**, using any suitable substrate/wafer bonding processes including, for example, hybrid bonding (as described above in detail), anodic bonding, and fusion (direct) bonding. In one example, fusion bonding may be performed between layers of silicon and silicon, silicon and silicon oxide, or silicon oxide and silicon oxide with pressure and heat. In another example, anodic bonding may be performed between layers of silicon oxide (in an ionic glass) and silicon with voltage, pressure, and heat. It is understood that depending on the bonding process, dielectric layers (e.g., silicon oxide layers) may be formed on one or both sides of bonding interface **1502**. For example, silicon oxide layers may be formed on the top surfaces of semiconductor structures **1000** and **1500** to allow SIO.sub.2—SiO.sub.2 bonding using fusion bonding. In some implementations, common plate **1028** of semiconductor structure **1000** are in contact with common plate **1028** of semiconductor structure **1500** at bonding interface **1502** and thus, may be viewed as a common electrode (e.g., common ground plate) of both semiconductor structures **1000** and **1500**.
(255) As illustrated in FIG. **15**C, silicon substrate **1002** (shown in FIG. **15**B) of semiconductor structure **1000** (on top of semiconductor structure **1500** after bonding) is removed to expose the undoped upper ends of semiconductor bodies **1012** (used to be the lower ends before flipping over). In some implementations, planarization processes (e.g., CMP) and/or etching processes are performed to remove silicon substrate **1002** of semiconductor structure **1000** until being stopped by silicon oxide layer **1004** and the upper ends of semiconductor bodies **1012** of semiconductor structure **1000**.
(256) As illustrated in FIG. **15**C, the exposed upper end of each semiconductor body **1012** of semiconductor structure **1000**, e.g., one of the two ends of semiconductor body **1012** in the vertical direction (the z-direction) that is away from semiconductor structure **1500**, is doped to form another source/drain **1023**. In some implementations, an implantation process and/or thermal diffusion process are performed to dope P-type dopants or N-type dopants to exposed upper ends of semiconductor bodies **1012** of semiconductor structure **1000** to form sources/drains **1023**. As a result, multi-gate vertical transistors having semiconductor body **1012**, sources/drains **1021** and **1023**, gate dielectric **1018**, and the gate electrode (part of conductive layer **1020**) are formed thereby in semiconductor structure **1000**, as shown in FIG. **15**C, according to some implementations. As described above, capacitors **1026** are thereby formed as well, and DRAM cells **1080** each having a multi-gate vertical transistor and a capacitor coupled to the multi-gate vertical transistor are thereby formed of semiconductor structure **1400**, as shown in FIG. **15**C, according to some implementations.
(257) As illustrated in FIG. **15**D, an interconnect layer **1032** can be formed above DRAM cells **1080** in semiconductor structure **1000**. Interconnect layer **1032** can include interconnects of MEOL and/or BEOL in a plurality of ILD layers to make electrical connections with DRAM cells **1080**. In some implementations, interconnect layer **1032** includes multiple ILD layers and interconnects

therein formed in multiple processes. For example, the interconnects in interconnect layers **1032** can include conductive materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, electroplating, electroless plating, or any combination thereof. Fabrication processes to form the interconnects can also include photolithography, CMP, wet/dry etch, or any other suitable processes. The ILD layers can include dielectric materials deposited on silicon oxide layer **1004** by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. The ILD layers and interconnects illustrated in FIG. **15**D can be collectively referred to as interconnect layer **1032**.

(258) A pad-out interconnect layer then can be formed on the bonded structure shown in FIG. **15**D as described above in detail with respect to operation **2314** in FIG. **23** and FIG. **10**M.

(259) Method **2300** may also be implemented by the fabrication process described in FIGS. **19**A-**19**M and **22** to form 3D memory device **1700** depicted in FIG. **17** having single-gate vertical transistors, as opposed to multiple-gate vertical transistors. Referring to FIG. **23**, method **2300** starts at operation **2302**, in which a peripheral circuit is formed on a first substrate. The first substrate can include a silicon substrate. In some implementations, an interconnect layer is formed above the peripheral circuit. The interconnect layer can include a plurality of interconnects in one or more ILD layers.

(260) As illustrated in FIG. **19**L, a plurality of transistors **1948** are formed on a silicon substrate **1944**. Transistors **1948** can be formed by a plurality of processes including, but not limited to, photolithography, dry/wet etch, thin film deposition, thermal growth, implantation, CMP, and any other suitable processes. In some implementations, doped regions are formed in silicon substrate **1944** by ion implantation and/or thermal diffusion, which function, for example, as the source and drain of transistors **1948**. In some implementations, isolation regions (e.g., STIs) are also formed in silicon substrate **1944** by wet/dry etch and thin film deposition. Transistors **1948** can form peripheral circuits **1946** on silicon substrate **1944**.

(261) As illustrated in FIG. **19**L, an interconnect layer **1950** can be formed above peripheral circuits **1946** having transistors **1948**. Interconnect layer **1950** can include interconnects of MEOL and/or BEOL in a plurality of ILD layers to make electrical connections with peripheral circuits **1946**. In some implementations, interconnect layer **1950** includes multiple ILD layers and interconnects therein formed in multiple processes. For example, the interconnects in interconnect layers **1950** can include conductive materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, electroplating, electroless plating, or any combination thereof. Fabrication processes to form the interconnects can also include photolithography, CMP, wet/dry etch, or any other suitable processes. The ILD layers can include dielectric materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. The ILD layers and interconnects illustrated in FIG. **19**L can be collectively referred to as interconnect layer **1950**.

(262) Method **2300** proceeds to operation **2304**, as illustrated in FIG. **23**, in which a first bonding layer is formed above the peripheral circuit (and the interconnect layer). The first bonding layer can include a first bonding contact. As illustrated in FIG. **19**L, a bonding layer **1952** is formed above interconnect layer **1950** and peripheral circuits **1946**. Bonding layer **1952** can include a plurality of bonding contacts surrounded by dielectrics. In some implementations, a dielectric layer (e.g., ILD layer) is deposited on the top surface of interconnect layer **1950** by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. The bonding contacts then can be formed through the dielectric layer and in contact with the interconnects in interconnect layer **1950** by first patterning contact holes through the dielectric layer using patterning process (e.g., photolithography and dry/wet etch of dielectric materials in the dielectric layer). The contact holes can be filled with a conductor (e.g., Cu). In some implementations, filling the contact holes includes depositing a barrier layer, an adhesion layer,

and/or a seed layer before depositing the conductor.

(263) Method **2300** proceeds to operation **2306**, as illustrated in FIG. **23**, in which an array of memory cells each including a vertical transistor and a storage unit is formed on a second substrate. The second substrate can include a carrier substrate. The storage unit can include a capacitor or a PCM element. In some implementations, a capacitor is formed to be coupled to the vertical transistor in the respective memory cell.

(264) For example, FIG. **25** illustrates a flowchart of a method **2500** for forming another array of memory cells each including a vertical transistor, according to some aspects of the present disclosure. At operation **2502** in FIG. **25**, a semiconductor pillar extending vertically in a substrate is formed. The substrate can be a silicon substrate. In some implementations, to form the semiconductor pillar, the substrate is etched in a first lateral direction to form a plurality of first trenches, a dielectric is deposited to fill the first trenches to form second trench isolations, and the substrate and the second trench isolations are etched in a second lateral direction to form a plurality of second trenches and the semiconductor pillar surrounded by the second trenches and the second trench isolations. In some implementations, a dielectric is deposited to partially fill the second trenches.

(265) As illustrated in FIG. **19**A, a plurality of parallel trenches **1904** are formed in the y-direction (e.g., the bit line direction) to form a plurality of parallel semiconductor walls **1905** in the y-direction. In some implementations, a lithography process is performed to pattern trenches **1904** and semiconductor walls **1905** using an etch mask (e.g., a photoresist mask and/or a hard mask), for example, based on the design of bit lines, and one or more dry etching and/or wet etching processes, such as RIE, are performed to etch trenches **1904** in a silicon substrate **1902**. Thus, semiconductor wall **1905** extending vertically in silicon substrate **1902** can be formed. The bottom of semiconductor wall **1905** can be below the top surface of silicon substrate **1902**. Since semiconductor walls **1905** are formed by etching silicon substrate **1902**, semiconductor walls **1905** can have the same material as silicon substrate **1902**, such as single crystalline silicon. FIG. **19**A illustrates both the side view (in the top portion of FIG. **19**A) of a cross-section along the x-direction (the word line direction, e.g., in the BB plane) and the plan view (in the bottom portion of FIG. **19**A) of a cross-section in the x-y plane (e.g., in the AA plane through semiconductor walls **1905**). The same drawing layout is arranged in FIG. **19**B as well.

(266) As illustrated in FIG. **19**B, trench isolations **1908** (e.g., STIs) are formed in trenches **1904**. In some implementations, a dielectric, such as silicon oxide, is deposited to fully fill trenches **1904** using one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. In some implementations, a planarization process, such as CMP, is performed to remove excess dielectric deposited beyond the top surface of silicon substrate **1902**. As a result, parallel semiconductor walls **1905** can be separated by trench isolations **1908**.

(267) As illustrated in FIG. **19**C, a plurality of parallel trenches **1910** are formed in the x-direction (e.g., the word line direction) to form an array of semiconductor pillars **1906** each extending vertically in silicon substrate **1902**. In some implementations, a lithography process is performed to pattern trenches **1910** to be perpendicular to trench isolations **1908** using an etch mask (e.g., a photoresist mask and/or a hard mask), for example, based on the design of word lines, and one or more dry etching and/or wet etching processes, such as RIE, are performed on silicon substrate **1902** and trench isolation **1908** to etch trenches **1910** in silicon substrate **1902**. As a result, semiconductor walls **1905** (shown in FIG. **19**B) can be cut by trenches **1910** to form an array of semiconductor pillars **1906** each extending vertically in silicon substrate **1902**. The bottom of semiconductor pillar **1906** can be below the top surface of silicon substrate **1902**. Since semiconductor pillars **1906** are formed by etching silicon substrate **1902**, semiconductor pillars **1906** can have the same material as silicon substrate **1902**, such as single crystalline silicon. FIG. **19**C illustrates both the side view (in the top portion of FIG. **19**C) of a cross-section along the y-direction (the bit line direction, e.g., in the CC plane) and the plan view (in the bottom portion of

FIG. **19**C) of a cross-section in the x-y plane (e.g., in the AA plane through semiconductor pillars **1906**). The same drawing layout is arranged in FIGS. **19**C-**19**G as well.

(268) As illustrated in FIG. **19**C, a dielectric layer **1912** is formed at the bottom of trench **1910**, for example, by depositing a dielectric, such as silicon oxide, to partially fill trench **1910**, using one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. The deposition conditions, such as deposition rate and/or time, can be controlled to control the thickness of dielectric layer **1912** and avoid fully filling trench **1910**. As a result, the bottom surface of trenches **1910** can be elevated to be above the bottom surface of semiconductor pillars **1906**. As shown in the plan view, the two opposite sides of semiconductor pillar **1906** in the y-direction are exposed by trenches **1910**, and the other two opposite sides of semiconductor pillar **1906** in the x-direction are in contact with trench isolation **1908**. In other words, semiconductor pillar **1906** is surrounded by trenches **1910** and trenches isolations **1908**.

(269) At operation **2504** in FIG. **25**, gate structures in contact with opposite sides of the semiconductor pillar are formed. In some implementations, to form the gate structures, gate dielectrics are formed over the opposite sides of the semiconductor pillar, and gate electrodes are formed over the gate dielectrics. In some implementations, to form the gate electrodes, conductive layers are deposited over the gate dielectrics, and the conductive layers are etched back.

(270) As illustrated in FIG. **19**D, gate dielectrics **1914** are formed over the two opposite sides of semiconductor pillars **1906** in the bit line direction (they-direction) exposed from trenches **1910**. As shown in the plan view, gate dielectrics **1914** can be parts of a continuous dielectric layer formed over sidewalls of each row of semiconductor pillars **1906** and trench isolations **1908**. In some implementations, gate dielectric **1914** is formed by depositing a layer of dielectric, such as silicon oxide, over the sidewalls of trenches **1910** using one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof, without fully filling trenches **1910**. It is understood that in some examples, gate dielectrics **1914** may not be parts of a continuous dielectric layer. For example, a wet oxidation and/or a dry oxidation process, such as in situ steam generation (ISSG) oxidation, is performed to form native oxide (e.g., silicon oxide) on semiconductor pillar **1906** (e.g., single crystalline silicon) as gate dielectric **1914**.

(271) As illustrated in FIG. **19**D, conductive layers **1916** are formed over gate dielectrics **1914** in trenches **1910**. In some implementations, conductive layers **1916** are formed by depositing one or more conductive materials, such as metal and/or metal compounds (e.g., W and TiN), over gate dielectrics **1914** using one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof, to partially fill trenches **1910**. For example, layers of TiN and W may be sequentially deposited to form conductive layer **1916**. A planarization process, e.g., CMP, can be performed to remove the excess conductive materials over the top surface of silicon substrate **1902**.

(272) As illustrated in FIG. **19**E, in some implementations, conductive layers **1916** are etched back, for example, using dry etch and/or wet etch (e.g., RIE), to form dents, such that the upper ends of conductive layers **1916** are below the top surface of semiconductor pillars **1906**. In some implementations, as gate dielectrics **1914** are not etched back, the upper ends of conductive layers **1916** are below the upper ends of gate dielectrics **1914** as well, which are flush with the top surface of semiconductor pillars **1906**. As a result, etched-back conductive layers **1916** can become word lines each extending in the word line direction (the x-direction), and parts of etched-back conductive layers **1916** that are facing semiconductor pillars **1906** can become gate electrodes. Gate structures each including a respective gate dielectric **1914** over the exposed side of semiconductor pillar **1906** and a respective gate electrode (part of conductive layer **1916**) over gate dielectric **1914** can be formed thereby. In some implementations, as shown in FIG. **19**E, a dielectric layer **1918** is formed in the remaining space of trenches **1910** as well as the dents (not shown) resulting from etching back of conductive layers **1916**, for example, by depositing a dielectric, such as silicon oxide, using one or more thin film deposition processes including, but not limited to,

CVD, PVD, ALD, or any combination thereof. It is understood that depending on the pitches of the word lines (the dimension of trenches **1910**), air gaps may be formed in dielectric layer **1918**.

(273) At operation **2506** in FIG. **25**, a first trench isolation extending vertically through the semiconductor pillar is formed to separate the semiconductor pillar into semiconductor bodies each in contact with a respective one of the gate structures. In some implementations, to form the first trench, the semiconductor pillar is etched in the second lateral direction to form a third trench, and a dielectric is deposited to fill the third trench.

(274) As illustrated in FIG. **19**F, a plurality of parallel trenches **1922** in the x-direction (e.g., the word line direction) are formed to form an array of semiconductor bodies **1920** each extending vertically in silicon substrate **1902**. In some implementations, a lithography process is performed to pattern trenches **1922** on semiconductor pillars **1906** (shown in FIG. **19**E) using an etch mask (e.g., a photoresist mask and/or a hard mask), and one or more dry etching and/or wet etching processes, such as RIE, are performed on semiconductor pillars **1906** and trench isolation **1908** to etch trenches **1922**. The etching can be controlled such that bottom of trenches **1922** is flush with or below the bottom surface of semiconductor pillars **1906**. As a result, each semiconductor pillar **1906** can be separated by a respective trench **1922** into two semiconductor bodies **1920** in the y-direction. Since semiconductor bodies **1920** are formed by etching silicon substrate **1902**, semiconductor bodies **1920** can have the same material as silicon substrate **1902**, such as single crystalline silicon. As shown in the plan view, each semiconductor body **1920** can be in contact with a gate structure having gate dielectric **1914** and gate electrode **1916** on one side of semiconductor body **1920** in the y-direction. The opposite side of semiconductor body **1920** can be exposed by trench **1922**. In some implementations, a mirror-symmetric arrangement of two semiconductor bodies **1920** and two gate structures thereof is achieved by forming trench **1922** across the middle of a respective semiconductor pillar **1906**.

(275) As illustrated in FIG. **19**G, a trench isolation **1926** is formed in trench **1922** (shown in FIG. **19**F), for example, by depositing a dielectric, such as silicon oxide, to fill trench **1922**, using one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. A planarization process can be performed to remove excess dielectric over the top surface of silicon substrate **1902**. It is understood that depending on the pitches of the semiconductor bodies **1920** (the dimension of trenches **1922**), air gaps may be formed in trench isolation **1926**. As shown in the plan view, parallel trench isolations **1926** each extending in the x-direction can form an array of semiconductor bodies **1920** in which a single side is in contact with a gate structure having gate dielectric **1914** and gate electrode **1916**.

(276) At operation **2508** in FIG. **25**, first ends of the semiconductor bodies away from the substrate are doped. As illustrated in FIG. **19**G, the exposed upper end of each semiconductor body **1920**, e.g., one of the two ends of semiconductor body **1920** in the vertical direction (the z-direction) that is away from silicon substrate **1902**, is doped to form a source/drain **1924** (e.g., a source terminal of a vertical transistor). In some implementations, an implantation process and/or thermal diffusion process are performed to dope P-type dopants or N-type dopants to exposed upper ends of semiconductor bodies **1920** to form sources/drains **1924**. In some implementations, a silicide layer is formed on source/drain **1924** by performing a silicidation process at the exposed upper ends of semiconductor bodies **1920**.

(277) At operation **2510** in FIG. **25**, storage units in contact with the semiconductor bodies, e.g., the doped first ends thereof, are formed. The storage unit can include a capacitor or a PCM element. In some implementations, to form a storage unit that is a capacitor, a first electrode is formed on the doped first end of the semiconductor body, a capacitor dielectric is formed on the first electrode, and a second electrode is formed on the capacitor dielectric.

(278) As illustrated in FIG. **19**H, one or more ILD layers are formed over the top surface of silicon substrate **1902**, for example, by depositing dielectrics using one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. Capacitor

contacts **1928**, first electrodes, capacitor dielectrics, and second electrodes of capacitors **1930**, and a common plate **1932** are subsequently formed in the ILD layers to be coupled to semiconductor bodies **1920**. In some implementations, capacitor contact **1928** is formed on a respective source/drain **1924**, e.g., the doped upper end of a respective semiconductor body **1920** by patterning and etching an electrode hole aligned with respective source/drain **1924** using lithography and etching processes and depositing conductive materials to fill the electrode hole using thin film deposition processes. In some implementations, common plate **1932** is formed on the second electrodes of capacitors **1930** by patterning and etching an electrode trench aligned with capacitors **1930** using lithography and etching processes and depositing conductive materials to fill the electrode trench using thin film deposition processes.

(279) At operation **2512** in FIG. **25**, the substrate is thinned to expose second ends of the semiconductor bodies opposite to the first end. As illustrated in FIG. **19**I, a carrier substrate **1934** (a.k.a. a handle substrate) is bonded onto the front side of silicon substrate **1902** on which devices are formed using any suitable bonding processes, such as anodic bonding, fusion bonding, transfer bonding, adhesive bonding, and eutectic bonding. The bonded structure can then be flipped upside down, such that silicon substrate **1902** become above carrier substrate **1934**.

(280) As illustrated in FIG. **19**J, silicon substrate **1902** (shown in FIG. **19**I) is thinned to expose the undoped upper ends of semiconductor bodies **1920** (used to be the lower ends before flipping over). In some implementations, planarization processes (e.g., CMP) and/or etching processes are performed to thin silicon substrate **1902** until being stopped by dielectric layer **1918** and the upper ends of semiconductor bodies **1920**.

(281) At operation **2514** in FIG. **25**, the exposed second ends of the semiconductor bodies are doped. As illustrated in FIG. **19**J, the exposed upper end of each semiconductor body **1920**, e.g., one of the two ends of semiconductor body **1920** in the vertical direction (the z-direction) that is away from carrier substrate **1934**, is doped to form another source/drain **1936** (e.g., a drain terminal of the vertical transistor). In some implementations, an implantation process and/or thermal diffusion process are performed to dope P-type dopants or N-type dopants to exposed upper ends of semiconductor bodies **1920** to form sources/drains **1936**. In some implementations, a silicide layer is formed on source/drain **1936** by performing a silicidation process at the exposed upper ends of semiconductor bodies **1920**. As a result, vertical transistors having semiconductor body **1920**, sources/drains **1924** and **1936**, gate dielectric **1914**, and the gate electrode (part of conductive layer **1916**) are formed thereby, as shown in FIG. **19**J, according to some implementations. As described above, capacitors **1930** each having the first and second electrodes and the capacitor dielectric are thereby formed as well, and DRAM cells **1980** each having a single-gate vertical transistor and a capacitor coupled to the single-gate vertical transistor are thereby formed, as shown in FIG. **19**J, according to some implementations.

(282) Referring back to FIG. **23**, method **2300** proceeds to operation **2308**, as illustrated in FIG. **23**, in which an interconnect layer including bit lines is formed above the array of memory cells. As illustrated in FIG. **19**K, an interconnect layer **1940** can be formed above DRAM cells **1980**. Interconnect layer **1940** can include interconnects of MEOL and/or BEOL in a plurality of ILD layers to make electrical connections with DRAM cells **1980**. In some implementations, interconnect layer **1940** includes multiple ILD layers and interconnects therein formed in multiple processes. For example, the interconnects in interconnect layers **1940** can include conductive materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, electroplating, electroless plating, or any combination thereof. Fabrication processes to form the interconnects can also include photolithography, CMP, wet/dry etch, or any other suitable processes. The ILD layers can include dielectric materials deposited using one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. The ILD layers and interconnects illustrated in FIG. **19**K can be collectively referred to as interconnect layer **1940**.

(283) As shown in FIG. **25**, at operation **2516**, to form the interconnect layer, a bit line is formed on the doped second end. As illustrated in FIG. **19**K, bit line **1938** can be formed on sources/drains **1936** by patterning and etching a trench aligned with respective source/drain **1936** using lithography and etching processes and depositing conductive materials to fill the trench using thin film deposition processes. In some implementations, forming bit line **1938** includes depositing a metal layer onto the exposed end of semiconductor body **1920**. As a result, bit line **1938** and capacitor **1930** can be formed on opposite sides of semiconductor body **1920** and coupled to opposite ends of semiconductor body **1920**. It is understood that additional local interconnects, such as word line contacts, capacitor contacts (e.g., a conductor), and bit line contacts (not shown in FIG. **19**K, e.g., a metal silicide contact) may be similarly formed as well. In some implementations, the bit line contact (e.g., a metal silicide contact) is formed on the exposed end of semiconductor body **1820**, and bit line **1938** is formed on the bit line contact.

(284) Method **2300** proceeds to operation **2310**, as illustrated in FIG. **23**, in which a second bonding layer is formed above the array of memory cells and the interconnect layer. The second bonding layer can include a second bonding contact. As illustrated in FIG. **19**K, a bonding layer **1942** is formed above interconnect layer **1940** and DRAM cells **1980**. Bonding layer **1942** can include a plurality of bonding contacts surrounded by dielectrics. In some implementations, a dielectric layer (e.g., ILD layer) is deposited on the top surface of interconnect layer **1940** by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. The bonding contacts can then be formed through the dielectric layer and in contact with the interconnects in interconnect layer **1940** by first patterning contact holes through the dielectric layer using patterning process (e.g., photolithography and dry/wet etch of dielectric materials in the dielectric layer). The contact holes can be filled with a conductor (e.g., Cu). In some implementations, filling the contact holes includes depositing a barrier layer, an adhesion layer, and/or a seed layer before depositing the conductor.

(285) Method **2300** proceeds to operation **2312**, as illustrated in FIG. **23**, in which the first semiconductor structure and the second semiconductor structure are bonded in a face-to-face manner, such that the first array of memory cells is coupled to the peripheral circuit across a bonding interface. The bonding can include hybrid bonding. In some implementations, the first bonding contact is in contact with the second bonding contact at the bonding interface after the bonding. In some implementations, the second semiconductor structure is above the first semiconductor structure after the bonding. In some implementations, the first semiconductor structure is above the second semiconductor structure after the bonding.

(286) As illustrated in FIG. **19**L, carrier substrate **1934** and components formed thereon (e.g., DRAM cells **1980**) are flipped upside down. As illustrated in FIG. **19**L, bonding layer **1942** facing down is bonded with bonding layer **1952** facing up, e.g., in a face-to-face manner, thereby forming a bonding interface **1954**. In some implementations, a treatment process, e.g., a plasma treatment, a wet treatment, and/or a thermal treatment, is applied to the bonding surfaces prior to the bonding. Although not shown in FIG. **19**L, silicon substrate **1944** and components formed thereon (e.g., peripheral circuits **1946**) can be flipped upside down, and bonding layer **1952** facing down can be bonded with bonding layer **1942** facing up, e.g., in a face-to-face manner, thereby forming bonding interface **1954**. After the bonding, the bonding contacts in bonding layer **1942** and the bonding contacts in bonding layer **1952** are aligned and in contact with one another, such that DRAM cells **1980** can be electrically connected to peripheral circuits **1946** across bonding interface **1954**. It is understood that in the bonded chip, DRAM cells **1980** may be either above or below peripheral circuits **1946**. Nevertheless, bonding interface **1954** can be formed vertically between peripheral circuits **1946** and DRAM cells **1980** after the bonding.

(287) Method **2300** proceeds to operation **2314**, as illustrated in FIG. **23**, in which a pad-out interconnect layer is formed on the backside of the first semiconductor structure or the second semiconductor structure. As illustrated in FIG. **19**M, a pad-out interconnect layer **1956** is formed

above on the backside of carrier substrate **1934**. Pad-out interconnect layer **1956** can include interconnects, such as pad contacts **1958**, formed in one or more ILD layers. Pad contacts **1958** can include conductive materials including, but not limited to, W, Co, Cu, Al, doped silicon, silicides, or any combination thereof. The ILD layers can include dielectric materials including, but not limited to, silicon oxide, silicon nitride, silicon oxynitride, low-k dielectrics, or any combination thereof. In some implementations, after the bonding, contacts **1960** are formed extending vertically through carrier substrate **1934**, for example, by wet/dry etching processes, followed by depositing conductive materials. Contacts **1960** can be in contact with the interconnects in pad-out interconnect layer **1956**. It is understood that in some examples, carrier substrate **1934** may be thinned or removed after bonding and prior to forming pad-out interconnect layer **1956** and contacts **1960**, for example, using planarization processes and/or etching processes.
(288) Although not shown, it is understood that in some examples, pad-out interconnect layer **1956** may be formed above on the backside of silicon substrate **1944**, and contacts **1960** may be formed extending vertically through silicon substrate **1944**. Silicon substrate **1944** may be thinned prior to forming pad-out interconnect layer **1956** and contacts **1960**, for example, using planarization processes and/or etching processes. Although not shown, it is further understood that in some examples, the fabrication processes described with respect to FIGS. **14**A-**14**E and **15**A-**15**D may be applied to form another array of DRAM cells **1980** in another semiconductor structure bonded to the semiconductor structure including DRAM cells **1980** described above with respect to FIGS. **19**A-**19**M.
(289) Method **2300** may further be implemented by the fabrication process described in FIGS. **22**A-**22**M and **26** to form 3D memory device **2100** depicted in FIG. **21** having double-gate vertical transistors, as opposed to single-gate vertical transistors. Referring to FIG. **23**, method **2300** starts at operation **2302**, in which a peripheral circuit is formed on a first substrate. The first substrate can include a silicon substrate. In some implementations, an interconnect layer is formed above the peripheral circuit. The interconnect layer can include a plurality of interconnects in one or more ILD layers.
(290) As illustrated in FIG. **22**L, a plurality of transistors **2248** are formed on a silicon substrate **2244**. Transistors **2248** can be formed by a plurality of processes including, but not limited to, photolithography, dry/wet etch, thin film deposition, thermal growth, implantation, CMP, and any other suitable processes. In some implementations, doped regions are formed in silicon substrate **2244** by ion implantation and/or thermal diffusion, which function, for example, as the source and drain of transistors **2248**. In some implementations, isolation regions (e.g., STIs) are also formed in silicon substrate **2244** by wet/dry etch and thin film deposition. Transistors **2248** can form peripheral circuits **2246** on silicon substrate **2244**.
(291) As illustrated in FIG. **22**L, an interconnect layer **2250** can be formed above peripheral circuits **2246** having transistors **2248**. Interconnect layer **2250** can include interconnects of MEOL and/or BEOL in a plurality of ILD layers to make electrical connections with peripheral circuits **2246**. In some implementations, interconnect layer **2250** includes multiple ILD layers and interconnects therein formed in multiple processes. For example, the interconnects in interconnect layers **2250** can include conductive materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, electroplating, electroless plating, or any combination thereof. Fabrication processes to form the interconnects can also include photolithography, CMP, wet/dry etch, or any other suitable processes. The ILD layers can include dielectric materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. The ILD layers and interconnects illustrated in FIG. **22**L can be collectively referred to as interconnect layer **2250**.
(292) Method **2300** proceeds to operation **2304**, as illustrated in FIG. **23**, in which a first bonding layer is formed above the peripheral circuit (and the interconnect layer). The first bonding layer can include a first bonding contact. As illustrated in FIG. **22**L, a bonding layer **2252** is formed above

interconnect layer **2250** and peripheral circuits **2246**. Bonding layer **2252** can include a plurality of bonding contacts surrounded by dielectrics. In some implementations, a dielectric layer (e.g., ILD layer) is deposited on the top surface of interconnect layer **2250** by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. The bonding contacts can then be formed through the dielectric layer and in contact with the interconnects in interconnect layer **2250** by first patterning contact holes through the dielectric layer using patterning process (e.g., photolithography and dry/wet etch of dielectric materials in the dielectric layer). The contact holes can be filled with a conductor (e.g., Cu). In some implementations, filling the contact holes includes depositing a barrier layer, an adhesion layer, and/or a seed layer before depositing the conductor.

(293) Method **2300** proceeds to operation **2306**, as illustrated in FIG. **23**, in which an array of memory cells each including a vertical transistor and a storage unit is formed on a second substrate. The second substrate can include a carrier substrate. The storage unit can include a capacitor or a PCM element. In some implementations, a capacitor is formed to be coupled to the vertical transistor in the respective memory cell.

(294) For example, FIG. **26** illustrates a flowchart of a method **2600** for forming still another array of memory cells each including a vertical transistor, according to some aspects of the present disclosure. At operation **2602** in FIG. **26**, a semiconductor body extending vertically in a substrate is formed. The substrate can be an SOI substrate including a handle layer, a buried oxide layer, and a device layer. In some implementations, to form the semiconductor body, the handle layer is etched in a first lateral direction to form first trenches, and the handle layer is etched in a second lateral direction to form second trenches, such that the two opposite sides of the semiconductor body is exposed by the second trenches. In some implementations, a dielectric is deposited to partially fill the second trenches.

(295) As illustrated in FIG. **22**A, a plurality of parallel trenches **2204** are formed in the y-direction (e.g., the bit line direction) to form a plurality of parallel semiconductor walls **2205** in the y-direction. In some implementations, a lithography process is performed to pattern trenches **2204** and semiconductor walls **2205** using an etch mask (e.g., a photoresist mask and/or a hard mask), for example, based on the design of bit lines, and one or more dry etching and/or wet etching processes, such as RIE, are performed to etch trenches **1904** in an SOI substrate **2201**. Thus, semiconductor wall **1905** extending vertically in SOI substrate **2201** can be formed. As shown in FIG. **22**A, SOI substrate **2201** can include a handle layer **2202**, a buried oxide layer **2203** on handle layer **2202**, and a device layer **2209** on buried oxide layer **2203**. In some implementations, buried oxide layer **2203** includes silicon oxide, and device layer **2209** includes single crystalline silicon. In some implementations, to form trenches **2204**, device layer **2209** is etched using RIE, stopped at buried oxide layer **2203**. That is, buried oxide layer **2203** can serve as the etch stop layer. It is understood that in some examples, device layer **2209** may not be part of an SOI substrate, but transferred and bonded onto buried oxide layer **2203** from another silicon substrate (not shown, e.g., an SOI substrate). It is also understood that in some examples, SOI substrate **2201** may be replaced with a silicon substrate, such as silicon substrate **1902** in FIG. **19**A; the etching of trenches **2204** may not be stopped by buried oxide layer **2203**, but by controlling the etching rate and/or duration, for example, as shown in FIG. **19**A

(296) Nevertheless, the bottom of semiconductor wall **2205** can be below the top surface of SOI substrate **2201**. Since semiconductor walls **2205** are formed by etching device layer **2209** of SOI substrate **2201**, semiconductor walls **2205** can have the same material as device layer **2209** of SOI substrate **2201**, such as single crystalline silicon. FIG. **22**A illustrates both the side view (in the top portion of FIG. **22**A) of a cross-section along the x-direction (the word line direction, e.g., in the BB plane) and the plan view (in the bottom portion of FIG. **22**A) of a cross-section in the x-y plane (e.g., in the AA plane through semiconductor walls **2205**).

(297) As illustrated in FIG. **22**B, a plurality of parallel trenches **2210** are formed in the x-direction

(e.g., the word line direction) to form an array of semiconductor bodies **2206** each extending vertically in SOI substrate **2201**. In some implementations, a lithography process is performed to pattern trenches **2210** to be perpendicular to trenches **2204** using an etch mask (e.g., a photoresist mask and/or a hard mask), for example, based on the design of word lines, and one or more dry etching and/or wet etching processes, such as RIE, are performed to etch trenches **2210** in device layer **2209** of SOI substrate **2201**. As a result, semiconductor walls **2205** (shown in FIG. **22**A) can be cut by trenches **2210** to form an array of semiconductor bodies **2206** each extending vertically in SOI substrate **2201**. The bottom of semiconductor body **2206** can be below the top surface of SOI substrate **2201**. Since semiconductor bodies **2206** are formed by etching device layer **2209** of SOI substrate **2201**, semiconductor bodies **2206** can have the same material as device layer **2209** of SOI substrate **2201**, such as single crystalline silicon. FIG. **22**B illustrates both the side view (in the top portion of FIG. **22**B) of a cross-section along the y-direction (the bit line direction, e.g., in the CC plane) and the plan view (in the bottom portion of FIG. **22**B) of a cross-section in the x-y plane (e.g., in the AA plane through semiconductor bodies **2206**). The same drawing layout is arranged in FIGS. **22**C-**22**G as well.

(298) It is understood that in some examples, trenches **2204** and **2210** may be formed in the same process, as opposed to two consecutive processes. For example, the same lithography process may be used to pattern and trenches **2204** and **2210**, followed by the same etching process. It is also understood that in some examples, trenches **2210** in the word line direction may be formed prior to the formation of trenches **2204** in the bit line direction. Nevertheless, after the formation of trenches **2204** and **2210**, semiconductor body **2206** can be formed, and all four sides of semiconductor body **2206** can be exposed by trenches **2204** and **2210**. In some implementations, two opposite sides of semiconductor body **2206** in the word line direction are exposed by trenches **2204**, and two opposite sides of semiconductor body **2206** in the bit line direction are exposed by trenches **2210**, As shown in the plan view. In other words, semiconductor body **2206** can be surrounded by trenches **2204** and **2210**.

(299) As illustrated in FIG. **22**C, a dielectric layer **2212** is formed at the bottom of trench **2210** (and trench **2204** in some examples), for example, by depositing a dielectric, such as silicon oxide, to partially fill trench **2210**, using one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. The deposition conditions, such as deposition rate and/or time, can be controlled to control the thickness of dielectric layer **2212** and avoid fully filling trench **2210**. As a result, the bottom surface of trenches **2210** can be elevated to be above the bottom surface of semiconductor bodies **2206**.

(300) At operation **2604** in FIG. **26**, a gate structure in contact with opposite sides of the semiconductor body is formed. In some implementations, to form the gate structure, a gate dielectric is formed over the opposite sides of the semiconductor body, and a gate electrode is formed over the gate dielectric. In some implementations, to form the gate electrode, a conductive layer is deposited over the gate dielectric, and the conductive layer is etched back.

(301) As illustrated in FIG. **22**D, a gate dielectric **2214** is formed over the two opposite sides of semiconductor body **2206** in the bit line direction (the y-direction) exposed from trenches **2210**. As shown in the plan view, gate dielectrics **2214** can be parts of a continuous dielectric layer formed over sidewalls of each row of semiconductor bodies **2206**. In some implementations, gate dielectric **2214** is formed by depositing a layer of dielectric, such as silicon oxide, over the sidewalls and top surfaces of semiconductor bodies **2206** using one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof, without fully filling trenches **2210**. It is understood that in some examples, gate dielectrics **2214** may not be parts of a continuous dielectric layer. For example, a wet oxidation and/or a dry oxidation process, such as ISSG oxidation, is performed to form native oxide (e.g., silicon oxide) on semiconductor bodies **2206** (e.g., single crystalline silicon) as gate dielectric **2214**.

(302) As illustrated in FIG. **22**D, a conductive layer **2216** is formed over gate dielectrics **2214**. In

some implementations, conductive layers **2216** are formed by depositing one or more conductive materials, such as metal and/or metal compounds (e.g., W and TiN), over gate dielectrics **2214** using one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof, to partially fill trenches **2210**. For example, layers of TiN and W may be sequentially deposited to form conductive layer **2216**. As shown in the side view, conductive layers **2216** can be a continuous layer in the bit line direction as the conductive materials can be deposited over the top surfaces of semiconductor bodies **2206** and the bottom surfaces of trenches **2210**.

(303) As illustrated in FIG. **22**E, in some implementations, parts of conductive layers **2216** at the bottom surfaces of trenches **2210** are removed to separate the continuous conductive layers **2216** into discrete pieces in the bit line direction, for example, using dry etch and/or we etch e.g., RIE) to form cuts **2211** on the bottom surfaces of trenches **2210**. In some implementations, parts of conductive layers **2216** at the top surfaces of semiconductor bodies **2206** are removed as well by the same etching process to expose gate dielectrics **2214** at the top surfaces of semiconductor bodies **2206**.

(304) As illustrated in FIG. **22**F, in some implementations, trench isolations **2218** are formed in trench **2210** (shown in FIG. **22**E), for example, by depositing a dielectric, such as silicon oxide, to fill trench **2210**, using one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. A planarization process (e.g., CMP and/or etching process) can be performed to remove excess dielectric over the top surface of semiconductor bodies **2206**. In some implementations, the planarization process removes parts of gate dielectrics **2214** over the top surfaces of semiconductor bodies **2206** as well to expose the top surfaces of semiconductor bodies **2206**. It is understood that depending on the pitches of the semiconductor bodies **2206** (the dimension of trenches **2210**), air gaps may be formed in trench isolation **2218**. As shown in the plan view, the deposition of the dielectric may fill the remaining spaces of trenches **2204** (shown in FIG. **22**E) as well to form isolations **2219** between adjacent semiconductor bodies **2206** in the word line direction (e.g., in the same row).

(305) As illustrated in FIG. **22**G, in some implementations, conductive layers **2216** are etched back, for example, using dry etch and/or wet etch (e.g., RIE), to form dents, such that the upper ends of conductive layers **2216** are below the top surface of semiconductor bodies **1906**. In some implementations, as gate dielectrics **2214** are not etched back, the upper ends of conductive layers **2216** are below the upper ends of gate dielectrics **2214** as well, which are flush with the top surface of semiconductor bodies **2206**. As a result, etched-back conductive layers **2216** can become word lines each extending in the word line direction (the x-direction), and parts of etched-back conductive layers **2216** that are facing semiconductor bodies **2206** can become gate electrodes. Gate structures each including a respective gate dielectric **2214** over the exposed two opposite sides (in the bit line direction) of semiconductor body **2206** and a respective gate electrode (part of conductive layer **2216**) over gate dielectric **2214** can be formed thereby.

(306) At operation **2608** in FIG. **26**, a first end of the semiconductor body away from the substrate is doped. As illustrated in FIG. **22**G, the exposed upper end (top surface) of each semiconductor body **2206**, e.g., one of the two ends of semiconductor body **2206** in the vertical direction (the z-direction) that is away from handle layer **2202** of SOI substrate **2201**, is doped to form a source/drain **2224** (e.g., a source terminal of a vertical transistor). In some implementations, an implantation process and/or thermal diffusion process are performed to dope P-type dopants or N-type dopants to exposed upper ends of semiconductor bodies **2206** to form sources/drains **2224**. In some implementations, a silicide layer is formed on source/drain **2224** by performing a silicidation process at the exposed upper ends of semiconductor bodies **2206**.

(307) At operation **2608** in FIG. **26**, a storage unit in contact with the semiconductor body, e.g., the doped first end thereof, is formed. The storage unit can include a capacitor or a PCM element. In some implementations, to form a storage unit that is a capacitor, a first electrode is formed on the doped first end of the semiconductor body, a capacitor dielectric is formed on the first electrode,

and a second electrode is formed on the capacitor dielectric.

(308) As illustrated in FIG. **22**H, one or more ILD layers are formed over the top surface of semiconductor bodies **2206**, for example, by depositing dielectrics using one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. Capacitor contacts **2228**, first electrodes, capacitor dielectrics, and second electrodes of capacitors **2230**, and a common plate **2232** are subsequently formed in the ILD layers to be coupled to semiconductor bodies **2206**. In some implementations, each capacitor contact **2228** is formed on a respective source/drain **2224**, e.g., the doped upper end of a respective semiconductor body **2206** by patterning and etching an electrode hole aligned with respective source/drain **2224** using lithography and etching processes and depositing conductive materials to fill the electrode hole using thin film deposition processes. In some implementations, common plate **2232** is formed on capacitors **2230** by patterning and etching an electrode trench aligned with capacitors dielectrics **2230** using lithography and etching processes and depositing conductive materials to fill the electrode trench using thin film deposition processes.

(309) At operation **2610** in FIG. **26**, the substrate is thinned to expose a second end of the semiconductor body opposite to the first end. As illustrated in FIG. **22**, a carrier substrate **2234** (a.k.a. a handle substrate) is bonded onto the front side of SOI substrate **2201** on which devices are formed using any suitable bonding processes, such as anodic bonding, fusion bonding, transfer bonding, adhesive bonding, and eutectic bonding. The bonded structure can then be flipped upside down, such that handle layer **2202** of SOI substrate **2201** become above carrier substrate **2234**.

(310) As illustrated in FIG. **22**J, SOI substrate **2201** is thinned to expose the undoped upper ends of semiconductor bodies **2206** (used to be the lower ends before flipping over). In some implementations, planarization processes (e.g., CMP) and/or etching processes are performed to remove handle layer **2202** and buried oxide layer **2203** (shown in FIG. **22**G) of SOI substrate **2201** until being stopped by the upper ends of semiconductor bodies **2206**.

(311) At operation **2612** in FIG. **26**, the exposed second end of the semiconductor body is doped. As illustrated in FIG. **22**J, the exposed upper end of each semiconductor body **2206**, e.g., one of the two ends of semiconductor body **2206** in the vertical direction (the z-direction) that is away from carrier substrate **2234**, is doped to form another source/drain **2236** (e.g., a drain terminal of the vertical transistor). In some implementations, an implantation process and/or thermal diffusion process are performed to dope P-type dopants or N-type dopants to exposed upper ends of semiconductor bodies **2206** to form sources/drains **2236**. In some implementations, a silicide layer is formed on source/drain **2236** by performing a silicidation process at the exposed upper ends of semiconductor bodies **2206**. As a result, vertical transistors having semiconductor body **2206**, sources/drains **2224** and **2236**, gate dielectric **2214**, and the gate electrode (part of conductive layer **2216**) are formed thereby, as shown in FIG. **22**J, according to some implementations. As described above, capacitors **2230** are thereby formed as well, and DRAM cells **2280** each having a double-gate vertical transistor and a capacitor coupled to the double-gate vertical transistor are thereby formed, as shown in FIG. **22**J, according to some implementations.

(312) Referring back to FIG. **23**, method **2300** proceeds to operation **2308**, as illustrated in FIG. **23**, in which an interconnect layer including bit lines is formed above the array of memory cells. As illustrated in FIG. **22**K, an interconnect layer **2240** can be formed above DRAM cells **2280**. Interconnect layer **2240** can include interconnects of MEOL and/or BEOL in a plurality of ILD layers to make electrical connections with DRAM cells **2280**. In some implementations, interconnect layer **2240** includes multiple ILD layers and interconnects therein formed in multiple processes. For example, the interconnects in interconnect layers **2240** can include conductive materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, electroplating, electroless plating, or any combination thereof. Fabrication processes to form the interconnects can also include photolithography, CMP, wet/dry etch, or any other suitable processes. The ILD layers can include dielectric materials deposited using one or

more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. The ILD layers and interconnects illustrated in FIG. **22**K can be collectively referred to as interconnect layer **2240**.

(313) As shown in FIG. **26**, at operation **2614**, to form the interconnect layer, a bit line is formed on the doped second end. As illustrated in FIG. **22**K, bit line **2238** can be formed on sources/drains **2236** by patterning and etching a trench aligned with respective source/drain **2236** using lithography and etching processes and depositing conductive materials to fill the trench using thin film deposition processes. In some implementations, forming bit line **2238** includes depositing a metal layer onto the exposed end of semiconductor body **2220**. As a result, bit line **2238** and capacitor **2230** can be formed on opposite sides of semiconductor body **2206** and coupled to opposite ends of semiconductor body **2206**. It is understood that additional local interconnects, such as word line contacts, capacitor contacts (e.g., a conductor), and bit line contacts (e.g., a metal silicide contact) may be similarly formed as well. In some implementations, the bit line contact (e.g., a metal silicide contact) is formed on the exposed end of semiconductor body **2220**, and bit line **2238** is formed on the bit line contact.

(314) Method **2300** proceeds to operation **2310**, as illustrated in FIG. **23**, in which a second bonding layer is formed above the array of memory cells and the interconnect layer. The second bonding layer can include a second bonding contact. As illustrated in FIG. **22**K, a bonding layer **2242** is formed above interconnect layer **2240** and DRAM cells **2280**. Bonding layer **2242** can include a plurality of bonding contacts surrounded by dielectrics. In some implementations, a dielectric layer (e.g., ILD layer) is deposited on the top surface of interconnect layer **2240** by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. The bonding contacts can then be formed through the dielectric layer and in contact with the interconnects in interconnect layer **2240** by first patterning contact holes through the dielectric layer using patterning process (e.g., photolithography and dry/wet etch of dielectric materials in the dielectric layer). The contact holes can be filled with a conductor (e.g., Cu). In some implementations, filling the contact holes includes depositing a barrier layer, an adhesion layer, and/or a seed layer before depositing the conductor.

(315) Method **2300** proceeds to operation **2312**, as illustrated in FIG. **23**, in which the first semiconductor structure and the second semiconductor structure are bonded in a face-to-face manner, such that the first array of memory cells is coupled to the peripheral circuit across a bonding interface. The bonding can include hybrid bonding. In some implementations, the first bonding contact is in contact with the second bonding contact at the bonding interface after the bonding. In some implementations, the second semiconductor structure is above the first semiconductor structure after the bonding. In some implementations, the first semiconductor structure is above the second semiconductor structure after the bonding.

(316) As illustrated in FIG. **22**L, carrier substrate **2234** and components formed thereon (e.g., DRAM cells **2280**) are flipped upside down. As illustrated in FIG. **22**L, bonding layer **2242** facing down is bonded with bonding layer **2252** facing up, e.g., in a face-to-face manner, thereby forming a bonding interface **2254**. In some implementations, a treatment process, e.g., a plasma treatment, a wet treatment, and/or a thermal treatment, is applied to the bonding surfaces prior to the bonding. Although not shown in FIG. **22**L, silicon substrate **2244** and components formed thereon (e.g., peripheral circuits **2246**) can be flipped upside down, and bonding layer **2252** facing down can be bonded with bonding layer **2242** facing up, e.g., in a face-to-face manner, thereby forming bonding interface **2254**. After the bonding, the bonding contacts in bonding layer **2242** and the bonding contacts in bonding layer **2252** are aligned and in contact with one another, such that DRAM cells **2280** can be electrically connected to peripheral circuits **2246** across bonding interface **2254**. It is understood that in the bonded chip, DRAM cells **2280** may be either above or below peripheral circuits **2246**. Nevertheless, bonding interface **2254** can be formed vertically between peripheral circuits **2246** and DRAM cells **2280** after the bonding.

(317) Method **2300** proceeds to operation **2314**, as illustrated in FIG. **23**, in which a pad-out interconnect layer is formed on the backside of the first semiconductor structure or the second semiconductor structure. As illustrated in FIG. **22**M, a pad-out interconnect layer **2256** is formed above on the backside of carrier substrate **2234**. Pad-out interconnect layer **2256** can include interconnects, such as pad contacts **2258**, formed in one or more ILD layers. Pad contacts **2258** can include conductive materials including, but not limited to, W, Co, Cu, Al, doped silicon, silicides, or any combination thereof. The ILD layers can include dielectric materials including, but not limited to, silicon oxide, silicon nitride, silicon oxynitride, low-k dielectrics, or any combination thereof. In some implementations, after the bonding, contacts **2260** are formed extending vertically through carrier substrate **2234**, for example, by wet/dry etching processes, followed by depositing conductive materials. Contacts **2260** can be in contact with the interconnects in pad-out interconnect layer **2256**. It is understood that in some examples, carrier substrate **2234** may be thinned or removed after bonding and prior to forming pad-out interconnect layer **2256** and contacts **2260**, for example, using planarization processes and/or etching processes.
(318) Although not shown, it is understood that in some examples, pad-out interconnect layer **2256** may be formed above on the backside of silicon substrate **2244**, and contacts **2260** may be formed extending vertically through silicon substrate **2244**. Silicon substrate **2244** may be thinned prior to forming pad-out interconnect layer **2256** and contacts **2260**, for example, using planarization processes and/or etching processes. Although not shown, it is further understood that in some examples, the fabrication processes described with respect to FIGS. **14**A-**14**E and **15**A-**15**D may be applied to form another array of DRAM cells **2280** in another semiconductor structure bonded to the semiconductor structure including DRAM cells **2280** described above with respect to FIGS. **22**A-**22**M.
(319) According to one aspect of the present disclosure, a 3D memory device includes a first semiconductor structure, a second semiconductor structure, a third semiconductor structure, a first bonding interface between the first semiconductor structure and the second semiconductor structure, and a second bonding interface between the second semiconductor structure and the third semiconductor structure. The first semiconductor structure includes a peripheral circuit. The second semiconductor structure includes a first array of memory cells. The third semiconductor structure includes a second array of memory cells. Each of the memory cells of the first and second arrays includes a vertical transistor extending in a first direction, and a storage unit coupled to the vertical transistor. The first array of memory cells is coupled to the peripheral circuit across the first bonding interface. The second array of memory cells is coupled to the peripheral circuit across the first bonding interface and the second bonding interfaces.
(320) In some implementations, the storage units in the first array of memory cells are disposed between the second bonding interface and the vertical transistors in the first array of memory cells, and the storage units in the second array of memory cells are disposed between the second bonding interface and the vertical transistors in the second array of memory cells.
(321) In some implementations, the third semiconductor structure further includes a pad-out interconnect layer, and the vertical transistors in the second array of memory cells are disposed between the storage units in the second array of memory cells and the pad-out interconnect layer.
(322) In some implementations, the first semiconductor structure further includes a first bonding layer including a first bonding contact, the second semiconductor structure further includes a second bonding layer including a second bonding contact, and the first bonding contact is in contact with the second bonding contact at the first bonding interface.
(323) In some implementations, the second and third semiconductor structures each further includes a plurality of bit lines each extending in a second direction perpendicular to the first direction, and a respective one of the bit lines and a respective storage unit are coupled to opposite ends of each one of the memory cells in the first direction.
(324) In some implementations, the second and third semiconductor structures each further

includes a plurality of word lines each extending in a third direction perpendicular to the first direction and the second direction. In some implementations, the vertical transistor includes a semiconductor body extending in the first direction, and a gate structure in contact with one or more sides of the semiconductor body.
(325) In some implementations, the gate structure includes a gate electrode, and a gate dielectric between the gate electrode and the semiconductor body in the second direction, and the gate dielectrics of two adjacent vertical transistors of the vertical transistors in the third direction are continuous.
(326) In some implementations, two adjacent vertical transistors of the vertical transistors in the second direction are mirror-symmetric to one another.
(327) In some implementations, the second semiconductor structure further includes a trench isolation extending in the third direction and disposed between the two adjacent vertical transistors in the second direction.
(328) In some implementations, the gate structure is in contact with one side of the semiconductor body in the second direction.
(329) In some implementations, the gate structure is in contact with two opposite sides of the semiconductor body in the second direction.
(330) In some implementations, the gate structure includes a gate electrode, and a gate dielectric between the gate electrode and the semiconductor body in the second and third directions, and the gate dielectrics of two adjacent vertical transistors of the vertical transistors in the third direction are separate.
(331) In some implementations, the vertical transistor is a GAA transistor in which the gate structure fully circumscribes the semiconductor body in a plan view.
(332) In some implementations, the gate structure partially circumscribes the semiconductor body in a plan view.
(333) In some implementations, the second semiconductor structure further includes a plurality of air gaps each disposed between adjacent word lines of the word lines in the second direction.
(334) In some implementations, the vertical transistor further includes a source and a drain disposed at two ends of the semiconductor body, respectively, in the first direction.
(335) In some implementations, one of the source and the drain of the vertical transistor is coupled to the storage unit in a respective memory cell.
(336) In some implementations, another one of the source and the drain of the vertical transistor is coupled to the respective bit line.
(337) In some implementations, two ends of the semiconductor body in the first direction extend beyond the gate structure, respectively.
(338) In some implementations, the bit lines in the second semiconductor structure are disposed between the vertical transistors in the second semiconductor structure and the first bonding interface.
(339) In some implementations, the vertical transistors in the third semiconductor structure are disposed between the bit lines in the third semiconductor structure and the second bonding interface.
(340) In some implementations, wherein the memory cells comprise at least a DRAM cell, a PCM cell, or a FRAM cell.
(341) According to another aspect of the present disclosure, a memory system includes a memory device configured to store data and a memory controller coupled to the memory device. The memory device includes a first semiconductor structure, a second semiconductor structure, a third semiconductor structure, a first bonding interface between the first semiconductor structure and the second semiconductor structure, and a second bonding interface between the second semiconductor structure and the third semiconductor structure. The first semiconductor structure includes a peripheral circuit. The second semiconductor structure includes a first array of memory cells. The

third semiconductor structure includes a second array of memory cells. Each of the memory cells of the first and second arrays includes a vertical transistor extending in a first direction, and a storage unit coupled to the vertical transistor. The first array of memory cells is coupled to the peripheral circuit across the first bonding interface. The second array of memory cells is coupled to the peripheral circuit across the first bonding interface and the second bonding interfaces. The memory controller is configured to control the first and second arrays of memory cells through the peripheral circuit.
(342) In some implementations, the memory system further includes a host coupled to the memory controller and configured to send or receive the data to or from the memory device.
(343) In some implementations, the memory cells comprise at least a DRAM cell, a PCM cell, or a FRAM cell.
(344) According to still another aspect of the present disclosure, a method for forming a 3D memory device is disclosed. A first semiconductor structure including a peripheral circuit is formed. A second semiconductor structure including a first array of memory cells is formed. Each of the memory cells of the first array includes a vertical transistor extending in the first direction, and a storage unit coupled to the vertical transistor. A third semiconductor structure including a second array of memory cells is formed. Each of the memory cells of the second array includes a vertical transistor extending in the first direction, and a storage unit coupled to the vertical transistor. The first semiconductor structure and the second semiconductor structure are bonded in a face-to-face manner, such that the first array of memory cells is coupled to the peripheral circuit across a bonding interface. The second semiconductor structure and the third semiconductor structure are bonded in a face-to-face manner.
(345) In some implementations the second and third semiconductor structures are bonded prior to bonding the first and second semiconductor structures.
(346) In some implementations, the second and third semiconductor structures are bonded after bonding the first and second semiconductor structures.
(347) In some implementations, a pad-out interconnect layer is formed on a backside of the third semiconductor structure after bonding the second and third semiconductor structures.
(348) In some implementations, to form the first semiconductor structure, a first bonding layer including a first bonding contact is formed. In some implementations, to form the second semiconductor structure, a second bonding layer including a second bonding contact is formed. In some implementations, the first bonding contact is in contact with the second bonding contact at the bonding interface after bonding the first and second semiconductor structures.
(349) In some implementations, to form the second semiconductor structure, the first array of memory cells is formed, and an interconnect layer including bit lines above the first array of memory cells prior to forming the second bonding layer.
(350) In some implementations, to form the first array of memory cells, a semiconductor body extending vertically is formed, a gate structure in contact with one or more sides of the semiconductor body is formed, and the storage unit in contact with one end of the semiconductor body vertically is formed.
(351) In some implementations, bonding the first and second semiconductor structures includes hybrid bonding.
(352) The foregoing description of the specific implementations can be readily modified and/or adapted for various applications. Therefore, such adaptations and modifications are intended to be within the meaning and range of equivalents of the disclosed implementations, based on the teaching and guidance presented herein.
(353) The breadth and scope of the present disclosure should not be limited by any of the above-described exemplary implementations, but should be defined only in accordance with the following claims and their equivalents.

Claims

1. A three-dimensional (3D) memory device, comprising: a first semiconductor structure comprising a peripheral circuit; a second semiconductor structure comprising a first array of memory cells; a third semiconductor structure comprising a second array of memory cells, wherein each of the memory cells of the first and second arrays comprises a vertical transistor extending in a first direction, and a storage unit coupled to the vertical transistor; a first bonding interface between the first semiconductor structure and the second semiconductor structure in the first direction, wherein the first array of memory cells is coupled to the peripheral circuit across the first bonding interface; and a second bonding interface between the second semiconductor structure and the third semiconductor structure in the first direction, wherein the second array of memory cells is coupled to the peripheral circuit across the first bonding interface and the second bonding interfaces, wherein: the peripheral circuit is configured to send a first signal to a first word line of the first array of memory cells in the second semiconductor structure through the first bonding interface; and the peripheral circuit is configured to send a second signal to a second word line of the second array of memory cells in the third semiconductor structure through the first bonding interface, the second bonding interface, and a contact that extends through the second semiconductor structure.
2. The 3D memory device of claim 1, wherein the storage units in the first array of memory cells are disposed between the second bonding interface and the vertical transistors in the first array of memory cells; the storage units in the second array of memory cells are disposed between the second bonding interface and the vertical transistors in the second array of memory cells; and the storage units in the first array of memory cells and the storage units in the second array of memory cells are symmetric with respect to the second bonding interface to arrange a common electrode at the second bonding interface.
3. The 3D memory device of claim 1, wherein the third semiconductor structure further comprises a pad-out interconnect layer; and the vertical transistors in the second array of memory cells are disposed between the storage units in the second array of memory cells and the pad-out interconnect layer.
4. The 3D memory device of claim 1, wherein the first semiconductor structure further comprises a first bonding layer comprising a first bonding contact, and the second semiconductor structure further comprises a second bonding layer comprising a second bonding contact; and the first bonding contact is in contact with the second bonding contact at the first bonding interface.
5. The 3D memory device of claim 1, wherein each of the second and third semiconductor structures further comprises a plurality of bit lines each extending in a second direction perpendicular to the first direction; and a respective one of the bit lines and a respective storage unit are coupled to opposite ends of each one of the memory cells in the first direction.
6. The 3D memory device of claim 5, wherein each of the second and third semiconductor structures further comprises a plurality of word lines each extending in a third direction perpendicular to the first direction and the second direction, the plurality of word lines in the second semiconductor structure comprising the first word line, and the plurality of word lines in the third semiconductor structure comprising the second word line; and the vertical transistor comprises: a semiconductor body extending in the first direction; and a gate structure in contact with one or more sides of the semiconductor body.
7. The 3D memory device of claim 6, wherein the gate structure comprises a gate electrode, and a gate dielectric between the gate electrode and the semiconductor body in the second direction; and the gate dielectrics of two adjacent vertical transistors of the vertical transistors in the third direction are continuous.
8. The 3D memory device of claim 7, wherein two adjacent vertical transistors of the vertical

transistors in the second direction are mirror-symmetric to one another.
9. The 3D memory device of claim 8, wherein the second semiconductor structure further comprises a trench isolation extending in the third direction and disposed between the two adjacent vertical transistors in the second direction.
10. The 3D memory device of claim 7, wherein the gate structure is in contact with one side of the semiconductor body in the second direction.
11. The 3D memory device of claim 7, wherein the gate structure is in contact with two opposite sides of the semiconductor body in the second direction.
12. The 3D memory device of claim 6, wherein the gate structure comprises a gate electrode, and a gate dielectric between the gate electrode and the semiconductor body in the second and third directions; and the gate dielectrics of two adjacent vertical transistors of the vertical transistors in the third direction are separate.
13. The 3D memory device of claim 12, wherein the vertical transistor is a gate-all-around (GAA) transistor in which the gate structure fully circumscribes the semiconductor body in a plan view.
14. The 3D memory device of claim 6, wherein the vertical transistor further comprises a source and a drain disposed at two ends of the semiconductor body, respectively, in the first direction.
15. The 3D memory device of claim 14, wherein one of the source and the drain of the vertical transistor is coupled to the storage unit in a respective memory cell; and another one of the source and the drain of the vertical transistor is coupled to the respective bit line.
16. The 3D memory device of claim 5, wherein the bit lines in the second semiconductor structure are disposed between the vertical transistors in the second semiconductor structure and the first bonding interface.
17. The 3D memory device of claim 5, wherein the vertical transistors in the third semiconductor structure are disposed between the bit lines in the third semiconductor structure and the second bonding interface.
18. A memory system, comprising: a memory device configured to store data, and comprising: a first semiconductor structure comprising a peripheral circuit; a second semiconductor structure comprising a first array of memory cells; a third semiconductor structure comprising a second array of memory cells, wherein each of the memory cells of the first and second arrays comprises a vertical transistor extending in a first direction, and a storage unit coupled to the vertical transistor; a first bonding interface between the first semiconductor structure and the second semiconductor structure in the first direction, wherein the first array of memory cells is coupled to the peripheral circuit across the first bonding interface; and a second bonding interface between the second semiconductor structure and the third semiconductor structure in the first direction, wherein the second array of memory cells is coupled to the peripheral circuit across the first bonding interface and the second bonding interfaces, wherein the storage units of the first array of memory cells and the storage units of the second array of memory cells are symmetric with respect to the second bonding interface; and a memory controller coupled to the memory device and configured to control the first and second arrays of memory cells through the peripheral circuit.
19. The memory device of claim 1, wherein a common electrode is arranged at the second bonding interface and connected with the storage units of the first array of memory cells and the storage units of the second array of memory cells.
20. The memory device of claim 1, wherein a common electrode is arranged at the second bonding interface and connected with the storage units of the first array of memory cells and the storage units of the second array of memory cells; and the peripheral circuit is further configured to send a third signal to the common electrode to control the first array of memory cells and the second array of memory cells.